

Cumulant geometry and controlled memory

Conditional covariance, relative determinants, and a quantitative modular bridge

A continuation of the investigation into the missing third-to-fourth cumulant feed

Main development. The omitted fourth response has an exact conditional mechanism: an incoming third-order source perturbs conditional covariances and their connected correlations with other neurons. Quadratic residuals and a relative determinant give complementary operator descriptions. A bounded Witten deformation connects the same source to a moving harmonic state. A sharp modular cross-ratio bound supplies conditional information contraction, and compatible Gramian projections turn observable compression into an explicit reduced recurrence. **Scope.**

Theorems, proofs, constructive approximations, and counterexamples at specified reference laws. These are not numerical results for the two reported networks. Their exact source calibration, conditional arrays, weights, and readout conventions were not supplied.

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Based on the supplied Note XIX and related programme notes.

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1 The mathematical target and the advances

The target is the deep response missing from the existing two-gain ledger. The retained fourth-cumulant index classes are (4) and (2, 2); the omitted classes are (3, 1), (2, 1, 1) and (1, 1, 1, 1). We seek their coherent weighted response to a declared incoming third-cumulant tensor. A second, distinct question is whether the resulting observables admit a small state description that remains valid across depth.

The supplied Note XIX reports the numerical deficit and the accuracy of the earlier ledger. This continuation treats those observations as project reports. It does not reproduce the runs, infer an unavailable covariance matrix, or silently choose a different unit-gain tangent.

1.1 The strongest results

Bridge	Exact result and its consequence
Conditional cumulants	A pair source creates a conditional covariance response d_{11}^{ij} . For four distinct output neurons, the fourth response is the sum over six pairs of $\kappa_X(d_{11}^{ij}, m_k, m_l)$. Fully distinct incoming triples and diagonal sources have their own explicit marked-block terms.
Quadratic observables	The fourth cumulant splits as $t^\top C^{-1}t + \ R\ _2^2 - 2v^2$. Under a source h , the optimal residual energy has derivative $\mathbb{E}[hR^2]$. This specifies the additional query beyond third moments.
Relative determinant	For a compression boundary $H = QAP$, the fourth logarithmic coefficient is $\ dH - Ha\ _{\text{HS}}^2 - 2\ H^*H\ _{\text{HS}}^2$. The probability vacuum identifies it with the ordinary fourth cumulant.
Witten response	The Gaussian Dirac index remains one under a bounded density deformation, while its harmonic state moves by $-\mathcal{D}_\perp^{-1}\dot{\mathcal{D}}\Omega$. This is the same displacement that generates the cumulant response.
Modular contraction	For a positive conditional table with log-cross-ratio diameter Δ , maximal correlation is at most $\tanh(\Delta/4)$, sharply. The diameter is twice an explicitly defined modular-generator leakage.
Conditional smoothing	Residual Gaussian noise gives exact Mehler multipliers. Hermite tails and clipped approximants provide certified evaluation; one latent factor alone does not imply a small exact covariance rank.
Compatible dynamics	Reachability-metric projections yield an explicit nonautonomous reduced recurrence, with full input/output error bounded by a sum of discarded past-to-future singular values. An observable gap gives uniform finite memory.

All these assertions are proved in the body. Each transition from a static formula to a dynamic approximation is stated with its own hypotheses. Counterexamples show why one scale variable, one-step fitted kernels, or unary smoothing cannot replace those hypotheses.

1.2 A self-contained definition of the omitted response

Let $H_i = (Z_i)_+$ and fix a downstream row w . At the reference law put

$$u_i = w_i(H_i - \mathbb{E}H_i), \quad S_j = \sum_i u_i^j, \quad C_{ij} = \mathbb{E}[u_i u_j],$$

$$V = \mathbb{E}S_1^2, \quad D = \text{tr } C, \quad R_C = \sum_{i \neq j} C_{ij}^2.$$

The sum in R_C is ordered. Define

$$P(u) = S_1^4 - 3S_2^2 + 2S_4.$$

Then the exact omitted coherent cumulant is

$$K_{\text{miss}}(w) = \mathbb{E}P(u) - 3(V^2 - D^2) + 6R_C. \quad (1)$$

To verify it, the retained fourth contraction is

$$K_{\text{ret}} = \mathbb{E}(3S_2^2 - 2S_4) - 3D^2 - 6R_C,$$

obtained by expanding the (4) and (2, 2) multiplicities. Subtract this from $\kappa_4(S_1) = \mathbb{E}S_1^4 - 3V^2$.

For a density score h with $\mathbb{E}h = 0$, set

$$b_i = \mathbb{E}[\partial_{u_i}P] = 4\mathbb{E}S_1^3 - 12\mathbb{E}[S_2u_i] + 8\mathbb{E}u_i^3.$$

With C_{off} denoting C with its diagonal set to zero, the full influence observable is

$$\begin{aligned} \Psi_{\text{miss}}(u) &= P(u) - b^\top u - 6VS_1^2 + 6DS_2 \\ &\quad + 12u^\top C_{\text{off}}u. \end{aligned} \quad (2)$$

Its response is

$$\delta_h K_{\text{miss}} = \mathbb{E}[h\Psi_{\text{miss}}]. \quad (3)$$

Indeed $\dot{u}_i = -\mathbb{E}[hu_i]$, so differentiation of $\mathbb{E}P$ contributes the term $-b^\top u$. The centered first moments imply $\dot{C}_{ij} = \mathbb{E}[hu_iu_j]$, $\dot{V} = \mathbb{E}[hS_1^2]$, and $\dot{D} = \mathbb{E}[hS_2]$. Differentiation of (1) now gives (2). For a Gaussian physical reference $N(\mu, \Sigma)$ with $\Sigma > 0$ and a symmetric incoming third tensor T , write $q = \Sigma^{-1}(Z - \mu)$, $Q = \Sigma^{-1}$ and

$$h_T = \frac{1}{6} \sum_{ijk} T_{ijk}(q_i q_j q_k - Q_{ij} q_k - Q_{ik} q_j - Q_{jk} q_i).$$

This is the density tangent $\frac{1}{6}T : \partial_\mu^3 p_{\mu, \Sigma}$ divided by the Gaussian density. Pair entries have $T_{ijj} = \Gamma_{ij}$ with three permutations; diagonal entries are $T_{iii} = \tau_i$. The tensor is transformed with all three indices when whitening. Its numerical calibration to one incoming gain unit remains part of the ledger specification.

If the outgoing quantity is $g_{\text{miss}} = K_{\text{miss}}/(3V^2)$, its complete source derivative is

$$\delta_h g_{\text{miss}} = \frac{\delta_h K_{\text{miss}}}{3V^2} - \frac{2K_{\text{miss}}\delta_h V}{3V^3}, \quad V > 0.$$

The conditional-reference section also controls this normalization term. Dividing a numerator response by a fixed reference denominator gives a different coordinate convention, which must be declared.

Notation. Symbols are local to their sections. In particular, X may denote a whitened input, a centered activation vector, or a Gaussian factor; each is defined when introduced. $\mathcal{D}$ is a Dirac operator, while D_ℓ is a dynamic state matrix. Operator Hilbert–Schmidt norms are denoted by $\|\cdot\|_{\text{HS}}$ or $\|\cdot\|_2$ when their operator domain is explicit. Scalar random-variable L^2 norms always refer to the declared law.

2 The source response as a weak Hodge pairing

Write $X \sim \gamma_n = N(0, I_n)$ and use probabilists' Hermite polynomials. For a symmetric three-tensor T , define

$$S_T(X) = \frac{1}{6} \sum_{ijk} T_{ijk} (X_i X_j X_k - \delta_{ij} X_k - \delta_{ik} X_j - \delta_{jk} X_i).$$

Then $\mathbb{E}S_T = 0$ and

$$\|S_T\|_2^2 = \frac{1}{6} \|T\|_F^2, \quad NS_T = 3S_T, \quad N = -\Delta + x \cdot \nabla.$$

The norm identity follows by Wick contraction: only the six bijections between the two Wick-ordered triples survive. The operator N is positive and self-adjoint on the Hermite basis, with eigenvalue k on chaos k .

The score is a genuine tangent of probability laws. For example,

$$\rho_\epsilon = \frac{(1 + \epsilon S_T/2)^2}{1 + \epsilon^2 \|S_T\|_2^2/4}$$

is nonnegative, integrates to one, and satisfies $\partial_\epsilon \mathbb{E}_{\rho_\epsilon} f|_0 = \mathbb{E}[S_T f]$ for every f with the required polynomial moments. This avoids assuming that the generally nonintegrable cubic exponential $e^{\epsilon S_T}$ defines a probability law.

Theorem 2.1 (First-derivative formula for the complete omitted response). *Let Ψ_{miss} be the fourth-cumulant influence observable defined in Section 1, composed with an affine Gaussian input and coordinatewise ReLU. At fixed weights and fixed source normalization,*

$$\Delta_T = \mathbb{E}[S_T \Psi_{\text{miss}}] = \frac{1}{3} \mathbb{E}[\nabla S_T \cdot \nabla \Psi_{\text{miss}}]. \quad (\text{SH.1})$$

In whitened coordinates,

$$\partial_a S_T = \frac{1}{2} \sum_{jk} T_{ajk} (X_j X_k - \delta_{jk}), \quad |\Delta_T| \leq \frac{\|T\|_F}{\sqrt{18}} \|\nabla \Psi_{\text{miss}}\|_2.$$

For physical coordinates $Z \sim N(\mu, \Sigma)$ and the corresponding physical density score h_T ,

$$\Delta_T = \frac{1}{3} \mathbb{E}[(\nabla_Z h_T)^\top \Sigma \nabla_Z \Psi_{\text{miss}}].$$

Proof. ReLU is globally Lipschitz, and Ψ_{miss} is a polynomial of degree at most four in centered ReLU outputs. Thus $\Psi_{\text{miss}} \in W^{1,p}(\gamma_n)$ for every finite p . Its weak first derivatives are the usual chain-rule derivatives with the gate indicators, outside a Gaussian null set. The Gaussian Dirichlet-form identity is

$$\langle Nf, g \rangle = \mathbb{E}[\nabla f \cdot \nabla g] \quad (f \in \text{Dom } N, g \in W^{1,2}).$$

It follows first for smooth functions by integration by parts and then for $g \in W^{1,2}$ by density. Apply it to $f = S_T$ and $g = \Psi_{\text{miss}}$. The norm estimate follows from Cauchy-Schwarz and $\|\nabla S_T\|_2^2 = 3\|S_T\|_2^2$. The physical formula follows by the affine change $Z = \mu + \Sigma^{1/2}X$, under which the third source remains in the third chaos. The derivative formula for S_T follows by differentiating the Wick polynomial and using symmetry of T . $\square$

This identity has a concrete analytic benefit: the evaluator needs first weak derivatives only. With $u_i = w_i((Z_i)_+ - \mathbb{E}(Z_i)_+)$,

$$\partial_{Z_i} \Psi_{\text{miss}} = w_i \mathbf{1}_{Z_i > 0} \partial_{u_i} \Psi_{\text{miss}}.$$

The polynomial $\partial_{u_i} \Psi_{\text{miss}}$ is cubic. All contributions that appear as gate-surface distributions in a third-derivative calculation are already included in the weak integration-by-parts identity. There is no omission of a surface term and no need to estimate a delta distribution.

The two unbiased integrands in (SH.1) need not have ordered variances. For example, in one dimension with $S = \text{He}_3/6$ and $\Psi_k(x) = \sin(kx)$, the variance of $S\Psi_k$ is uniformly bounded, whereas that of $(1/3)S'\Psi'_k$ grows quadratically in k . Indeed, $S' = \text{He}_2/2$, the mean tends to zero, and $\mathbb{E}[\text{He}_2(X)^2 \cos^2(kX)] \rightarrow 1$ by the Gaussian Fourier transform. Thus first-derivative evaluation is not a universal variance reduction theorem.

2.1 A connected fifth cumulant, with one distinguished source

Proposition 2.2 (Marked-cumulant identity). *For any centered score S and fixed observables $Y_1, \dots, Y_d$ with the required product moments,*

$$\boxed{\delta_S \kappa(Y_1, \dots, Y_d) = \kappa(S, Y_1, \dots, Y_d)}. \quad (\text{SH.2})$$

In particular each omitted fourth sector is a connected fifth cumulant with one distinguished incoming score.

Proof. Work with the formal moment polynomial through degree one in s and degree d in $t_1, \dots, t_d$. The coefficient linear in s in

$$\log \mathbb{E} \exp\{sS + \sum_j t_j Y_j\}$$

is $\mathbb{E}[S e^{\sum_j t_j Y_j}] / \mathbb{E}[e^{\sum_j t_j Y_j}]$. The same formal expression is the derivative of the log moment polynomial under a density path whose first derivative is S . Extracting the $t_1 \cdots t_d$ coefficient proves the identity. This proof uses finitely many moments, not existence of a joint exponential moment for the cubic score. $\square$

The identity is useful for checking centering terms and connected diagrams. It does not mean that the missing response is an arbitrary fifth-order input cumulant: the distinguished variable is the specified third-Hermite score, so its contractions are fixed by T .

3 One Poisson–Stein field carries both cumulant channels

For a mean-zero function $F \in L^2(\gamma_n)$ define $N^{-1}F = \sum_{k \geq 1} k^{-1} \Pi_k F$. Its gradient belongs to L^2 , and for $G \in W^{1,2}$,

$$\mathbb{E}[FG] = \mathbb{E}[\nabla N^{-1}F \cdot \nabla G]. \quad (\text{SH.3})$$

This is the covariance identity behind Malliavin–Stein cumulant formulas; see Nourdin–Peccati, Lemma 3.1, Remark 3.2 and Theorem 4.3 [3]. Here it is proved immediately by Hermite expansion and the same closed Dirichlet form used above.

Theorem 3.1 (A shared field for the third and fourth cumulants). *Let F be a centered globally Lipschitz function of a finite-dimensional standard Gaussian vector, let $v = \mathbb{E}F^2$, and put*

$$\tau_F = \nabla F \cdot \nabla N^{-1}F.$$

Then τ_F is bounded by the square of a Lipschitz constant in absolute value, $\mathbb{E}\tau_F = v$, and

$$\boxed{\kappa_3(F) = 2 \text{Cov}(F, \tau_F), \quad \kappa_4(F) = 3 \text{Cov}(F^2, \tau_F)}. \quad (\text{SH.4})$$

Thus the two channels are two different readouts of the same random field $\tau_F - v$.

Proof. The Mehler semigroup $P_t = e^{-tN}$ satisfies $\nabla P_t F = e^{-t} P_t \nabla F$ in the weak sense. Integration gives

$$\nabla N^{-1} F = \int_0^\infty e^{-t} P_t \nabla F dt.$$

Since P_t averages and $|\nabla F| \leq L$ almost everywhere, $|\nabla N^{-1} F| \leq L$, proving $|\tau_F| \leq L^2$. Apply (SH.3) with $G = F, F^2, F^3$ respectively. Polynomial functions of a Gaussian Lipschitz variable have the required Sobolev integrability. One obtains

$$v = \mathbb{E}\tau_F, \quad \mathbb{E}F^3 = 2\mathbb{E}[F\tau_F], \quad \mathbb{E}F^4 = 3\mathbb{E}[F^2\tau_F].$$

Subtract $3v^2$ in the last identity, and use $\mathbb{E}F = 0$. □

For centered Lipschitz variables $F_1, \dots, F_n$, define

$$\tau_{ij} = \nabla F_i \cdot \nabla N^{-1} F_j.$$

Then $\mathbb{E}\tau_{ij} = \mathbb{E}[F_i F_j] = C_{ij}$ and

$$\kappa(F_i, F_j, F_k) = \mathbb{E}[F_i \tau_{jk} + F_j \tau_{ik}], \quad (4)$$

$$\kappa(F_i, F_j, F_k, F_l) = \text{Cov}(F_i F_j, \tau_{kl}) + \text{Cov}(F_i F_k, \tau_{jl}) + \text{Cov}(F_j F_k, \tau_{il}). \quad (5)$$

To prove these statements, apply (SH.3) with its first variable F_k or F_l and its second variable the corresponding product; the derivative of that product supplies each summand. Subtract the three pairings for the fourth identity. The displayed asymmetry is a choice of integration-by-parts argument, not an asymmetry of the cumulant tensor. The matrix τ is not claimed to be symmetric or positive semidefinite pointwise.

3.1 An explicit gate formula

Let $Z_i = \mu_i + \sigma_i v_i^\top X$ with $|v_i| = 1$, $\sigma_i > 0$, and $\Sigma_{ij} = \sigma_i \sigma_j v_i^\top v_j$. Put $F_i = w_i((Z_i)_+ - \mathbb{E}(Z_i)_+)$, $\alpha_i = \mu_i/\sigma_i$, $\xi_i = v_i^\top X$, and $G_i = \mathbf{1}_{Z_i > 0}$. Then

$$\tau_{ij}(X) = w_i w_j \Sigma_{ij} G_i(X) J_j(\xi_j), \quad J_j(\xi) = \int_0^1 \Phi\left(\frac{\alpha_j + q\xi}{\sqrt{1 - q^2}}\right) dq. \quad (\text{SH.5})$$

Indeed $\nabla F_i = w_i \sigma_i v_i G_i$. Conditional Gaussian averaging gives $P_t G_j = \Phi((\alpha_j + e^{-t}\xi_j)/\sqrt{1 - e^{-2t}})$. Substitution $q = e^{-t}$ in the Poisson gradient proves the formula. The integrand is bounded by one; its limiting value at $q = 1$ has no effect on the integral.

Equations (4)–(SH.5) replace a formal fourth-order reservoir by covariances between quadratic activation observables and explicit pairwise gate fields. Their coherent contractions still require the actual covariance and weight geometry. A small number of global gain amplitudes does not imply a small number of such fields.

A derivative that must not be silently dropped. These identities are exact at the Gaussian reference. If the law changes along a non-Gaussian score, the same fixed field τ_F need not satisfy the integration-by-parts identity for the changed law. Consequently one cannot differentiate (SH.4) while holding τ_F fixed and identify that with the incoming third-source response. The marked cumulant formula (SH.2), the influence (SH.1), or a properly varied Stein operator must be used. The conditional calculation below supplies one such exact response route.

4 A fully solvable Gaussian quadratic bridge

The first two nonconstant Hermite components of a ReLU observable are a linear-plus-quadratic Gaussian function. We can solve its entire third-score response generator, rather than guessing the channel coupling.

Let $A \in \mathbb{R}^n$, let B be symmetric, and set

$$Y = A^\top X + \frac{1}{2}(X^\top B X - \text{tr } B).$$

For symmetric T , write $T[a, b, c] = \sum T_{ijk} a_i b_j c_k$ and $T[a, M] = \sum T_{ijk} a_i M_{jk}$.

Theorem 4.1 (Exact third-source susceptibility of a Gaussian quadratic). *Define $R_t = (I - tB)^{-1}$ and $m_t = tR_t A$. For t in a neighborhood of zero,*

$$\delta_T \log \mathbb{E} e^{tY} = \frac{1}{6} T[m_t, m_t, m_t] + \frac{1}{2} T[m_t, R_t - I]. \quad (\text{SH.6})$$

In particular,

$$\delta_T \kappa_1(Y) = 0, \quad (6)$$

$$\delta_T \kappa_2(Y) = T[A, B], \quad (7)$$

$$\delta_T \kappa_3(Y) = T[A, A, A] + 3T[A, B^2] + 3T[BA, B], \quad (8)$$

$$\delta_T \kappa_4(Y) = 12T[A, A, BA] + 12\{T[A, B^3] + T[BA, B^2] + T[B^2 A, B]\}. \quad (9)$$

The first fourth-order term is linear in the quadratic component and the remaining terms are cubic in it.

Proof. Tilting the Gaussian density by e^{tY} gives a Gaussian with mean $m_t = t(I - tB)^{-1}A$ and covariance $R_t = (I - tB)^{-1}$ whenever $I - tB > 0$. Differentiating the log expectation along the score gives the expectation of S_T under this tilted Gaussian. Its third moment is $m_i m_j m_k + m_i R_{jk} + m_j R_{ik} + m_k R_{ij}$. Subtracting the three Wick-centering terms and using symmetry of T gives (SH.6). All the integrals are finite near $t = 0$, including the score times the quadratic exponential, so the density and t derivatives are justified by dominated convergence.

Expand

$$m_t = tA + t^2 BA + t^3 B^2 A + t^4 B^3 A + \dots, \quad R_t - I = tB + t^2 B^2 + t^3 B^3 + \dots.$$

The coefficient of t^j , multiplied by $j!$, is $\delta_T \kappa_j(Y)$. At degree four the first term in (SH.6) contributes $T[A, A, BA]/2$ and the second contributes $(T[A, B^3] + T[BA, B^2] + T[B^2 A, B])/2$. This proves every displayed coefficient. $\square$

For one variable and $T = 1$, the fourth susceptibility is $12a^3b + 36ab^3$. At $B = 0$ the third source changes the third cumulant but does not feed the fourth. The first feed is produced by the interaction between the third source and a quadratic activation component. Cubic powers of that component supply an additional exact contribution.

For the centered ReLU row $F = \sum_i w_i((Z_i)_+ - \mathbb{E}(Z_i)_+)$ above, its orthogonal projection on chaoses one and two has exactly this form with

$$A = \sum_i w_i \sigma_i \Phi(\alpha_i) v_i, \quad B = \sum_i w_i \sigma_i \varphi(\alpha_i) v_i v_i^\top.$$

To verify this, Gaussian integration by parts gives $\mathbb{E}[(Z_i)_+ \text{He}_1(\xi_i)] = \sigma_i \Phi(\alpha_i)$ and $\mathbb{E}[(Z_i)_+ \text{He}_2(\xi_i)] = \sigma_i \varphi(\alpha_i)$; the second Hermite projection has the factor $1/2!$ already present in the definition of Y .

The identity is exact for the projected polynomial. It is not, without a tail estimate, the fourth response of the full ReLU row. For any joint fourth tensor, polarization of the homogeneous quartic response in w recovers its components, after which one may retain the three required index-multiplicity classes. This is an algebraic recovery statement, not an assertion that expanding all components is economical.

The conditional Mehler bounds later in this note give one way to certify a low-Hermite approximation. They must be applied to the response's actual conditional moments and jets, not merely to the conditional mean. Related neural-network cumulant mappings and quadratic activation calculations appear in Fischer et al., especially Appendix C [9]; no global novelty claim is made for using quadratic Gaussian observables.

5 A useful index bridge: deformation of the harmonic state

The Gaussian de Rham Dirac operator acts on $\mathcal{H} = L^2(\gamma_n) \otimes \Lambda^\bullet \mathbb{C}^n$:

$$\mathcal{D} = \sum_j \{\varepsilon_j \partial_j + \iota_j(-\partial_j + x_j)\}.$$

Here ε_j is exterior multiplication by the j th basis vector and $\iota_j = \varepsilon_j^*$ is contraction. Fermion parity grades the space. On polynomial forms,

$$\mathcal{D}^2 = N + N_f.$$

The Hermite–exterior basis therefore gives a self-adjoint closure with compact resolvent, one zero mode $\Omega = 1$ in even degree, and $\text{ind } \mathcal{D}^+ = 1$. Its positive-energy eigenspaces are paired by $\mathcal{D}$. This is the previously established Gaussian spectral model, stated here to fix the operator and its domain.

There is a nontrivial response calculation inside this same Fredholm geometry. It concerns how the normalized harmonic representative moves. Let h be a real, smooth, bounded function with bounded gradient and $\mathbb{E}h = 0$, and set

$$\rho_\epsilon = \frac{e^{\epsilon h}}{\mathbb{E}e^{\epsilon h}}, \quad \Omega_\epsilon = \sqrt{\rho_\epsilon}.$$

Theorem 5.1 (Witten deformation and its reduced-resolvent response). *On the fixed Hilbert space $\mathcal{H}$, the de Rham Dirac operator for the weighted probability law $\rho_\epsilon \gamma_n$, conjugated by multiplication by $\sqrt{\rho_\epsilon}$, is*

$$\boxed{\mathcal{D}_\epsilon = \mathcal{D} - \frac{\epsilon}{2} c_+(dh), \quad c_+(dh) = \sum_j (\varepsilon_j + \iota_j) \partial_j h.} \quad (\text{SH.7})$$

It is self-adjoint on $\text{Dom } \mathcal{D}$, has compact resolvent, and $\text{ind } \mathcal{D}_\epsilon^+ = 1$ for every finite ϵ . Its normalized zero mode includes Ω_ϵ , and at $\epsilon = 0$,

$$\dot{\Omega} = \frac{h}{2} = -\mathcal{D}_\perp^{-1} \dot{\mathcal{D}} \Omega. \quad (\text{SH.8})$$

Here $\mathcal{D}_\perp^{-1}$ is the inverse on the orthogonal complement of the unperturbed zero mode. For a real-valued observable Ψ with $\Psi, h\Psi \in L^2(\gamma_n)$,

$$\boxed{\delta_h \mathbb{E} \Psi = -2 \text{Re} \langle \mathcal{D}_\perp^{-1} \dot{\mathcal{D}} \Omega, M_\Psi \Omega \rangle.} \quad (\text{SH.9})$$

Proof. The adjoint of $d = \sum_j \varepsilon_j \partial_j$ in the weighted measure is $d_\rho^* = d_\gamma^* - \sum_j \iota_j \partial_j \log \rho$. Conjugation by $\sqrt{\rho}$ sends d to $d - \frac{1}{2} \sum_j \varepsilon_j \partial_j \log \rho$ and sends d_ρ^* to $d_\gamma^* - \frac{1}{2} \sum_j \iota_j \partial_j \log \rho$. Since $\partial_j \log \rho_\epsilon = \epsilon \partial_j h$,

this proves (SH.7) on smooth forms. The functions $\rho_\epsilon^{1/2}$ and $\rho_\epsilon^{-1/2}$ are bounded and have bounded gradients. Multiplication by them preserves the first-order graph domains: the commutator with the Gaussian Dirac operator is bounded. Thus the conjugation identifies the closed weighted operator with the displayed perturbation, not just their expressions on a smooth core.

Clifford anticommutation gives $c_+(dh)^2 = |\nabla h|^2 I$, so the perturbation is bounded and self-adjoint. Bounded self-adjoint perturbation preserves self-adjointness on the same domain. The resolvent identity expresses a perturbed resolvent as a bounded multiple of an unperturbed compact resolvent, so it remains compact. The operators are odd. Their chiral parts, considered as maps from the fixed graph domain to the opposite parity space, form a continuous family of Fredholm operators. Each is Fredholm because compact resolvent isolates zero with finite multiplicity and gives closed range; their index is consequently constant and equals one.

On zero-forms, $d\Omega_\epsilon = (\epsilon/2) dh \Omega_\epsilon$, which cancels the perturbation. Thus $\mathcal{D}_\epsilon \Omega_\epsilon = 0$. Normalization and $\mathbb{E}h = 0$ give $\dot{\Omega} = h/2$ and $\langle \Omega, \dot{\Omega} \rangle = 0$. Differentiating the zero-mode equation yields $\mathcal{D}\dot{\Omega} = -\dot{\mathcal{D}}\Omega$. The unperturbed operator is invertible on $\Omega^\perp$ with bounded inverse, because its nonzero squared eigenvalues are positive integers. This proves (SH.8). Finally, $\mathbb{E}_{\rho_\epsilon} \Psi = \langle \Omega_\epsilon, M_\Psi \Omega_\epsilon \rangle$. Differentiation proves (SH.9). $\square$

The deformation is the same differential-operator mechanism as Witten's deformed de Rham complex [2], here proved in the weighted Gaussian setting with explicit hypotheses. The Fredholm index remains one while the harmonic representative and all its observable matrix elements can vary. The reduced resolvent measures that variation and retains information that the integer index does not record.

For the unbounded cubic score S_T , one must not substitute $h = S_T$ in the global exponential family above. Instead choose smooth cutoffs $h_R = \chi_R S_T - \mathbb{E}[\chi_R S_T]$. Gaussian polynomial integrability gives $h_R \rightarrow S_T$ in $W^{1,2}$, for standard cutoffs with uniformly controlled scaled derivatives. Each h_R satisfies the bounded theorem. Its tangent $h_R/2$, the response $\mathbb{E}[h_R \Psi]$, and the Dirichlet pairing converge to those of S_T whenever $\Psi \in W^{1,2}$. This is a statement about a limit of valid tangent formulas; it asserts neither normalizability of e^{eS_T} nor compact resolvent of a formal unbounded cubic deformation.

5.1 Fisher metric with zero Berry curvature

For a finite family of bounded scores h_a use the positive normalized vacuum $\Omega_\theta = (e^{\sum_a \theta_a h_a} / \mathbb{E} e^{\sum_a \theta_a h_a})^{1/2}$. At zero, with $Q = I - |\Omega\rangle\langle\Omega|$,

$$\langle \partial_a \Omega, Q \partial_b \Omega \rangle = \frac{1}{4} \text{Cov}(h_a, h_b). \quad (\text{SH.10})$$

This follows by differentiating the normalized square root. The Berry connection $i\langle \Omega_\theta, \partial_a \Omega_\theta \rangle$ vanishes identically in this real positive gauge, so its curvature is zero. The real metric is nevertheless generally nonzero. At the Gaussian third-score tangent, cutoff convergence gives

$$g(T, U) = \frac{1}{24} \langle T, U \rangle_F.$$

Thus vanishing curvature or a fixed line-bundle index does not force small susceptibility: the metric measures the continuously variable score size.

If one retains only a sigma-algebra $\mathcal{G}$, its induced score is $\mathbb{E}[h \mid \mathcal{G}]$. Orthogonal projection gives the exact Fisher loss

$$\|h\|_2^2 - \|\mathbb{E}[h \mid \mathcal{G}]\|_2^2 = \|h - \mathbb{E}[h \mid \mathcal{G}]\|_2^2.$$

The conditional expectation is onto a genuine algebra of observables; a projection onto two moment influences is a different, smaller linear operation. The information-contraction and memory results

below state which restriction is being used. No quantum recovery theorem is inferred from this elementary identity.

6 Relative determinants and the geometry of the missing fourth cumulant

The previous investigation located the source response in a weighted third Hermite spectral sector and showed why the ordinary Gaussian Dirac index does not determine its amplitude. Here a different part of the operator framework becomes useful. Compression fails to preserve multiplication and exponentiation. A relative Fredholm determinant measures the latter failure; its Taylor coefficients are cumulants in the canonical probability representation. At fourth order those coefficients admit an exact formula in terms of two boundary operators. This leads to a quadratic-observable residual, a positive moment-completion problem, and an explicit source mode that is invisible to every retained fourth-cumulant entry.

Three projections must be distinguished. The probability vacuum projection is $P_0 = |1\rangle\langle 1|$ in $L^2(\mu)$. A moment approximation uses a projection P_V onto a space of observables. The dynamic closure uses the layer-dependent retained projection from the preceding investigation. They share a compression calculus, but are not the same projection. No identification between a neuron-index mask and any of these Hilbert-space projections is assumed.

6.1 A relative Fredholm determinant arising from a compression extension

Let P be an orthogonal projection on a Hilbert space $\mathcal{H}$, and put $Q = I - P$. Denote the Schatten ideals by $\mathcal{S}_1$ and $\mathcal{S}_2$. The algebra

$$\mathcal{A}_P^{(2)} = \{A \in \mathcal{B}(\mathcal{H}) : [P, A] \in \mathcal{S}_2\}$$

is closed under products and adjoints, by the commutator Leibniz rule. If $A, B \in \mathcal{A}_P^{(2)}$, then

$$PAPBP - PABP = -PAQBP \in \mathcal{S}_1.$$

Thus compression is multiplicative modulo trace-class operators. This is the precise algebraic structure used here. For the Hardy projection and Laurent multiplication, its off-diagonal corners are the usual Hankel operators and this identity is the Toeplitz product formula. No Hardy-space or Toeplitz model of the network itself is being asserted.

Theorem R.1 (relative counting determinant). Suppose $A = A^* \in \mathcal{B}(\mathcal{H})$ and $H = QAP \in \mathcal{S}_2$. Write, on $P\mathcal{H} \oplus Q\mathcal{H}$,

$$A = \begin{pmatrix} a & H^* \\ H & d \end{pmatrix}, \quad a = PAP, \quad d = QAQ.$$

Then

$$\mathcal{D}_{P,A}(t) = \det_{P\mathcal{H}}(e^{-ta}Pe^{tA}P) \tag{10}$$

is a well-defined Fredholm determinant for every complex t , with $\mathcal{D}_{P,A}(0) = 1$. It has an analytic logarithm near zero. For real t , the operator inside the determinant is invertible. If $M = \|A\|$, then

$$\|Pe^{tA}P - e^{ta}\|_1 \leq \frac{|t|^2}{2}e^{|t|M}\|H\|_2^2 \quad (t \in \mathbb{R}).$$

When P has finite rank,

$$\log \mathcal{D}_{P,A}(t) = \log \det_{P\mathcal{H}}(Pe^{tA}P) - t \operatorname{Tr} a.$$

Proof. Put $C(t) = Pe^{tA}P$ and $O(t) = Qe^{tA}P$, viewed as block operators. Their equations are $C' = aC + H^*O$ and $O' = HC + dO$, with $C(0) = I$ and $O(0) = 0$. For $t \geq 0$, variation of constants gives

$$C(t) - e^{ta} = \int_0^t \int_0^s e^{(t-s)a} H^* e^{(s-u)d} HC(u) du ds.$$

The integrand is trace class because it contains two Hilbert–Schmidt factors, and its trace norm is at most $e^{tM} \|H\|_2^2$. The same argument applied to $-A$ handles negative t . Integrating along straight complex line segments proves trace-norm analyticity for complex t . Multiplication by e^{-ta} therefore produces an operator of the form $I + \mathcal{S}_1$, to which the Fredholm determinant applies. It is one at zero, hence has a local analytic logarithm. For real t , $Pe^{tA}P \geq e^{-|t|M}I$ on $P\mathcal{H}$, so both factors are invertible. The finite-rank formula is ordinary determinant multiplicativity. $\square$

The determinant removes the part of exponentiation that would occur if the retained subspace were invariant. Every remaining term contains a trip into $Q\mathcal{H}$ and a return. This is a relative analytic invariant, not an integer index. It is defined without putting a desired scalar into an arbitrarily chosen rank-one operator.

Theorem R.2 (the first boundary cumulants). Set

$$L = dH - Ha, \quad c_m = \left. \frac{d^m}{dt^m} \log \mathcal{D}_{P,A}(t) \right|_{t=0}.$$

Then

$$c_2 = \text{Tr}(H^*H) = \|H\|_2^2, \quad (11)$$

$$c_3 = \text{Tr}(H^*L), \quad (12)$$

$$c_4 = \boxed{\|L\|_2^2 - 2\|H^*H\|_2^2}. \quad (13)$$

In particular,

$$|c_3| \leq 2M\|H\|_2^2, \quad -2M^2\|H\|_2^2 \leq c_4 \leq 4M^2\|H\|_2^2.$$

Here M may be replaced by $\|A - \alpha I\|$ for any real α .

Proof. First suppose P has finite rank and put $M_j = PA^jP$. Expanding $\text{Tr} \log(I + \sum_{j \geq 1} t^j M_j / j!)$ gives

$$\begin{aligned} c_2 &= \text{Tr}(M_2 - a^2), \\ c_3 &= \text{Tr}(M_3 - 3aM_2 + 2a^3), \\ c_4 &= \text{Tr}(M_4 - 4aM_3 - 3M_2^2 + 12a^2M_2 - 6a^4). \end{aligned}$$

Writing $B = H^*H$, block multiplication gives

$$M_2 = a^2 + B, \quad M_3 = a^3 + aB + Ba + H^*dH,$$

and, after using cyclicity of the finite trace,

$$\text{Tr} M_4 = \text{Tr}(a^4 + 3a^2B + B^2 + 2aH^*dH + H^*d^2H).$$

Substitution cancels every term having no boundary crossing. The fourth coefficient is

$$\text{Tr}(H^*d^2H - 2aH^*dH + a^2H^*H - 2(H^*H)^2),$$

which is (13). For a general projection, expand $e^{-ta}Pe^{tA}P - I$ in the trace norm supplied by Theorem R.1. The purely diagonal words cancel algebraically before any trace is taken. Every remaining word contains at least two Hilbert–Schmidt boundary factors and is trace class. Cyclic cancellation under these defined traces then gives the same formulas. Finally $\|L\|_2 \leq 2M\|H\|_2$ and $\|H^*H\|_2^2 \leq \|H\|_2^2\|H\|_2^2 \leq M^2\|H\|_2^2$. Shifting A leaves both H and $dH - Ha$ unchanged. $\square$

The fourth coefficient is a difference of two nonnegative metric quantities. It is not determined by the rank of H , and need not have either sign. If $dH = Ha$, it is exactly $-2\|H^*H\|_2^2$. Gaussian cancellation instead requires $\|dH - Ha\|_2^2 = 2\|H^*H\|_2^2$.

For differentiable bounded families, with H differentiable in $\mathcal{S}_2$, the exact first variation is

$$\dot{c}_4 = 2 \operatorname{Re} \operatorname{Tr}(L^* \dot{L}) - 8 \operatorname{Re} \operatorname{Tr}((H^* H) H^* \dot{H}), \quad (14)$$

where $\dot{L} = \dot{d}H + d\dot{H} - \dot{H}a - H\dot{a}$. This follows by differentiating the two squared norms. It is a transgression formula for this relative determinant, rather than a Chern–Simons identity asserted without a specified cocycle.

6.2 The canonical comparison with the probability law and its source

Let μ be the actual reference probability law, let $\mathcal{H} = L^2(\mu)$, let $\Omega = 1$, and put $P_0 = |\Omega\rangle\langle\Omega|$. An observable Y acts by the canonical multiplication operator $A = M_Y$. Whenever the displayed exponential moment exists,

$$\det_{P_0\mathcal{H}}(P_0 e^{tA} P_0) = \mathbb{E}_\mu e^{tY}.$$

Thus $c_m = \kappa_m(Y)$ for $m \geq 2$. For a Gaussian reference and a finite ReLU network, Y has at most linear growth, so these exponential moments exist for every real t . For the fourth-order identities alone, $Y \in L^4(\mu)$ suffices: one may use the finite moment jet, or truncate Y and pass to the limit. No claim that neurons are fermionic particles is needed. The rank-one determinant is simply the canonical state applied to the multiplication algebra.

The general finite-rank determinant belongs to the same counting-statistics calculus used for determinantal point processes. What is useful here is the compression identity and its boundary expansion, not the transfer of a determinantal random-matrix ensemble to the network.

Theorem R.3 (varying the vacuum produces the cumulant source). Let $h \in L^2(\mu)$ be real with $\mathbb{E}h = 0$. Use the normalized vector

$$\Omega_\epsilon = \frac{1 + \epsilon h/2}{\sqrt{1 + \epsilon^2 \mathbb{E}h^2/4}}, \quad P_\epsilon = |\Omega_\epsilon\rangle\langle\Omega_\epsilon|.$$

Its state on multiplication operators is integration against $d\mu_\epsilon = \Omega_\epsilon^2 d\mu$, a probability measure. Assume $\mathbb{E}[(1 + h^2)(1 + |Y|^4)] < \infty$. Wherever the corresponding weighted exponential moments exist locally, one has

$$\partial_\epsilon|_0 \log \frac{\det(P_\epsilon e^{tM_Y} P_\epsilon|_{P_\epsilon\mathcal{H}})}{\det(P_0 e^{tM_Y} P_0|_{P_0\mathcal{H}})} = \frac{\mathbb{E}[he^{tY}]}{\mathbb{E}e^{tY}}.$$

Consequently, in the finite formal moment sense when necessary,

$$\partial_\epsilon \kappa_4^{\mu_\epsilon}(Y)|_0 = \operatorname{cum}_\mu(h, Y, Y, Y, Y).$$

Writing $X = Y - \mathbb{E}Y$, $v = \mathbb{E}X^2$, and $m_3 = \mathbb{E}X^3$, this is

$$\mathbb{E}[h(X^4 - 6vX^2 - 4m_3X)]. \quad (15)$$

Proof. The determinant equals $\mathbb{E}[\Omega_\epsilon^2 e^{tY}]$. Its derivative at zero is $\mathbb{E}[he^{tY}]$. The logarithm therefore gives the quotient. Introducing an additional formal variable s , $\partial_s \log \mathbb{E}e^{sh+tY}|_{s=0}$ is the same quotient. Its fourth t derivative is the stated joint cumulant. Alternatively, differentiate $\kappa_4(Y) = \mathbb{E}(Y - m)^4 - 3v^2$. Here $\dot{m} = \mathbb{E}hX$, $\dot{v} = \mathbb{E}hX^2$, and differentiating the moving center in the fourth moment contributes $-4m_3\mathbb{E}hX$. This gives (15). $\square$

For the prescribed Gaussian third tangent, $h = h_T$ is exactly the Hermite score from the preceding note. Polarizing the four Y arguments and summing only the omitted multiplicity classes yields $\mathbb{E}[h_T \Psi_{\text{miss}}]$. This is an explicit comparison between the relative determinant and the target response, including the moving centers. It introduces no different incoming source direction.

The source also has a concrete noncommutative implementation. Define the skew-adjoint rank-two operator

$$K = \frac{1}{2}(|h\rangle\langle\Omega| - |\Omega\rangle\langle h|).$$

Then $\dot{P} = [K, P_0]$. In a unitarily transported frame with P_0 fixed, the observable has tangent $\dot{A} = [A, K]$. This can be substituted into (14). At a centered scalar reference, putting $b = X$, $c = X^2 - v$, and $\beta = \mathbb{E}hX$, the block calculation gives

$$\dot{H} = \frac{1}{2}Q_0(Xh), \quad \dot{a} = \beta, \quad \dot{d} = -\frac{1}{2}(|X\rangle\langle h| + |h\rangle\langle X|),$$

and

$$\dot{L} = \frac{1}{2}Q_0(X^2h) - 2\beta X - \frac{1}{2}vh.$$

Substituting these expressions into (14) reproduces (15). For unbounded X, h , this statement concerns the displayed finite vectors and moments; bounded operator differentiability is not silently imposed on multiplication by a Hermite polynomial.

What the relative index and phase record in this construction. Both P_0 and P_ϵ have rank one. Their pair index is zero, including when expressed as the trace of an odd power of their finite-rank difference. For this real normalized section, $\langle\Omega_\epsilon, \partial_\epsilon\Omega_\epsilon\rangle = 0$. Hence the natural Berry connection of this particular source path is zero, while the derivative in Theorem R.3 can be nonzero. This excludes that specific phase or relative-index pairing as an amplitude formula; it does not exclude every possible measured index or every possible spectral-shift construction.

There is a related symmetric/antisymmetric distinction. For finite-rank P and commuting bounded observables A, B ,

$$\text{Tr}(PAQB P - PBQAP) = \text{Tr}(P[A, B]P) - \text{Tr}[PAP, PBP] = 0.$$

The symmetric boundary pairing, such as $\|QAP\|_2^2$, can nevertheless be positive. Thus this finite-rank antisymmetric cyclic defect does not recover even the variance. Infinite Hardy projections can have a nonzero trace of a compressed commutator; that requires new infinite-dimensional comparison data, absent from the present probability vacuum.

6.3 A quadratic residual is exactly the first lost even cumulant

Theorem R.4 (Krylov compression and the first lost moment). Let $A = A^*$, let Ω be a unit vector in $\text{Dom } A^k$, and let $V \subset \text{Dom } A$ be a finite-dimensional subspace containing

$$\Omega, A\Omega, \dots, A^{k-1}\Omega.$$

Let P_V be its orthogonal projection and $B = P_V A P_V|_V$. The vacuum moments of A and B , and therefore their cumulants, agree through order $2k - 1$. At the next order,

$$\boxed{\kappa_{2k}(A; \Omega) - \kappa_{2k}(B; \Omega) = \|(I - P_V)A^k\Omega\|^2.} \quad (16)$$

Moments of the unbounded A are interpreted by its spectral measure.

Proof. Induction gives $B^j\Omega = A^j\Omega$ for $j \leq k - 1$, and $B^k\Omega = P_V A^k\Omega$. Every moment of degree at most $2k - 2$ can be written as an inner product between two such powers of degrees at most $k - 1$. The moment of degree $2k - 1$ is $\langle A^{k-1}\Omega, A^k\Omega \rangle$, and projecting the second factor does not change

it because the first is in V . The difference at degree $2k$ is $\|A^k\Omega\|^2 - \|P_V A^k\Omega\|^2$. Cumulants are triangular polynomials in the moments, with coefficient one on the highest moment. Since all lower moments match, their $2k$ th cumulant difference is this same squared norm. $\square$

For $A = M_Y$ and $k = 2$, any V containing $1, Y$ preserves the first three moments. Its fourth-cumulant loss is exactly $\|(I - P_V)Y^2\|_2^2$. This is the elementary moment mechanism behind Jacobi/Lanczos compression and Gaussian quadrature. It is stronger than an index count: it supplies a signed, exact loss, and specifies the missing observable. Adding Y^2 to V removes this fourth-order loss. This local fact is not a proof that its future pullbacks lie in a small space.

Corollary R.5 (the exact contribution of the third moments). Let $X = (X_1, \dots, X_n)$ be centered with finite fourth moments and positive definite covariance C . Fix w , and set

$$Y = w^\top X, \quad V_Y = w^\top C w, \quad q_w = Y^2 - V_Y, \quad t_w = \mathbb{E}[X q_w].$$

Thus $(t_w)_i = \text{cum}(X_i, Y, Y)$ is a third-cumulant contraction. Let

$$R_w = q_w - X^\top C^{-1} t_w.$$

Then

$$\boxed{\kappa_4(Y) = t_w^\top C^{-1} t_w + \|R_w\|_2^2 - 2V_Y^2.} \quad (17)$$

Along any differentiable family for which these moments are differentiable and C stays positive definite,

$$\boxed{\dot{\kappa}_4 = 2\dot{t}_w^\top C^{-1} t_w - t_w^\top C^{-1} \dot{C} C^{-1} t_w + \frac{d}{d\epsilon} \|R_w\|_{L^2(\mu_\epsilon)}^2 - 4V_Y \dot{V}_Y.} \quad (18)$$

All means, t_w , R_w , and the norm in the latter identity are recomputed in the changing law.

Proof. The orthogonal projection of q_w onto the span of $X_1, \dots, X_n$ is $X^\top C^{-1} t_w$, by the normal equations. Pythagoras gives $\|q_w\|_2^2 = t_w^\top C^{-1} t_w + \|R_w\|_2^2$. But $\|q_w\|_2^2 = \mathbb{E}Y^4 - V_Y^2 = \kappa_4(Y) + 2V_Y^2$. Differentiate this identity and use $\partial_\epsilon C^{-1} = -C^{-1} \dot{C} C^{-1}$. $\square$

This exposes a genuine third-to-fourth contribution without asserting that it exhausts the response. The undetermined term is the derivative of the quadratic residual's norm. It can neither be set to zero nor determined from the third cumulants by positivity alone. With only Y retained, the same proof gives the familiar sharp inequality

$$\kappa_4(Y) \geq \frac{\kappa_3(Y)^2}{V_Y} - 2V_Y^2.$$

Equality means that the centered quadratic observable is already linear in the retained observables. If C is singular, the corresponding statements hold on its effective range, with the Moore–Penrose inverse; the covariance Gram matrix forces t_w into that range.

6.4 The omitted sectors are missing entries of a positive pair Gram matrix

Index the unordered neuron pairs by $\mathcal{P} = \{(i, j) : i \leq j\}$, and define centered quadratic observables and their Gram matrix by

$$B_{ij} = X_i X_j - C_{ij}, \quad G_{ij,kl} = \mathbb{E}[B_{ij} B_{kl}].$$

The exact identity is

$$\boxed{G_{ij,kl} = \kappa_{ijkl} + C_{ik} C_{jl} + C_{il} C_{jk}.} \quad (19)$$

In particular $G \succeq 0$. If $T_{i,jk} = \kappa_{ijk}$ is the rectangular third-cumulant matrix, then

$$\begin{pmatrix} C & T \\ T^\top & G \end{pmatrix} \succeq 0, \quad G \succeq T^\top C^{-1} T \quad (C > 0).$$

Both statements are Gram positivity; the second is its Schur complement.

The retained (4) and (2, 2) data determine the following entries:

$$\begin{aligned} G_{ii,ii} &= \kappa_{iiii} + 2C_{ii}^2, \\ G_{ii,jj} &= \kappa_{iijj} + 2C_{ij}^2, \\ G_{ij,ij} &= \kappa_{iijj} + C_{ii}C_{jj} + C_{ij}^2 \quad (i < j). \end{aligned}$$

Thus every diagonal of G and its whole block of diagonal-pair observables are known. The omitted (3, 1), (2, 1, 1), (1, 1, 1, 1) sectors are the remaining off-diagonal entries after the displayed Wick terms have been subtracted. This identifies the missing coherence as actual inner products of quadratic observables, rather than an unspecified fourth-order array.

Set $v_{ii} = w_i^2$ and $v_{ij} = 2w_i w_j$ for $i < j$. Then $q_w = \sum_{i \leq j} v_{ij} B_{ij}$ and

$$K_{\text{miss}}(w) = v^\top G v - 2V_Y^2 - K_{\text{ret}}(w),$$

where

$$K_{\text{ret}}(w) = \sum_i w_i^4 \kappa_{iiii} + 6 \sum_{i < j} w_i^2 w_j^2 \kappa_{iijj}.$$

Theorem R.6 (a rigorous completion interval from retained data). Let $\mathcal{L}(G) = b$ collect the known entries just listed and, if desired, the affine constraints expressing full symmetry of $G_{ij,kl} + C_{ij}C_{kl}$. Define

$$\mathcal{F} = \{G = G^\top \succeq 0 : \mathcal{L}(G) = b\}.$$

If the retained data come from a probability law, $\mathcal{F}$ is nonempty and compact. Therefore

$$\min_{G \in \mathcal{F}} v^\top G v - 2V_Y^2 - K_{\text{ret}} \leq K_{\text{miss}} \leq \max_{G \in \mathcal{F}} v^\top G v - 2V_Y^2 - K_{\text{ret}}.$$

A vector λ with $\mathcal{L}^* \lambda - vv^\top \succeq 0$ gives the certified upper bound $v^\top G v \leq \lambda^\top b$. If $vv^\top - \mathcal{L}^* \lambda \succeq 0$, it gives the corresponding lower bound. Known third moments can tighten the feasible set by the joint Gram constraint above.

Proof. The true Gram matrix is feasible. Every diagonal of G is fixed, and positivity implies $|G_{ab}| \leq \sqrt{G_{aa}G_{bb}}$. Hence the feasible set is bounded; it is closed, thus compact. The objective is continuous. For an upper certificate, the trace of the product of two positive semidefinite matrices is nonnegative, so

$$0 \leq \text{Tr}[(\mathcal{L}^* \lambda - vv^\top)G] = \lambda^\top b - v^\top G v.$$

The lower certificate is identical with the inequality reversed. $\square$

These are outer bounds on probability-law completions. Not every positive matrix obeying the displayed affine constraints is realized by products of random variables. No moment-realizability theorem or strong semidefinite duality hypothesis is being smuggled into the result. The optimization also has a quadratic-pair dimension; it is a potential certificate, not an asserted inexpensive network implementation.

For a source response, a further Gram matrix makes the necessary extra information explicit. If $hB_{ij} \in L^2$, define

$$J_{ab} = \mathbb{E}[hB_a B_b], \quad R_{ab} = \mathbb{E}[h^2 B_a B_b].$$

Then

$$\begin{pmatrix} G & J \\ J & R \end{pmatrix} \succeq 0,$$

because it is the joint Gram matrix of B and hB . Keeping the baseline centering in these definitions, put $\beta = \mathbb{E}hY$ and $\dot{V}_Y = \mathbb{E}hY^2$. The exact response is

$$\boxed{\dot{K}_{\text{miss}} = v^\top Jv - 4V_Y \dot{V}_Y - 4\kappa_3(Y)\beta - \dot{K}_{\text{ret}}.} \quad (20)$$

Indeed $v^\top Jv = \mathbb{E}[h(Y^2 - V_Y)^2] = \mathbb{E}hY^4 - 2V_Y \dot{V}_Y$, and Theorem R.3 supplies the other terms. The retained derivative on the right is its complete, recentered derivative. Thus it can be taken from a correctly specified existing ledger. Weighted Gram bounds or their dual certificates can constrain the first variation once bounds on R are supplied. Baseline positivity of G alone does not supply them. In the Gaussian Hermite/ReLU setting the required mixed moments are finite, but finiteness does not make their values known.

6.5 Two falsifying examples locate the first uncertain comparison

Proposition R.7 (an invisible retained-sector source mode). Let Z_1, Z_2 be independent standard Gaussians, $H_i = (Z_i)_+$, and $c = (2\pi)^{-1/2}$. Consider the legitimate Gaussian third-cumulant score

$$h = \frac{t}{2} \left[(Z_1^2 - 1)Z_2 - Z_1(Z_2^2 - 1) \right].$$

It has $T_{112} = T_{121} = T_{211} = t$ and $T_{122} = T_{212} = T_{221} = -t$, with all other entries zero. The source derivatives of both means, the complete covariance matrix, both marginal fourth cumulants, and the retained κ_{1122} all vanish. Nevertheless,

$$\dot{\kappa}_{1112} = 3tc^3, \quad \dot{\kappa}_{1222} = -3tc^3.$$

For readout weights (a, b) , its omitted response is

$$\boxed{\dot{K}_{\text{miss}}(a, b) = 12tc^3 ab(a^2 - b^2).} \quad (21)$$

Proof. Every marginal derivative vanishes because the score has zero conditional expectation given either coordinate. The score is antisymmetric under exchange of Z_1, Z_2 , so every symmetric mixed moment has zero derivative. This handles the mixed covariance and κ_{1122} ; the changing means and covariances have already been shown to have zero derivative. For $U = Z_+ - c$, let

$$A_j = \mathbb{E}[U^j(Z^2 - 1)], \quad B_j = \mathbb{E}[U^j Z].$$

The elementary half-normal moments give

$$A_1 = c, \quad B_1 = \frac{1}{2}, \quad A_3 = 3c + 3c^3, \quad B_3 = \frac{3}{2} - \frac{9}{2}c^2.$$

For example these follow by expanding U^j and using $\mathbb{E}Z_+ = c$, $\mathbb{E}Z_+^2 = 1/2$, $\mathbb{E}Z_+^3 = 2c$, $\mathbb{E}Z_+^4 = 3/2$, and $\mathbb{E}Z_+^5 = 8c$. Independence now gives

$$\dot{\kappa}_{1112} = \frac{t}{2}(A_3 B_1 - B_3 A_1) = 3tc^3.$$

Exchange gives the second identity, and the two $(3, 1)$ multiplicities have coefficients $4a^3b$ and $4ab^3$. $\square$

The two source directions t and $-t$ have the same score norm $|t|$ and the same retained response, but opposite omitted responses. This is a counterexample for a family allowing arbitrary pair-source tensors. It does not assert that the network's one prescribed unit-gain tensor lies in this antisymmetric direction. Its implication is precise: no universal formula can reconstruct the omitted feed from these retained derivatives alone, even at a nondegenerate Gaussian reference.

Proposition R.8 (the tensor mask is not a positive compression). Let Z be standard Gaussian, $H_1 = Z_+$, and $H_2 = (-Z)_+$. For the row $(1, -1)$ the output is exactly Z and has zero fourth cumulant. Yet the omitted $(3, 1)$ contraction is

$$K_{\text{miss}} = -8c^2(1 - 6c^2) < 0.$$

Proof. The coordinates have mean c , variance $1/2 - c^2$, and covariance $-c^2$. Their positive powers have zero mixed raw moment. Expanding the centered product gives

$$\mathbb{E}[(H_1 - c)^3(H_2 - c)] = -\frac{1}{2}c^2 - 3c^4.$$

Subtracting the three covariance pairings yields $\kappa_{1112} = c^2 - 6c^4$; symmetry gives the same κ_{1222} . The two readout coefficients are both -4 . Finally $6c^2 = 3/\pi < 1$. $\square$

The loss in Theorem R.4 is nonnegative, whereas this multiplicity-mask loss is negative. Consequently the pair mask cannot universally be the Krylov compression in that theorem. The theorem may guide a better state representation, but applying it to the existing mask requires a new, network-specific comparison. The example has a singular two-dimensional preactivation covariance because both coordinates use one Gaussian; this does not affect its exact one-dimensional probability calculation. It also gives nondegenerate examples: replace the preactivations by $Z + \delta E_1$ and $-Z + \delta E_2$, where E_1, E_2, Z are independent standard Gaussians and $\delta > 0$. Their covariance is positive definite. ReLU’s Lipschitz bound and Gaussian domination make the fourth moments continuous as $\delta \downarrow 0$. The strictly negative omitted contraction therefore remains negative for all sufficiently small positive δ .

6.6 The dynamic bridge, the productive next implication, and its scope

For an exact deterministic layer F with $\nu = F_{\#}\mu$, pullback $U : L^2(\nu) \rightarrow L^2(\mu)$ is an isometry and satisfies

$$UM_f = M_{f \circ F}U, \quad U1 = 1.$$

These identities intertwine the probability vacua and the multiplication representations. Therefore all the determinant, Gram, and moment identities are preserved under exact pullback. This is the honest bridge from the static cumulant construction to depth. It creates no contraction by itself. An efficient deep theory still needs control of the retained query space after these pullbacks, or of the readout-observable residual memory.

The new route suggests a sharper candidate for that control. Instead of treating the omitted fourth tensor as an undifferentiated object, study the few quadratic residuals R_w selected by the future readouts and their source derivatives. Their covariance Gram matrix is positive. Their part predicted by current linear observables is fixed by third moments. Their remaining norms and angles are the genuinely additional data. An exact or low-rank evolution law for those residuals would supply a small fourth channel for a substantive reason. Neither an index nor the two observed network trajectories proves such a law.

The first uncertain implication is whether the actual omitted-sector influence and its future pullbacks admit a small, controlled family of quadratic residual Gram entries. The current work proves the influence observables and their static decomposition. It does not prove that their transport uses a small or inexpensive family of Gram entries. Proposition R.7 prevents resolving that implication by a universal identity in retained fourth data alone. Proposition R.8 prevents resolving it by silently identifying the index mask with an orthogonal compression.

6.7 Primary-source check and research status

The operator formulas above are proved here. The literature check verifies the following adjacent source-to-target dictionary, without a novelty claim:

1. I. Klich, *Full Counting Statistics: An elementary derivation of Levitov's formula*, arXiv:cond-mat/0209642v1, 27 September 2002, <https://arxiv.org/pdf/cond-mat/0209642>. Equation (8) proves a Fock trace/determinant identity for a finite-dimensional single-particle space, and equation (13) gives the counting determinant under the stated noninteracting-fermion set-up. The source itself explicitly restricts that derivation to a finite single-particle Hilbert space. Here only the determinant calculus is compared, with a separately proved probability-vacuum specialization. No infinite Fock-space theorem or physical fermion model is imported.
2. A. Soshnikov, *Gaussian limit for determinantal random point fields*, arXiv:math/0006037v3, 21 May 2001, <https://arxiv.org/pdf/math/0006037>. Lemma 1, equation (14), is the cyclic trace expansion for cumulants. Its cumulants use the ordinary factorial convention. No asymptotic Gaussian limit is applied to the network.
3. J. Breuer and M. Duits, *Central Limit Theorems for Biorthogonal Ensembles and Asymptotics of Recurrence Coefficients*, arXiv:1309.6224v2, 8 October 2013, <https://arxiv.org/pdf/1309.6224>. Equation (3.6) is the counting determinant; equation (3.11) is the Toeplitz/Hankel product identity. Lemma 4.5 centers the determinant, and Lemma 4.6 controls its cumulant coefficients by Hilbert–Schmidt commutators. Their C_m is $\kappa_m/m!$, so it must not be identified with the present c_m . The banded/right-limit hypotheses of their later limit theorems have not been established here. Equations R.1–R.4 and their bounds were derived directly, without depending on a transcription of their constants.
4. J. Avron, R. Seiler and B. Simon, *The Index of a Pair of Projections*, Journal of Functional Analysis 120 (1994), 220–237, DOI <https://doi.org/10.1006/jfan.1994.1031>. The primary author copy is https://phsites.technion.ac.il/avron/wp-content/uploads/sites/3/2013/05/J_Func_Anal_120_220_1994_index_pair.pdf. The authenticated article and publisher abstract state the odd-power trace/index formula under the stated Schatten hypothesis. The finite-rank rank-one index-zero assertion used here is independently elementary; it does not rely on extending that theorem beyond its hypotheses.

The search covered the exact counting-determinant, Toeplitz/Hankel, commutator-cumulant, and pair-projection terminology and the above primary sources, checked on 7 October 2026. It was a theorem-translation and route investigation, not a comprehensive novelty search. No source cache was initialized, and no claim of source-PDF retention is made.

Route status. The relative determinant, its fourth boundary coefficient and source variation, the Krylov loss, the third-moment Schur decomposition, the Gram completion bounds, and both counterexamples are established by the proofs above. The suggested inexpensive evolution of the selected quadratic residuals remains conditional. No weight arrays, exact source-tangent implementation, or new network measurements were available to test that comparison. This route neither computes the two networks' deep rate nor proposes changing the fitted chain table before the other defects are included.

7 Conditional geometry: the omitted feed, heat smoothing, and error certificates

This section strengthens the exact factor-Gaussian evaluation route in three directions. First, it identifies the conditional cumulants that create each omitted multiplicity class. Second, conditional Gaussian averaging is an actual Mehler semigroup, which gives explicit geometric Hermite tails. Third, changing a Gaussian reference covariance or its prescribed source tangent has a certified effect on the missing response. The latter certificate uses at most four-coordinate Gaussian geometry and univariate ReLU moments. It does not assume classical differentiability of ReLU.

The results concern specified Gaussian reference laws and their tangents. They do not certify the error between a Gaussian reference and the actual non-Gaussian network state, nor the sufficiency of two transported gains.

7.1 Conditional cumulants identify the three missing mechanisms

Suppose

$$Z_i = \mu_i + B_i X + s_i \epsilon_i, \quad X \sim N(0, I_r), \quad \epsilon \sim N(0, I_n), \quad X \perp \epsilon, \quad s_i > 0,$$

and let $H_i = (Z_i)_+$. Conditional on X , the H_i are independent. Set

$$m_i = \mathbb{E}[H_i | X], \quad v_i = \text{Var}(H_i | X), \quad c_{i3} = \kappa_3(H_i | X).$$

These are random functions of X . Cumulants and covariances with subscript X below are taken only over the factor X .

Theorem CG.1 (conditional decomposition of omitted sectors). For distinct indices as indicated,

$$\kappa(H_i, H_i, H_i, H_j) = \kappa_X(m_i, m_i, m_i, m_j) + 3\kappa_X(v_i, m_i, m_j) + \text{Cov}_X(c_{i3}, m_j), \quad (22)$$

$$\kappa(H_i, H_i, H_j, H_k) = \kappa_X(m_i, m_i, m_j, m_k) + \kappa_X(v_i, m_j, m_k), \quad (23)$$

$$\kappa(H_i, H_j, H_k, H_l) = \kappa_X(m_i, m_j, m_k, m_l). \quad (24)$$

Proof. For any finite family $Y_1, \dots, Y_d$, the conditional cumulant identity is

$$\kappa(Y_1, \dots, Y_d) = \sum_{\pi \in \mathcal{P}_d} \kappa_X(\kappa(Y_j : j \in A | X) : A \in \pi). \quad (\text{CG.1a})$$

A one-argument outer cumulant means expectation. To prove the identity, use formal moment polynomials through total degree d in $\log \mathbb{E}_X \exp\{\log \mathbb{E}[e^{\sum t_j Y_j} | X]\}$. Expanding the outer logarithm in cumulants, the coefficient of $t_1 \cdots t_d$ is indexed by a partition of the labels, with one conditional cumulant per block. Only moments through degree d are used. This is also the conditional cumulant identity of Brillinger; the derivation here fixes all conventions.

Under conditional independence, a conditional cumulant vanishes if its block contains two different neuron indices. In class $(3, 1)$ the surviving partitions are the four-singleton partition, the three choices of a pair of copies of i , and the block of three copies of i . In class $(2, 1, 1)$ only the singleton partition and the pair of copies of i survive. In class $(1, 1, 1, 1)$ only the singleton partition survives. $\square$

The three omitted classes therefore involve different factor observables: conditional skew coupled to a conditional mean, conditional variance coupled to two conditional means, and a connected fourth cumulant of conditional means. A scalar factor can carry many linearly independent such observables.

7.2 A pair source creates conditional covariance, even when the baseline is independent

It is not valid to apply a third mean derivative directly to (22)–(24) and call it the pure third-source response. A third mean derivative of a cumulant includes extra products of lower density derivatives. We instead lift the specified source to the joint factor space and differentiate the conditional cumulant identity once.

For a symmetric pure third source with pair entries $T_{ij} = \Gamma_{ij}$, define the conditional density score

$$h_T^{\text{res}}(\epsilon) = \sum_{i \neq j} \frac{\Gamma_{ij}}{2s_i^2 s_j} \text{He}_2(\epsilon_i) \text{He}_1(\epsilon_j). \quad (\text{CG.2a})$$

Its conditional mean given X is zero, so the factor law is unchanged to first order. Integrating the joint density variation over X gives $\frac{1}{6}T : \partial_{\mu}^3 p_{\mu, \Sigma}$ for the marginal Gaussian law of Z . Thus this is exactly the specified incoming pure third tangent. It can be realized by the normalized squared-score positive density path.

At each factor value, let $M_i(t) = \mathbb{E}_{\epsilon} e^{tH_i}$ and define the formal series

$$A_{i,q}(t) = \frac{\partial_{\mu_i}^q M_i(t)}{M_i(t)} = \sum_{a \geq 1} a_{i,q,a} \frac{t^a}{a!}, \quad q = 1, 2. \quad (\text{CG.2b})$$

The derivative holds the factor value and residual variance fixed. These coefficients are finite polynomials in univariate half moments and their mean derivatives. They are not assumed to be ordinary mean derivatives of the univariate cumulant for $q = 2$.

For $i \neq j$ set

$$d_{ab}^{ij} = \frac{1}{2} (\Gamma_{ij} a_{i,2,a} a_{j,1,b} + \Gamma_{ji} a_{i,1,a} a_{j,2,b}), \quad a, b \geq 1. \quad (\text{CG.2c})$$

This is the first variation of $\kappa(H_i^{[a]}, H_j^{[b]} | X)$, where the superscript denotes repeated arguments, not powers. In particular $d_{11}^{ij} = d_{11}^{ji}$. All conditional one-neuron cumulants have zero variation under the pair source, and every conditional cumulant involving at least three distinct neurons also has zero variation.

Theorem CG.2 (exact pair-source feed by conditional mechanisms). For the pair source above,

$$\delta_T \kappa(H_i, H_i, H_i, H_j) = \mathbb{E}_X d_{31}^{ij} + 3 \text{Cov}_X(d_{21}^{ij}, m_i) + 3 \text{Cov}_X(d_{11}^{ij}, v_i) + 3 \kappa_X(d_{11}^{ij}, m_i, m_i), \quad (25)$$

$$\begin{aligned} \delta_T \kappa(H_i, H_i, H_j, H_k) &= \text{Cov}_X(d_{21}^{ij}, m_k) + \text{Cov}_X(d_{21}^{ik}, m_j) + \text{Cov}_X(d_{11}^{jk}, v_i) \\ &\quad + 2 \kappa_X(d_{11}^{ij}, m_i, m_k) + 2 \kappa_X(d_{11}^{ik}, m_i, m_j) + \kappa_X(d_{11}^{jk}, m_i, m_i), \end{aligned} \quad (26)$$

$$\delta_T \kappa(H_i, H_j, H_k, H_l) = \sum_{\{a,b\} \subset \{i,j,k,l\}} \kappa_X(d_{11}^{ab}, m_c, m_d), \quad (27)$$

where in the last line $\{c, d\}$ is the complementary pair. There are six terms in that sum.

Proof. Conditional independence factorizes the tilted baseline density. For one ordered source pair its conditional logarithmic moment variation is $\frac{1}{2} \Gamma_{ij} A_{i,2}(t_i) A_{j,1}(t_j)$. This proves (CG.2c) and the conditional support assertions.

Because the factor law is fixed, differentiating (CG.1a) gives

$$\delta \kappa(Y_1, \dots, Y_d) = \sum_{\pi \in \mathcal{P}_d} \sum_{A \in \pi} \kappa_X(\delta \kappa(Y_j : j \in A | X), \kappa(Y_j : j \in C | X) : C \in \pi \setminus \{A\}). \quad (\text{CG.2d})$$

An unmarked block must contain a single neuron index; the marked block must contain exactly two. In class (3, 1), the marked blocks can have types (3, 1), (2, 1), or (1, 1). Their complements give respectively an expectation, three covariances with m_i , three covariances with v_i , and three third cumulants with two copies of m_i .

In class (2, 1, 1), a marked triple can be ijj or iik ; a marked pair jk leaves either the block ii or its two singletons; and a marked pair ij or ik can be selected in two ways. In the all-distinct class, only a marked pair with two singleton complements survives. These enumerations give all displayed coefficients. $\square$

The all-distinct fourth response has a particularly precise mechanism: it is an outer third cumulant of a conditional covariance response with two conditional means. At zero factor rank the outer covariances and higher cumulants vanish. Then a pair source can feed (3, 1) directly, but not (2, 1, 1) or (1, 1, 1, 1).

Diagonal third sources are included by adding $\sum_i \tau_i \text{He}_3(\epsilon_i)/(6s_i^3)$ to the residual score. They change only the conditional one-neuron cumulants; their omitted response follows by the ordinary first derivative of (22)–(24) with those conditional fields. This first-variation statement is distinct from taking three mean derivatives of the already-formed outer cumulant.

Corollary CG.2a (all-distinct feed for the full third tensor). For a general symmetric T , write $\tau_i = T_{iii}$ and $\Gamma_{ij} = T_{iij}$ as before, and retain its all-distinct entries as well. Define

$$s_i^T = \frac{\tau_i}{6} \partial_{\mu_i}^3 m_i, \quad e_{ijk}^T = T_{ijk}(\partial_{\mu_i} m_i)(\partial_{\mu_j} m_j)(\partial_{\mu_k} m_k) \quad (i, j, k \text{ distinct}).$$

Here the superscript distinguishes the conditional mean response s_i^T from the residual standard deviation s_i . For four distinct output labels $I = \{i, j, k, l\}$,

$$\begin{aligned} \delta_T \kappa(H_i, H_j, H_k, H_l) &= \sum_{a \in I} \kappa_X(s_a^T, m_b, m_c, m_d) \\ &+ \sum_{\{a, b\} \subset I} \kappa_X(d_{11}^{ab}, m_c, m_d) + \sum_{\{a, b, c\} \subset I} \text{Cov}_X(e_{abc}^T, m_d). \end{aligned} \quad (28)$$

Complements determine the unspecified labels in each sum. There are four, six, and four terms, respectively.

Proof. Extend the residual score by the diagonal terms already described and by

$$\sum_{i < j < k} \frac{T_{ijk}}{s_i s_j s_k} \epsilon_i \epsilon_j \epsilon_k.$$

The coefficient of an unordered distinct triple is one because its six permutations cancel the 1/6 in the source convention. The associated conditional logarithmic moment variation is

$$T_{ijk} A_{i,1}(t_i) A_{j,1}(t_j) A_{k,1}(t_k).$$

Its coefficient linear in each t_i, t_j, t_k is e_{ijk}^T . No conditional fourth cumulant with four distinct neuron indices can be created by this third-order score. In the marked conditional partition identity, the marked block therefore has size one, two, or three. Every unmarked block is a singleton because all output labels are distinct. This gives the displayed three sums and all multiplicities. $\square$

In particular, restoring all-distinct incoming components of the full Gaussian scale tangent adds the last covariance sum. It is a different operation from correcting only the omitted outgoing sectors while holding the existing pair-restricted input source fixed.

7.3 The exact Mehler smoothing hidden in the factor representation

Write

$$\sigma_i^2 = \|B_i\|^2 + s_i^2, \quad r_i = \frac{\|B_i\|}{\sigma_i} < 1, \quad \alpha_i = \frac{\mu_i}{\sigma_i}.$$

When $B_i \neq 0$, let $e_i = B_i/\|B_i\|$ and $U_i = e_i X$. Then $U_i \sim N(0, 1)$ and

$$Z_i = \mu_i + \sigma_i(r_i U_i + \sqrt{1 - r_i^2} \epsilon_i).$$

For $0 < r < 1$, define

$$(P_r f)(u) = \mathbb{E} f(ru + \sqrt{1 - r^2} \epsilon).$$

If N is the Gaussian number operator, this is exactly $P_r = e^{-tN}$ with $t = -\log r$. In particular, $P_r(\text{He}_k / \sqrt{k!}) = r^k \text{He}_k / \sqrt{k!}$.

Proof of the spectral multiplier. For $h_k = \text{He}_k / \sqrt{k!}$, use the generating function

$$e^{ty - t^2/2} = \sum_k h_k(y) t^k / \sqrt{k!}.$$

Conditional Gaussian integration replaces t by rt in the same generating function evaluated at u . Comparing coefficients proves $P_r h_k = r^k h_k$. Gaussian polynomials are dense in $L^2(\gamma_1)$, and conditional expectation is an L^2 contraction, so the identity extends to the full Hermite expansion. $\square$

For example, $m_i(X) = P_{r_i} f_i(U_i)$ with $f_i(y) = (\mu_i + \sigma_i y)_+$. Thus positive residual noise applies actual heat smoothing before the factor observables are combined. The geometric factor r_i^k is available even though an unsmoothed gate has only limited Gaussian Sobolev regularity.

Let Π_k denote projection onto total factor Hermite degree k .

Theorem CG.3 (conditional-mean spectrum and covariance certificate). For $k \geq 2$,

$$\|\Pi_k m_i\|_2^2 = \sigma_i^2 r_i^{2k} \varphi(\alpha_i)^2 \frac{\text{He}_{k-2}(\alpha_i)^2}{k!}. \quad (29)$$

For $k = 1$, the energy is $\sigma_i^2 r_i^2 \Phi(\alpha_i)^2$. Consequently, for $K \geq 0$,

$$\|(I - \Pi_{\leq K}) m_i\|_2^2 \leq r_i^{2K+2} \text{Var}(H_i). \quad (30)$$

Let $m^{\leq K}$ be the componentwise projection, and let $G_K = \text{Cov}_X(m^{\leq K})$. Then

$$\text{Cov}(H) = \text{diag}(\mathbb{E}_X v_i) + G_K + E_K, \quad E_K \succeq 0, \quad \text{rank } G_K \leq \binom{r + K}{K} - 1,$$

$$\|E_K\|_{\text{op}} \leq \text{tr } E_K \leq \sum_i r_i^{2K+2} \text{Var}(H_i). \quad (31)$$

For any downstream matrix W ,

$$W \text{Cov}(H) W^\top = W \text{diag}(\mathbb{E} v_i) W^\top + W G_K W^\top + W E_K W^\top,$$

and $\|W E_K W^\top\|_{\text{op}} \leq \|W\|_{\text{op}}^2 \text{tr } E_K$. For a single row w , the sharper bound

$$w^\top E_K w \leq \left(\sum_i |w_i| r_i^{K+1} \sqrt{\text{Var}(H_i)} \right)^2 \quad (\text{CG.3a})$$

also holds.

Proof. For $k \geq 2$, distributional Gaussian integration by parts gives the univariate Hermite coefficient

$$\langle f_i, h_k \rangle = \sigma_i (-1)^{k-2} \varphi(\alpha_i) \frac{\text{He}_{k-2}(\alpha_i)}{\sqrt{k!}}.$$

Indeed, the k th mean derivative of a ReLU expectation is $\sigma_i^{1-k} (-1)^{k-2} \text{He}_{k-2}(\alpha_i) \varphi(\alpha_i)$. For $k = 1$ the coefficient is $\sigma_i \Phi(\alpha_i)$. The Mehler multiplier now proves the energy formula, since $h_k(e_i X)$ has unit $L^2(\gamma_r)$ norm and lies in total degree k . Bounding each multiplier in the tail by r_i^{K+1} proves (30).

Conditional total covariance gives $\text{Cov}(H) = \text{diag}(\mathbb{E}v_i) + \text{Cov}_X(m)$. Orthogonality of the common Hermite projection splits the last Gram matrix into $G_K + E_K$ with both terms positive semidefinite. The nonconstant polynomial space of degree at most K has dimension $\binom{r+K}{K} - 1$. The trace is the sum of squared component tails. The row estimate follows from the triangle inequality in L^2 . $\square$

The dense matrix $W \text{diag}(\mathbb{E}v_i) W^\top$ remains in this identity. Fixed weights need not make it diagonal. Moreover, the next preactivation law is not asserted to be Gaussian or conditionally independent. The theorem certifies covariance compression at this reference layer; it does not assert invariance of a factor-Gaussian family through depth.

Proposition CG.4 (one factor can have arbitrarily large exact rank). Fix $s > 0$, take $B_i = 1$, and choose distinct μ_i . Then $\text{Cov}_X(m_1, \dots, m_n)$ is positive definite for every n .

Proof. If $\sum_i a_i m_i(X)$ is constant almost surely, continuity and the full support of the Gaussian imply the same equality for every real x . Twice differentiating gives $\sum_i a_i s^{-1} \varphi((x + \mu_i)/s) = 0$. Multiplying by $\exp(x^2/(2s^2))$ leaves a linear combination of exponentials with distinct exponents $-\mu_i/s^2$. Taking derivatives of orders $0, \dots, n-1$ at zero gives a Vandermonde system, hence all $a_i = 0$. This is exactly strict positive definiteness of the covariance Gram matrix. $\square$

Thus the useful conclusion is a quantitative approximation rank derived from residual noise and Hermite degree, not an exact rank inferred from the number of global factors.

7.4 Moment jets admit certified polynomial replacement

Fix the reference centering $\bar{h}_i = \mathbb{E}H_i$ and weight w_i . For $k \leq 4$ define

$$f_{ik}(y) = \{w_i[(\mu_i + \sigma_i y)_+ - \bar{h}_i]\}^k, \quad f_{ik,j} = \langle f_{ik}, h_j \rangle.$$

Let $\nu_{ik}^{(q)}(X)$ be the q th auxiliary mean derivative of the conditional moment, with $\bar{h}_i$ held fixed, for $0 \leq q \leq 3$.

Theorem CG.5 (exact jet spectrum). For $B_i \neq 0$,

$$\nu_{ik}^{(q)}(X) = \sigma_i^{-q} \sum_{m \geq 0} r_i^m \sqrt{\frac{(m+q)!}{m!}} f_{ik,m+q} h_m(e_i X). \quad (32)$$

The formula is an L^2 identity. If $q = 0$, its constant term is $\mathbb{E}f_{ik}$; for $q \geq 1$ the displayed formula also specifies the constant response term. With $p_{ikq}^K = \Pi_{\leq K} \nu_{ik}^{(q)}$,

$$\|\nu_{ik}^{(q)} - p_{ikq}^K\|_2^2 \leq \sigma_i^{-2q} c_{K,q}(r_i) \text{Var}(f_{ik}), \quad c_{K,q}(r) = \sup_{m \geq K+1} r^{2m} \frac{(m+q)!}{m!}. \quad (33)$$

For $q = 0$, $c_{K,0} = r^{2K+2}$. For $q \geq 1$, the supremum is the maximum of a unimodal explicit integer sequence: the ratio of consecutive terms is $r^2(m+q+1)/(m+1)$. In particular it can be evaluated without an infinite search. Alternatively, for any $r < a < 1$,

$$c_{K,q}(r) \leq a^{2K+2} \frac{q!}{[1 - (r/a)^2]^{q+1}}. \quad (\text{CG.5a})$$

If $B_i = 0$, all these conditional jets are constants and all tails vanish.

Proof. For fixed positive r_i , the q th auxiliary mean derivative is $\sigma_i^{-q} P_{r_i}(f_{ik}^{(q)})(e_i X)$, with the derivative understood distributionally. The normalized Hermite derivative is $h_{m+q}^{(q)} = \sqrt{(m+q)!/m!} h_m$. Equivalently, differentiating the smoothed function gives $\sigma_i^{-q} r_i^{-q} \partial_u^q P_{r_i} f_{ik}$, which yields the same series. The geometric multiplier makes the differentiated series square summable because the factorial ratio is a polynomial of degree q .

Parseval gives the exact squared tail, and replacing the multiplier by its supremum leaves a subseries of $\sum_{j \geq 1} |f_{ik,j}|^2$. For (CG.5a), use $\sum_{m \geq 0} (m+q)! x^m / m! = q! / (1-x)^{q+1}$ and bound each summand by the sum, with $x = (r/a)^2$. $\square$

All coefficients $f_{ik,j}$ needed up to a finite degree are univariate Gaussian half moments against a polynomial. Expanding the polynomial and using the half-moment recurrence computes them exactly up to the evaluations of Φ and φ .

Theorem CG.6 (finite polynomial integration with an explicit error). Suppose a scalar factor integrand has a finite expression

$$R(X) = \sum_{\alpha} c_{\alpha} \prod_{j=1}^{d_{\alpha}} F_{\alpha j}(X), \quad d_{\alpha} \leq 4, \quad (\text{CG.6a})$$

in primitives $F_{\alpha j} = \nu_{ik}^{(q)}$, with known baseline constants. The exact missing pair-source integrand from the factor-node formula has this form. Replace each primitive by its degree- K projection and call the result R_K . Then R_K is a polynomial of total factor degree at most $4K$, so its Gaussian expectation is a finite Gaussian moment calculation. For a product of $d \geq 2$ factors, let $\epsilon_i = \|F_i - p_i\|_2$ and $p = 2(d-1)$. Then

$$\left| \mathbb{E} \prod_{i=1}^d F_i - \mathbb{E} \prod_{i=1}^d p_i \right| \leq \sum_{i=1}^d \epsilon_i \prod_{j < i} \|p_j\|_p \prod_{j > i} \|F_j\|_p. \quad (34)$$

For $d = 1$ use $|\mathbb{E}(F_1 - p_1)| \leq \epsilon_1$. Sum these bounds against $|c_{\alpha}|$ to certify $|\mathbb{E}R - \mathbb{E}R_K|$.

Every quantity on the right is explicitly evaluable. The ϵ_i have the preceding univariate tail bounds. For a primitive with residual standard deviation s_i and $p \geq 2$,

$$\|\nu_{ik}^{(q)}\|_p \leq s_i^{-q} \|f_{ik}\|_{2p} \|\text{He}_q\|_{2p}. \quad (\text{CG.6b})$$

The norm of p_{ikq}^K is an ordinary univariate Gaussian polynomial norm, because it is a polynomial in the single ridge coordinate $e_i X$. For the even exponents $p = 2, 4, 6$ used in (34), it is a finite Gaussian moment polynomial.

Proof. Expand the difference of products by replacing one factor at a time. Apply Cauchy–Schwarz to the error factor and the product of all remaining factors, then Hölder with $d-1$ equal exponents. For (CG.6b), the residual Gaussian score identity gives $\nu_{ik}^{(q)} = s_i^{-q} \mathbb{E}_{\epsilon} [f_{ik}(r_i U_i + \sqrt{1 - r_i^2} \epsilon_i) \text{He}_q(\epsilon_i) | X]$. Conditional Jensen followed by Hölder bounds its L^p norm by the stated univariate norms; independence between these two factors is not required.

The influence has degree four in the activations. Conditional independence replaces each raw product by a product of at most four unary conditional moments, and the source derivative distributes across those same factors. Thus no monomial requires more than four primitives. Projection replaces each primitive by a degree- K polynomial. All polynomial moments are computed using $\mathbb{E}X^{2a} = (2a - 1)!!$ and odd Gaussian moments zero, coordinatewise in a standard factor basis. $\square$

This is a certificate at each finite K . It does not assert uniform small cost as r or the desired precision grows, or infer L^p convergence solely from an L^2 tail. Polynomial coefficient arithmetic in a fixed basis can use the existing $O(n^2)$ factor-node circuit, with an additional cost depending on r and K , instead of materializing all neuron quadruples. The finite error bound itself determines whether a chosen degree suffices.

7.5 At most four coordinates govern Gaussian-reference stability

Let $(\mu(s), \Sigma(s), T(s))$, $0 \leq s \leq 1$, be a continuously differentiable path with $\Sigma(s) > 0$ and symmetric third tensor $T(s)$. For a fixed row w , let

$$N(s) = \delta_{T(s)} K_{\text{miss}}(N(\mu(s), \Sigma(s)); w),$$

where the pure third tangent and all recentering terms are defined at the current reference law. Write

$$M_i(s) = \|H_i - \mathbb{E}H_i\|_{L^8}, \quad a_i(s) = |w_i| M_i(s).$$

These are univariate moments determined by $\mu_i(s)$ and $\Sigma_{ii}(s)$. For nonnegative a define

$$\mathcal{W}_{\text{miss}}(a) = \left(\sum_i a_i \right)^4 - 3 \left(\sum_i a_i^2 \right)^2 + 2 \sum_i a_i^4. \quad (\text{CG.7a})$$

This is precisely the sum of the nonnegative weight products over omitted ordered quadruples.

For each nonempty $S \subset \{1, \dots, n\}$ with $|S| \leq 4$, restrict all parameters to S and define

$$\begin{aligned} \tau_S &= \|(\Sigma_S^{-1/2})^{\otimes 3} T_S\|_F, & \dot{\tau}_S &= \|(\Sigma_S^{-1/2})^{\otimes 3} \dot{T}_S\|_F, \\ \beta_S &= \|\Sigma_S^{-1/2} \dot{\mu}_S\|_2, & \eta_S &= \|\Sigma_S^{-1/2} \dot{\Sigma}_S \Sigma_S^{-1/2}\|_F. \end{aligned}$$

The dot in $\dot{\tau}_S$ denotes the displayed norm of $\dot{T}_S$; it is not the derivative of the norm τ_S . Put

$$\mathcal{B}_S = \frac{11}{\sqrt{6}} \dot{\tau}_S + 11 \tau_S \sqrt{\frac{2}{3} \beta_S^2 + \frac{5}{6} \eta_S^2} + \frac{38}{\sqrt{6}} \tau_S \sqrt{\beta_S^2 + \frac{1}{2} \eta_S^2}. \quad (\text{CG.7b})$$

Theorem CG.7 (certified reference and source perturbation). The response $N(s)$ is continuously differentiable, and

$$\boxed{|N(1) - N(0)| \leq \int_0^1 \mathcal{W}_{\text{miss}}(a(s)) \max_{1 \leq |S| \leq 4} \mathcal{B}_S(s) ds.} \quad (35)$$

Also

$$|N(s)| \leq \frac{11}{\sqrt{6}} \mathcal{W}_{\text{miss}}(a(s)) \max_{1 \leq |S| \leq 4} \tau_S(s). \quad (\text{CG.7c})$$

One may replace the maximum by the tighter sum over omitted quadruples, using the corresponding support S for each term. No eigenvalue of the full n -dimensional covariance is needed if the restricted geometries are available.

Proof. Locality. Every joint fourth cumulant in the missing contraction is a functional of at most four activation coordinates. Its moment products depend only on the corresponding Gaussian marginal. The density tangent on this marginal is determined exactly by T_S : integrating a mean derivative with an index outside S gives zero. This is why only the restricted parameters enter.

Gaussian density scores. At one path point and on one support S , let p be the Gaussian density. For a symmetric tensor A of degree k , the Gaussian Hermite isometry gives

$$\left\| \frac{A : \partial_\mu^k p}{p} \right\|_2 = \sqrt{k!} \|(\Sigma^{-1/2})^{\otimes k} A\|_F. \quad (\text{CG.7d})$$

To prove it, whiten the Gaussian, expand in the normalized multivariate Hermite basis, and use orthogonality; the $k!$ accounts for ordered tensor indices. Symmetrization is an orthogonal projection for the Frobenius norm.

The source score is $h_T = (T : \partial_\mu^3 p)/(6p)$ and has norm $\tau_S/\sqrt{6}$. The reference-path score has orthogonal first and second chaos parts, so its norm is $\sqrt{\beta_S^2 + \eta_S^2/2}$. Holding T fixed at this point, the joint source/reference density derivative has fourth and fifth chaos parts

$$\frac{1}{6p}(T \otimes \dot{\mu}) : \partial_\mu^4 p + \frac{1}{12p}(T \otimes \dot{\Sigma}) : \partial_\mu^5 p. \quad (\text{CG.7e})$$

Their squared norm is at most $\tau_S^2(2\beta_S^2/3 + 5\eta_S^2/6)$. Indeed $\sqrt{4!}/6 = \sqrt{2/3}$ and $\sqrt{5!}/12 = \sqrt{5/6}$, and the two degrees are orthogonal. The moving source adds the third-chaos score $h_{\dot{T}}$.

Centered marked partitions. For four centered scalar variables $Y_1, \dots, Y_4$, differentiate the moment-partition formula

$$\kappa(Y_1, Y_2, Y_3, Y_4) = \sum_{\pi \in \mathcal{P}_4} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{A \in \pi} \mathbb{E} \prod_{j \in A} Y_j.$$

Centering constants can be fixed at the current reference point: cumulants of order at least two are invariant under adding deterministic constants. For one density variation, only one-block terms, the three pair-pair partitions with either block marked, and the four singleton-triple partitions with their singleton marked survive. Their absolute coefficient count is $1 + 6 + 4 = 11$.

For a mixed second density variation, terms with both marks in one block have the same coefficient count 11. Terms with marks in different blocks have count 6 from pair-pair partitions, 8 from singleton-triple partitions, and 24 from singleton-singleton-pair partitions (6 partitions, two assignments, coefficient 2). Their total is 38. Every unmarked singleton vanishes.

For a score h and a block A , Cauchy–Schwarz and Hölder give

$$\left| \mathbb{E} \left[h \prod_{j \in A} Y_j \right] \right| \leq \|h\|_2 \prod_{j \in A} \|Y_j\|_{2|A|} \leq \|h\|_2 \prod_{j \in A} \|Y_j\|_8.$$

Unmarked blocks obey the same product bound without a score norm. Combining the marked-partition counts with the three score norms gives $\mathcal{B}_S \prod_j \|Y_j\|_8$. Summing the weighted omitted quadruples and integrating proves (35). The one-variation count gives (CG.7c).

Nonsmoothness. On a compact positive-definite covariance path, every required density derivative is a polynomial times a Gaussian with uniform integrable domination against any product of four ReLUs. Differentiation therefore acts on the smooth density, not on a presumed classical fifth derivative of ReLU. The identity $\dot{p} = \dot{\mu} : \partial_\mu p + \frac{1}{2} \dot{\Sigma} : \partial_\mu^2 p$ follows by differentiating the Gaussian density or its characteristic function. These facts justify all moment derivatives and their continuity. $\square$

This argument is a quantitative specialization of Gaussian density differentiation (the distributional form of Price’s theorem); it supplies its own constants and proof for the cumulant response rather than invoking an unspecified Lipschitz estimate.

7.6 Computable simplifications and moving gain normalization

Every M_i is explicit. If $H = (\mu + \sigma G)_+$ and $m = \mu\Phi(\mu/\sigma) + \sigma\varphi(\mu/\sigma)$, then

$$M_i^8 = \mathbb{E}(H - m)^8 = (-m)^8 + \sum_{j=1}^8 \binom{8}{j} (-m)^{8-j} \mathbb{E}H^j.$$

The half moments satisfy $M_1^{\text{half}} = \mu\Phi + \sigma\varphi$ and $M_j^{\text{half}} = \mu M_{j-1}^{\text{half}} + (j-1)\sigma^2 M_{j-2}^{\text{half}}$, with $M_0^{\text{half}} = \Phi$ for that recurrence. A simple mean-independent bound is

$$\|H - \mathbb{E}H\|_8 \leq 1680^{1/8} \sigma. \quad (\text{CG.8a})$$

To prove it, let H' be an independent copy. Conditional Jensen gives $\mathbb{E}|H - \mathbb{E}H|^8 \leq \mathbb{E}|H - H'|^8$; the ReLU contraction then bounds this by $\sigma^8 \mathbb{E}|G - G'|^8 = 1680\sigma^8$.

For a same-diagonal covariance approximation, set $D_\sigma = \text{diag}(\sqrt{\Sigma_{ii}})$ and $R = D_\sigma^{-1} \Sigma D_\sigma^{-1}$. If every restricted correlation matrix of order at most four has least eigenvalue at least $\lambda > 0$, then

$$\tau_S \leq \lambda^{-3/2} \|\bar{T}_S\|_F, \quad \eta_S \leq \lambda^{-1} \|\bar{\Sigma}_S\|_F,$$

where bars divide each tensor index by its marginal standard deviation. For a fixed source and mean, (35) becomes

$$|N(1) - N(0)| \leq C_\Sigma \int_0^1 \mathcal{W}_{\text{miss}}(a) \max_{|S| \leq 4} (\tau_S \eta_S) ds, \quad C_\Sigma = 11\sqrt{\frac{5}{6}} + \frac{38}{\sqrt{12}}. \quad (\text{CG.8b})$$

If $|\bar{T}_{ijk}| \leq t_*$ and $|\bar{\Sigma}_{ij}| \leq e_*$, then $\tau_S \eta_S \leq 32\lambda^{-5/2} t_* e_*$ because $|S| \leq 4$. For a pair-restricted source the factor 8 in the tensor-norm estimate can be replaced by $\sqrt{40}$. The easily checked sufficient condition $\max_{i \neq j} |R_{ij}| \leq \rho < 1/3$ gives $\lambda \geq 1 - 3\rho$ by the quadratic-form bound $x^\top R_S x \geq [1 - (|S| - 1)\rho] \|x\|^2$. Failure of this sufficient condition is not evidence that the covariance is poorly conditioned; the actual restricted eigenvalues may still be used.

Corollary CG.8 (normalization and coherent projection are part of the certificate).

Let $v(s) = \text{Var}(\sum_i w_i H_i) > 0$, and express the unnormalized missing response in the current baseline gain units as $r(s) = N(s)/(3v(s)^2)$. This is not in general the complete source derivative of $K_{\text{miss}}/(3v^2)$: a pure third input source can change the output variance. If $B_N(s)$ denotes the integrand bound on $|N'(s)|$ in (35), then

$$|r(1) - r(0)| \leq \int_0^1 \left[\frac{B_N(s)}{3v(s)^2} + \frac{2|N(s)|}{3v(s)^3} B_v(s) \right] ds, \quad (\text{CG.8c})$$

where an evaluable variance-derivative bound is

$$B_v(s) = \left(\sum_i |w_i| \|H_i - \mathbb{E}H_i\|_4 \right)^2 \max_{1 \leq |S| \leq 2} \sqrt{\beta_S^2 + \eta_S^2/2}.$$

The baseline $v(s)$ itself needs only univariate and bivariate Gaussian ReLU moments. It can be replaced by a certified positive lower bound.

More generally, for a vector of row responses $\mathbf{N}(s)$ and a moving coherent projection $a(s)$,

$$\left| \frac{d}{ds} (a^\top \mathbf{N}) \right| \leq \sum_k |a_k| B_{N_k} + \sum_k |a'_k| |N_k|. \quad (\text{CG.8d})$$

For the Euclidean least-squares projection $a = b/\|b\|^2$ with $b \neq 0$,

$$\|a\| = \|b\|^{-1}, \quad \|a'\| = \|b'\|/\|b\|^2.$$

The moving incoming gain normalization is included through the actual $T(s)$ and $\dot{T}(s)$ in Theorem CG.7.

For clarity, put $K(s) = K_{\text{miss}}(s)$ and $v_T(s) = \delta_{T(s)}v(s)$. The complete normalized source response is

$$\tilde{r}(s) = \delta_{T(s)} \frac{K_{\text{miss}}}{3v^2} = \frac{N(s)}{3v(s)^2} - \frac{2K(s)v_T(s)}{3v(s)^3}. \quad (\text{CG.8e})$$

If $B_K \geq |K'|$ and $B_{v_T} \geq |v_T'|$, its reference-path derivative has the fully explicit bound

$$|\tilde{r}'| \leq \frac{B_N}{3v^2} + \frac{2|N|B_v}{3v^3} + \frac{2(B_K|v_T| + |K|B_{v_T})}{3v^3} + \frac{2|Kv_T|B_v}{v^4}.$$

All the added quantities admit the same local certificates. Define

$$q_S = \sqrt{\beta_S^2 + \eta_S^2/2}, \quad u_S = \tau_S/\sqrt{6}, \quad j_S = \tau_S \sqrt{2\beta_S^2/3 + 5\eta_S^2/6},$$

and let $a_i^{(p)} = |w_i| \|H_i - \mathbb{E}H_i\|_p$. Then valid choices are

$$\begin{aligned} |K| &\leq 4\mathcal{W}_{\text{miss}}(a^{(4)}), & B_K &= 11\mathcal{W}_{\text{miss}}(a^{(8)}) \max_{|S| \leq 4} q_S, \\ |v_T| &\leq \left(\sum_i a_i^{(4)} \right)^2 \max_{|S| \leq 2} u_S, & B_{v_T} &= \left(\sum_i a_i^{(4)} \right)^2 \max_{|S| \leq 2} \left(\frac{\dot{\tau}_S}{\sqrt{6}} + j_S + 2u_S q_S \right). \end{aligned}$$

Thus both meanings of gain transport can be compared without silently freezing the output normalization.

Proof. The quotient and product rules give the displayed derivatives. The variance first variation is the score applied to a centered quadratic observable. Apply the same two-coordinate locality and Cauchy–Schwarz argument as above, now with fourth moments. Finally, $a' = (I - 2uu^\top)b'/\|b\|^2$, $u = b/\|b\|$; the factor in parentheses is an orthogonal reflection. For the full normalized response apply the quotient rule once in the source and once along the reference path. A centered fourth cumulant has one raw fourth moment and three covariance products, giving the baseline bound with coefficient four. Its first derivative has the already proved coefficient eleven. A centered second cumulant has first-derivative coefficient one and mixed-derivative coefficients one for a joint score and two for distinct score marks. These give the remaining displayed bounds. $\square$

Positive covariance conditioning and a nonvanishing output normalization cannot be dropped without replacement. For example, take two independent $N(0, \sigma^2)$ coordinates, a fixed pair source $T_{112} = t$, and $w = (1, 1)$. Scaling the exact independent-Gaussian example gives $N = 3t\varphi(0)\sigma$. Its derivative with respect to σ^2 is $3t\varphi(0)/(2\sigma)$, which is unbounded as $\sigma \downarrow 0$. The normalized response is proportional to $t\sigma^{-3}$. This example holds the physical source fixed; a source recalibrated with variance has its additional $\dot{T}$ term and is a different comparison.

7.7 What the route establishes for a depth programme

The factor reduction now has a controlled approximation version. Given a candidate diagonal-plus-factor covariance, the moment and jet expansion provides a finite polynomial response with an explicit integration error. The Gaussian-reference interpolation theorem then bounds the error caused by the covariance, mean, source, and gain-projection changes used to obtain that representation. Both error budgets are expressed in supplied Gaussian parameters, weights, univariate moments, and restricted matrix norms.

The conditional formulas also identify what must propagate: for a pair source, the all-distinct fourth feed is controlled by connected outer cumulants of conditional covariance responses and conditional

means. These are richer observables than a single factor amplitude. Residual Gaussian noise gives their unary building blocks geometric spectral tails; it does not prove that every nonlinear combination or every future readout closes in two coordinates.

What remains for a particular network is to exhibit a representation with small enough restricted covariance error, sufficient residual smoothing, and a resulting certified response error that survives the readout-weighted depth propagation. None of these numerical hypotheses is inferred from the reported existence of one scale register.

Sources and evidence scope. The conditional partition identity is from David R. Brillinger, *The calculation of cumulants via conditioning*, Annals of the Institute of Statistical Mathematics **21** (1969), 215–218, https://www.ism.ac.jp/editsec/aism/pdf/021_1_0215.pdf. The distributional Gaussian covariance differentiation framework is verified in Felix Voigtlaender, *A general version of Price's theorem*, arXiv:1710.03576v2 (2020), <https://arxiv.org/pdf/1710.03576>. The specialized conditional source identities, explicit response constants, Hermite certificates, and comparison bounds above are derived in this section. No claim of global novelty is made. No network parameter arrays or Monte Carlo data were used.

8 A leading-coefficient theorem for weak common factors

Fix a finite width, residual standard deviations $s_i > 0$, means μ_i , loading vectors $b_i \in \mathbb{R}^r$, weights, and the physical pair-source coefficients Γ_{ij} . Scale only the loadings:

$$Z_i(\varepsilon) = \mu_i + \varepsilon b_i^\top X + s_i \epsilon_i, \quad X, \epsilon \text{ independent standard Gaussian vectors.}$$

The covariance is $\text{diag}(s_i^2) + \varepsilon^2 B B^\top$. The expansions below are for this fixed incoming source, with the pure third-density tangent defined at each Gaussian reference.

Define the univariate conditional fields

$$f_i(a) = \mathbb{E}(a + s_i G)_+, \quad V_i(a) = \text{Var}((a + s_i G)_+), \quad U_i(a) = \kappa_3((a + s_i G)_+).$$

All are explicit from univariate half moments. With $z = a/s_i$,

$$f'_i = \Phi(z), \quad f''_i = \varphi(z)/s_i, \quad f'''_i = -z\varphi(z)/s_i^2, \quad f''''_i = (z^2 - 1)\varphi(z)/s_i^3.$$

Put $A_i = V''_i + 2(f'_i)^2 = 2f'_i - 2f_i f''_i$. In each expression below, a neuron function is evaluated at $\mu_i + b_i^\top x$. Define smooth factor functions

$$\begin{aligned} m_i(x) &= f_i(\mu_i + b_i^\top x), & v_i(x) &= V_i(\mu_i + b_i^\top x), \\ D_{11}^{ij}(x) &= \tfrac{1}{2}\Gamma_{ij}f''_i f'_j + \tfrac{1}{2}\Gamma_{ji}f'_i f''_j, \\ D_{21}^{ij}(x) &= \tfrac{1}{2}\Gamma_{ij}A_i f'_j + \tfrac{1}{2}\Gamma_{ji}V'_i f''_j, \\ D_{31}^{ij}(x) &= \tfrac{1}{2}\Gamma_{ij}[U''_i + 6f'_i V'_i]f'_j + \tfrac{1}{2}\Gamma_{ji}U'_i f''_j. \end{aligned}$$

These are the conditional pair-source cumulant responses of CG.2, evaluated at $x = \varepsilon X$. For example, the identity $M''_i/M_i = (\log M_i)'' + [(\log M_i)']^2$ gives the coefficients A_i and $U''_i + 6f'_i V'_i$ in its second and third moment slots. Primes in this identity differentiate the Gaussian mean parameter.

For any smooth factor function F , let $g_F = \nabla F(0)$ and $H_F = \nabla^2 F(0)$, and define

$$\mathfrak{Z}(F, G, H) = g_G^\top H_F g_H + g_F^\top H_G g_H + g_F^\top H_H g_G.$$

Lemma WF.1 (the three-vertex Gaussian tree coefficient). If F, G, H have derivatives through order six of polynomial growth, then

$$\begin{aligned}\text{Cov}(F(\varepsilon X), G(\varepsilon X)) &= \varepsilon^2 g_F^\top g_G + O(\varepsilon^4), \\ \kappa(F(\varepsilon X), G(\varepsilon X), H(\varepsilon X)) &= \varepsilon^4 \mathfrak{T}(F, G, H) + O(\varepsilon^6).\end{aligned}$$

Proof. Gaussian domination justifies differentiating the moment expressions six times near zero. Their expectations are even functions of ε because X and $-X$ have the same law. The centered expansion is

$$F(\varepsilon X) - \mathbb{E}F(\varepsilon X) = \varepsilon g_F^\top X + \frac{1}{2} \varepsilon^2 (X^\top H_F X - \text{tr } H_F) + O_{L^p}(\varepsilon^3)$$

for each fixed finite p . The covariance coefficient follows immediately. For the third cumulant, the cubic term is a Gaussian odd moment and vanishes. The quartic terms have one quadratic and two linear factors. Wick's fourth-moment identity gives

$$\mathbb{E}[(u^\top X)(v^\top X)(X^\top H X - \text{tr } H)] = 2u^\top H v.$$

The factor $1/2$ in the Taylor term cancels this two. Summing the three choices of quadratic vertex gives $\mathfrak{T}$. Evenness and the bounded sixth derivative give the stated remainders. $\square$

Theorem WF.2 (leading feed in each omitted sector). Let $\Delta_{31}^{ij}(\varepsilon)$, $\Delta_{211}^{ijk}(\varepsilon)$, and $\Delta_{1111}^{ijkl}(\varepsilon)$ be the pure pair-source responses of the indicated outgoing joint fourth cumulants. Indices are distinct except for the displayed repeated index. Then

$$\Delta_{31}^{ij}(\varepsilon) = D_{31}^{ij}(0) + O(\varepsilon^2), \tag{WF.1}$$

$$\Delta_{211}^{ijk}(\varepsilon) = \varepsilon^2 \left[g_{D_{21}^{ij}}^\top g_{m_k} + g_{D_{21}^{ik}}^\top g_{m_j} + g_{D_{21}^{jk}}^\top g_{v_i} \right] + O(\varepsilon^4), \tag{WF.2}$$

$$\Delta_{1111}^{ijkl}(\varepsilon) = \varepsilon^4 \sum_{\{a,b\} \subset \{i,j,k,l\}} \mathfrak{T}(D_{11}^{ab}, m_c, m_d) + O(\varepsilon^6). \tag{WF.3}$$

Complements specify $\{c, d\}$. The last coefficient has six symmetric tree terms, each itself containing three choices of the Hessian vertex.

Proof. Apply the exact conditional pair-source identities CG.2 at each ε . In $(3, 1)$ the conditional expectation of D_{31} has limit $D_{31}(0)$, and its other terms are covariances or third cumulants of constant fields at zero. In $(2, 1, 1)$ the three outer covariances give exactly the quadratic coefficient shown; all outer third cumulants are $O(\varepsilon^4)$. In the all-distinct class there are exactly six outer third cumulants. Apply Lemma WF.1 to each.

The displayed functions are built from Gaussian half moments and their mean derivatives with fixed $s_i > 0$. They and all required derivatives have polynomial growth. Thus the lemma's domination and remainder hypotheses hold. No asymptotic statement in width is used. $\square$

The coefficient is explicit in means, residual variances, loadings and source entries. For example, with all scalar derivatives now evaluated at their respective μ_i ,

$$g_{m_i} = f'_i b_i, \quad H_{m_i} = f''_i b_i b_i^\top, \quad g_{v_i} = V'_i b_i,$$

$$\begin{aligned}g_{D_{11}^{ij}} &= \frac{1}{2} \Gamma_{ij} (f_i''' f_j' b_i + f_i'' f_j'' b_j) + \frac{1}{2} \Gamma_{ji} (f_i'' f_j'' b_i + f_i' f_j''' b_j), \\ H_{D_{11}^{ij}} &= \frac{1}{2} \Gamma_{ij} [f_i'''' f_j' b_i b_i^\top + f_i''' f_j'' (b_i b_j^\top + b_j b_i^\top) + f_i'' f_j''' b_j b_j^\top] \\ &\quad + \frac{1}{2} \Gamma_{ji} [f_i'''' f_j' b_i b_i^\top + f_i''' f_j'' (b_i b_j^\top + b_j b_i^\top) + f_i' f_j'''' b_j b_j^\top], \\ g_{D_{21}^{ij}} &= \frac{1}{2} \Gamma_{ij} (A_i' f_j' b_i + A_i f_j'' b_j) + \frac{1}{2} \Gamma_{ji} (V_i'' f_j'' b_i + V_i' f_j''' b_j).\end{aligned}$$

Every contraction therefore reduces to loading inner products $b_i^\top b_j$, the common-factor covariance entries. Keeping only affine approximations to m_c, m_d would omit the last two terms of $\mathfrak{T}$ at the same order ε^4 .

The respective contraction multiplicities are

$$4w_i^3w_j, \quad 12w_i^2w_jw_k \ (j < k), \quad 24w_iw_jw_kw_l \ (i < j < k < l).$$

The theorem gives a leading order for each component. Cancellations, multiplicities, and the dependence of weight sums on width can change their relative sizes after contraction. Coefficients may vanish in special configurations. There is no universal sign or width-uniform bound.

Finally, a full incoming scale tensor must be distinguished from the fixed pair source. If an all-distinct source entry has the form $T_{abc}(\varepsilon) = \varepsilon^2 \Upsilon_{abc}$, its additional term from CG.2a is

$$\begin{aligned} & \text{Cov}_X \left(\varepsilon^2 \Upsilon_{abc} \prod_{i \in \{a,b,c\}} f'_i(\mu_i + \varepsilon b_i^\top X), m_d(\varepsilon X) \right) \\ &= \varepsilon^4 \Upsilon_{abc} \left[\sum_{i \in \{a,b,c\}} f''_i b_i \prod_{j \in \{a,b,c\} \setminus \{i\}} f'_j \right]^\top (f'_d b_d) + O(\varepsilon^6). \end{aligned}$$

Thus these incoming triples can contribute at the same leading order as the pair-source all-distinct feed. A Gaussian scale tangent has exactly this order in its all-distinct entries under the stated covariance scaling. Its diagonal and pair calibration also moves with covariance and must be matched separately. The present expansion does not silently replace the ledger's incoming source.

9 Clipped Hermite approximation closes the product-convergence gap

Geometric L^2 tails of the unary conditional moments do not alone imply convergence of products of their unbounded Hermite partial sums. A clipped approximation removes that gap for every strictly positive residual Gaussian variance. The resulting integral is no longer a polynomial Wick calculation. This section gives both a deterministic quadrature certificate and a bounded-sampling certificate for that new integral, with no claim of uniform cost as the factor dimension grows or residual smoothing vanishes.

9.1 An elementary interpolation inequality for clipping

For $M > 0$, define

$$C_M(t) = \max\{-M, \min\{t, M\}\}.$$

It is one-Lipschitz and has range $[-M, M]$. All L^p norms below are on the same probability space; the approximated variables need not be independent.

Theorem 9.1 (Clipping converts an L^2 certificate into an L^p certificate). *For real $g \in L^p$, $p \geq 2$ finite, and $h \in L^2$,*

$$\|g - C_M(h)\|_p \leq \|g - C_M(g)\|_p + (2M)^{1-2/p} \|g - h\|_2^{2/p}. \quad (36)$$

Proof. Set $d = C_M(g) - C_M(h)$. Pointwise, $|d| \leq 2M$ and $|d| \leq |g - h|$. Hence

$$\mathbb{E}|d|^p \leq (2M)^{p-2} \mathbb{E}|g - h|^2.$$

Apply the triangle inequality to $g - C_M(h) = g - C_M(g) + d$. The case $p = 2$ has the factor $(2M)^0 = 1$. $\square$

This inequality does not claim that the original partial sums converge in L^p . It proves convergence for a different, explicitly bounded sequence.

9.2 Every conditional ReLU moment jet has explicit polynomial growth

Write one residual representation as

$$H = (\mu + bU + s\epsilon)_+, \quad U, \epsilon \sim N(0, 1) \text{ independent}, \quad b \geq 0, \quad s > 0.$$

For $k \in \{1, 2, 3, 4\}$, $q \in \{0, 1, 2, 3\}$, a fixed centering constant $\bar{h}$, and a fixed weight w , let

$$g_{kq}(u) = \partial_\eta^q \mathbb{E}_\epsilon \{w[(\mu + \eta + bu + s\epsilon)_+ - \bar{h}]^k\} \Big|_{\eta=0}.$$

The derivative holds $\bar{h}, b, s$ fixed. Put $A = |\mu| + |\bar{h}|$ and

$$C_{kq} = |w|^k s^{-q} 3^{k-1} \sqrt{q!} \left[A^k + b^k + s^k \sqrt{(2k-1)!!} \right]. \quad (37)$$

Lemma 9.2 (A growth bound for all required jets). *The moment jet is well defined and satisfies*

$$|g_{kq}(u)| \leq C_{kq}(1 + |u|^k), \quad u \in \mathbb{R}. \quad (38)$$

Constants or zero primitives can be kept exactly.

Proof. Differentiating the residual Gaussian density gives

$$g_{kq}(u) = s^{-q} \mathbb{E}_\epsilon \left[\{w[(\mu + bu + s\epsilon)_+ - \bar{h}]\}^k \text{He}_q(\epsilon) \right].$$

Density differentiation is justified by Gaussian domination of every polynomial factor on compact parameter intervals. Since

$$|(\mu + bu + s\epsilon)_+ - \bar{h}| \leq A + b|u| + s|\epsilon|,$$

use $(x + y + z)^k \leq 3^{k-1}(x^k + y^k + z^k)$ for nonnegative arguments. The elementary Hermite identity $\mathbb{E} \text{He}_q(\epsilon)^2 = q!$ and Cauchy–Schwarz imply

$$\mathbb{E} |\text{He}_q| \leq \sqrt{q!}, \quad \mathbb{E} [|\epsilon|^k |\text{He}_q|] \leq \sqrt{(2k-1)!! q!}.$$

Thus the bound before consolidating constants is

$$|w|^k s^{-q} 3^{k-1} \sqrt{q!} \left[A^k + b^k |u|^k + s^k \sqrt{(2k-1)!!} \right],$$

which is bounded by (38). $\square$

The constant is conservative and can be replaced by a sharper certified one. Its explicit dependence on s^{-q} is consequential: the theorem is not uniform when residual smoothing tends to zero.

9.3 A complete, all-smoothing-parameters convergence theorem

Let $h_j = \text{He}_j / \sqrt{j!}$ be the normalized Hermite basis. For one of the preceding jets put

$$\sigma^2 = b^2 + s^2, \quad r = b/\sigma < 1, \quad f(y) = \{w[(\mu + \sigma y)_+ - \bar{h}]\}^k.$$

When $b > 0$, the degree- K projection of $g = g_{kq}$ is

$$p_K(u) = \sigma^{-q} \sum_{m=0}^K r^m \sqrt{\frac{(m+q)!}{m!}} \langle f, h_{m+q} \rangle h_m(u).$$

As in CG.5, Gaussian smoothing and Hermite differentiation give

$$\|g - p_K\|_2^2 = \sigma^{-2q} \sum_{m>K} r^{2m} \frac{(m+q)!}{m!} |\langle f, h_{m+q} \rangle|^2.$$

In particular, for any $r < \theta < 1$,

$$\|g - p_K\|_2 \leq A_g \theta^{K+1}, \quad A_g = \frac{\sigma^{-q} \sqrt{q! \text{Var}(f)}}{[1 - (r/\theta)^2]^{(q+1)/2}}. \quad (39)$$

Indeed, bound the multiplier using $\sum_{m \geq 0} (m+q)! x^m / m! = q! / (1-x)^{q+1}$. The case $q = 0$ also has the sharper direct bound $\sqrt{\text{Var}(f)} r^{K+1}$. If $b = 0$, g is constant in the factor and is retained exactly.

We give explicit Gaussian clipping tails. For an integer $j \geq 0$ and $R \geq 0$, write

$$I_j(R) = \mathbb{E}[|U|^j \mathbf{1}_{\{|U|>R\}}] = 2 \int_R^\infty u^j \varphi(u) du.$$

The computable recurrence is

$$I_0(R) = 2\Phi(-R), \quad I_1(R) = 2\varphi(R), \quad I_j(R) = 2R^{j-1}\varphi(R) + (j-1)I_{j-2}(R) \quad (j \geq 2). \quad (40)$$

It follows directly by integration by parts.

For $R \geq 1$, define $c_0 = c_1 = 2$ and $c_j = 2 + (j - 1)c_{j-2}$ for $j \geq 2$. The Gaussian tail bound $I_0(R) \leq 2\varphi(R)/R$ and induction give

$$I_j(R) \leq c_j R^{j-1} \varphi(R) \quad (j \geq 0).$$

For positive integers d and $p \geq 2$, put

$$B_{d,p} = \left[\frac{1}{\sqrt{2\pi}} \sum_{j=0}^p \binom{p}{j} c_{dj} \right]^{1/p}.$$

Theorem 9.3 (Explicit clipped Hermite convergence for every $r < 1$). *Suppose a real primitive $g(U)$ satisfies $|g(u)| \leq C(1 + |u|^d)$ with $C > 0$, integer $d \geq 1$, and its Hermite partial sums satisfy $\|g - p_K\|_2 \leq A_g \theta^{K+1}$ for some $0 < \theta < 1$. Define*

$$R_K^2 = 1 + 4|\log \theta|(K + 1), \quad M_K = C(1 + R_K^d), \quad \hat{g}_K = C_{M_K}(p_K). \quad (41)$$

For every integer $p \geq 2$,

$$\|g - \hat{g}_K\|_p \leq \theta^{2(K+1)/p} \left[C B_{d,p} e^{-1/(2p)} R_K^{d-1/p} + (2C(1 + R_K^d))^{1-2/p} A_g^{2/p} \right]. \quad (42)$$

In particular, the sequence converges in every fixed finite integer L^p norm. For the moment jets above, this holds for every $s > 0$ and every $r < 1$. For $p = 4$ and $d \leq 4$, the error is at most a constant times $K^{15/8} \theta^{(K+1)/2}$ for large K , at fixed primitive parameters.

Proof. For $M = C(1 + R^d)$, $|g| > M$ implies $|U| > R$, and

$$\|g - C_M(g)\|_p^p \leq C^p \sum_{j=0}^p \binom{p}{j} I_{dj}(R) \leq C^p B_{d,p}^p R^{dp-1} e^{-R^2/2}.$$

Taking the p th root and applying Theorem 9.1 gives the result, because $e^{-R_K^2/(2p)} = e^{-1/(2p)} \theta^{2(K+1)/p}$. Both remaining factors grow polynomially in K , while $\theta < 1$. The worst exponent at $p = 4, d \leq 4$ is $(d - 1/4)/2 \leq 15/8$. $\square$

One common θ can be chosen for any finite set of primitives, provided $\max_i r_i < \theta < 1$. This is not a width-uniform assertion when the maximum r_i approaches one with width.

9.4 Products and the full fourth-cumulant response

Let $X \sim N(0, I_\tau)$, and let a finite response expression be

$$\mathcal{R}(X) = \sum_{\alpha} c_{\alpha} \prod_{j=1}^{d_{\alpha}} g_{\alpha j}(e_{\alpha j}^{\top} X), \quad d_{\alpha} \leq 4, \quad \|e_{\alpha j}\| = 1. \quad (43)$$

Constants can be represented separately. Different ridge coordinates may be arbitrarily correlated. Use a common K and θ in Theorem 9.3, and replace each primitive by its clipped approximation to obtain $\hat{\mathcal{R}}_K$. The reference coefficients and normalizers in this expression are fixed. Factor-dependent reciprocal normalizers would require additional growth and integrability bounds before they could be included. Let $E_i(K)$ be its certified L^4 error and let G_i be a certified L^4 norm bound for the exact primitive. For example,

$$G_i \leq C_i \left[1 + ((4d_i - 1)!)^{1/4} \right]$$

follows from its growth bound.

Theorem 9.4 (Certified convergence of the response expectation). *The expectation error has the explicit bound*

$$|\mathbb{E}\mathcal{R} - \mathbb{E}\widehat{\mathcal{R}}_K| \leq \mathcal{E}_K := \sum_{\alpha} |c_{\alpha}| \sum_{i=1}^{d_{\alpha}} E_{\alpha i}(K) \prod_{j < i} [G_{\alpha j} + E_{\alpha j}(K)] \prod_{j > i} G_{\alpha j}. \quad (44)$$

Every fixed finite expression of this kind satisfies $\mathcal{E}_K \rightarrow 0$ for all strictly positive residual smoothing parameters. For the jets of degree $d_i \leq 4$ and a fixed finite expression, $\mathcal{E}_K = O(K^{15/8}\theta^{(K+1)/2})$.

Proof. Telescope each product by replacing one factor at a time. Apply Holder with four equal exponents, inserting factors equal to one if the product has fewer than four factors. The exact factors have norm at most G_i , and the approximants have norm at most $G_i + E_i(K)$ by the triangle inequality. This proves the bound without independence. Each $E_i(K)$ tends to zero, and there are only finitely many factors and monomials. Their remaining norms are uniformly bounded for sufficiently large K , proving the rate. $\square$

The missing-response influence is a polynomial of activation degree at most four with reference coefficients held fixed. Conditional independence expresses its expectation as products of unary conditional moments whose *moment degrees* add to at most four. Distributing the prescribed source derivatives changes the jet orders q , but does not increase those moment labels k . Thus the specific response expression has the additional structure

$$\sum_{j=1}^{d_{\alpha}} k_{\alpha j} \leq 4, \quad g_{\alpha j} = \nu_{\cdot, k_{\alpha j}}^{(q_{\alpha j})}. \quad (45)$$

This statement applies after the physical influence and source derivative have been derived. The jet entries are then separate primitive observables to approximate. Differentiating a clipped zeroth-order approximation to create its higher jets would be a different operation, with extra clipping boundaries, and is not the construction used here.

The clipped response has the global deterministic bound

$$|\widehat{\mathcal{R}}_K(X)| \leq B_K := \sum_{\alpha} |c_{\alpha}| \prod_j M_{\alpha j, K}. \quad (46)$$

For (45), $R_K \geq 1$ gives

$$B_K \leq C_{\mathcal{R}} R_K^4, \quad C_{\mathcal{R}} = \sum_{\alpha} |c_{\alpha}| 2^{d_{\alpha}} \prod_j C_{\alpha j}, \quad (47)$$

with constant monomials included by the empty-product convention. Consequently $B_K = O(K^2)$ at fixed problem parameters. Bounds can also be propagated through the existing arithmetic circuit without explicitly expanding all neuron-index monomials.

9.5 Integrating the clipped expression with a certificate

Clipping is nonlinear. In general $\widehat{\mathcal{R}}_K$ is not a polynomial, so its integral is not given by the finite Wick calculation available for the unclipped replacement. There are nevertheless two explicit ways to certify the remaining integral.

Deterministic Gaussian cubature. Let $\mathfrak{r} \geq 1$ and choose a cube $[-A, A]^{\mathfrak{r}}$, $A > 0$. For each ridge polynomial $p_{i,K}$, compute a derivative bound

$$L_{i,K,A} \geq \max_{|t| \leq A\sqrt{\mathfrak{r}}} |p'_{i,K}(t)|.$$

For monomial coefficients $p(t) = \sum_j b_j t^j$, the explicit choice $\sum_{j \geq 1} j |b_j| (A\sqrt{\mathfrak{r}})^{j-1}$ suffices. Since clipping is one-Lipschitz, its ridge restriction has this Lipschitz constant on the cube. Product telescoping therefore gives the response Lipschitz bound

$$L_{K,A} = \sum_{\alpha} |c_{\alpha}| \sum_i L_{\alpha i, K, A} \prod_{j \neq i} M_{\alpha j, K}.$$

Partition the cube into boxes of side at most h , evaluate $\hat{\mathcal{R}}_K$ at each box center, and weight by the exact product Gaussian mass of that box. Call the resulting sum $Q_{K,A,h}$. Then

$$|\mathbb{E}\mathcal{R} - Q_{K,A,h}| \leq \mathcal{E}_K + \frac{1}{2} L_{K,A} \sqrt{\mathfrak{r}} h + 2\mathfrak{r} B_K \Phi(-A). \quad (48)$$

Indeed, the center-to-point distance in a box is at most $\sqrt{\mathfrak{r}}h/2$, and the Gaussian mass outside the cube is at most $2\mathfrak{r}\Phi(-A)$ by the union bound. These prove the last two terms; Theorem 9.4 gives the first. The Gaussian box masses are products of univariate CDF differences. Numerical error in their evaluation and in polynomial coefficients must be budgeted separately.

This is a deterministic existence-and-certification result for every fixed finite factor dimension. Its grid size can grow exponentially with $\mathfrak{r}$; it is not a dimension-free quadrature theorem.

A bounded-sampling certificate. Draw independent $X_1, \dots, X_N \sim N(0, I_{\mathfrak{r}})$ and put

$$\overline{\mathcal{R}}_{K,N} = N^{-1} \sum_{a=1}^N \hat{\mathcal{R}}_K(X_a).$$

For $0 < \delta < 1$,

$$\Pr \left\{ |\mathbb{E}\mathcal{R} - \overline{\mathcal{R}}_{K,N}| \leq \mathcal{E}_K + B_K \sqrt{\frac{2 \log(2/\delta)}{N}} \right\} \geq 1 - \delta. \quad (49)$$

For completeness, if $Y \in [-B, B]$, the log moment-generating function of $Y - \mathbb{E}Y$ has second derivative equal to the variance under an exponential tilt. Every such variance is at most B^2 , because $\text{Var}(Y) \leq \mathbb{E}(Y - 0)^2 \leq B^2$. Integrating the second derivative twice gives $\log \mathbb{E}e^{t(Y - \mathbb{E}Y)} \leq t^2 B^2 / 2$. Independence, the exponential Markov inequality, and optimization in t give the upper tail $\exp[-N\varepsilon^2/(2B^2)]$, and applying the same argument to $-Y$ gives the two-sided factor two. Add the deterministic clipping bias to prove (49).

For a fixed response satisfying (45), choose K so that $\mathcal{E}_K \leq \varepsilon/2$. As $\varepsilon \downarrow 0$ this requires $K = O(\log(1/\varepsilon))$, with constants depending on the fixed smoothing parameters and response coefficients. Taking

$$N \geq 8B_K^2 \varepsilon^{-2} \log(2/\delta)$$

then gives total error at most ε with probability at least $1 - \delta$. Since $B_K = O(K^2)$, this sufficient sample count is

$$O\left(\varepsilon^{-2} [\log(1/\varepsilon)]^4 \log(1/\delta)\right)$$

at fixed system parameters. This counts samples of a bounded, derived factor integrand; it does not claim that the constants are small, that building the response circuit is free, or that unprovided network data have been simulated.

What this completes and what it does not. The clipped construction establishes arbitrary-accuracy integration of the specified Gaussian-reference response for every positive residual variance, with an explicit deterministic bias bound and two integration certificates. It supplies the product-convergence statement that an L^2 primitive tail alone did not provide. It gives up exact polynomial integration and does not preserve compatibility between independently clipped jet orders. That compatibility is unnecessary when approximating the already-derived response expression.

Approximation of the covariance, movement of the source tensor and gain normalizer, errors in baseline constants, and propagation through the network remain separately quantified terms. The covariance-reference certificate and the readout-weighted memory theorem can be combined with the local error above only after their reference laws, source conventions, and readout metrics have been matched. No assertion of a two-gain closure or a measured deep-network rate follows from clipping by itself.

10 A sharp modular-defect bound for conditional memory

The finite modular bridge can be made quantitative. A cross-ratio of a positive joint table is exactly twice a matrix-unit defect of its modular generator. The same cross-ratio bounds maximal correlation, with the sharp dimension-free constant $\tanh(\Delta_{\text{cr}}/4)$. If the table belongs to an actual conditional Markov family, this bounds Fisher transport and the memory kernels obtained by eliminating omitted observables. This is a comparison between specified objects, with an explicit remaining hypothesis: the proposed coarse transition must really compose as the relevant channel.

Throughout this section, the modular parameter t is *not network depth*. Modular evolution is a state-dependent automorphism group of a matrix algebra. Depth evolution below is a separately specified Markov transition between probability spaces. The theorem compares a static modular-generator defect with the norm of that transition; it does not identify their two evolutions.

10.1 A finite joint state and its two conditional operators

Let $\mathbf{P}$ and $\mathbf{F}$ be nonempty finite alphabets and let $\pi = (\pi_{pf})$ be a joint probability table. Initially assume every cell is strictly positive. Probability observables and scores are real valued. Write

$$\mu_p = \sum_f \pi_{pf}, \quad \nu_f = \sum_p \pi_{pf}, \quad \Delta_{\text{cr}} = \max_{p,p',f,f'} \left| \log \frac{\pi_{pf}\pi_{p'f'}}{\pi_{pf'}\pi_{p'f}} \right|.$$

The conditional operator on future observables and its Hilbert adjoint are

$$\begin{aligned} (Kg)(p) &= \sum_f \frac{\pi_{pf}}{\mu_p} g(f), & K : L^2(\nu) &\longrightarrow L^2(\mu), \\ (K^*h)(f) &= \sum_p \frac{\pi_{pf}}{\nu_f} h(p), & K^* : L^2(\mu) &\longrightarrow L^2(\nu). \end{aligned}$$

Thus K acts backward on observables while K^* transports a perturbation score forward. Both preserve constants and map centered functions to centered functions. The adjoint identity follows by expanding $\sum_{p,f} \pi_{pf} h(p) g(f)$. Define the maximal correlation by

$$r(\pi) = \sup \left\{ |\mathbb{E}[h(P)g(F)]| : \mathbb{E}h = \mathbb{E}g = 0, \quad \mathbb{E}h^2 = \mathbb{E}g^2 = 1 \right\}.$$

If one alphabet has only one point, set $r(\pi) = 0$. Otherwise this is $\|K\|_{L_0^2(\nu) \rightarrow L_0^2(\mu)} = \|K^*\|_{L_0^2(\mu) \rightarrow L_0^2(\nu)}$.

Theorem MM.1 (sharp cross-ratio contraction). For every strictly positive finite joint table,

$$r(\pi) \leq q(\pi) := \tanh\left(\frac{\Delta_{\text{cr}}}{4}\right). \quad (50)$$

The constant is dimension free and is attained by positive binary tables.

Proof. First let a, b be two positive probability vectors, put $L_j = a_j/b_j$, and set $m = \min_j L_j$, $M = \max_j L_j$. Since $\sum_j b_j L_j = 1$, $m \leq 1 \leq M$. If $m = M$ the two vectors are equal. Otherwise the chord bound for the convex function $x \mapsto |x - 1|$ on $[m, M]$ gives

$$\text{TV}(a, b) = \frac{1}{2} \mathbb{E}_b |L - 1| \leq \frac{(M - 1)(1 - m)}{M - m}.$$

Writing $R = M/m > 1$, the right side equals

$$\frac{R + 1 - Rm - m^{-1}}{R - 1} \leq \frac{R + 1 - 2\sqrt{R}}{R - 1} = \frac{\sqrt{R} - 1}{\sqrt{R} + 1} = \tanh\left(\frac{\log R}{4}\right),$$

where the inequality is the arithmetic–geometric mean inequality.

For two rows $K_p, K_{p'}$, the oscillation of $\log(K_p/K_{p'})$ is at most Δ_{cr} : the row normalizers cancel in a cross-ratio. Therefore every row-pair total variation distance is at most $q = q(\pi)$. For a real function g , subtracting the midpoint of its range gives

$$|(Kg)(p) - (Kg)(p')| \leq \text{TV}(K_p, K_{p'}) \text{osc}(g), \quad \text{osc}(Kg) \leq q \text{osc}(g).$$

The reverse conditional kernel K^* has the same cross-ratio diameter, so the same oscillation bound holds for K^* .

The Markov operator KK^* on $L^2(\mu)$ is self-adjoint and positive semidefinite, and it contracts oscillations by q^2 . On the centered finite-dimensional subspace its largest eigenvalue is $r(\pi)^2$. If this subspace is nontrivial, choose a real eigenfunction h for that eigenvalue. It is nonconstant and hence has positive oscillation. Thus

$$r(\pi)^2 \text{osc}(h) = \text{osc}(KK^*h) \leq q^2 \text{osc}(h).$$

This proves the bound, including the zero-eigenvalue case.

For sharpness, take $0 \leq r < 1$ and

$$\pi = \frac{1}{4} \begin{pmatrix} 1+r & 1-r \\ 1-r & 1+r \end{pmatrix}.$$

Both marginals are uniform, the centered conditional eigenvalue is r , and $\Delta_{\text{cr}} = 2 \log((1+r)/(1-r)) = 4 \operatorname{arctanh} r$. $\square$

This proof supplies the L^2 statement directly; it does not silently replace the norm of a conditional operator by its contraction coefficient in Hilbert’s projective metric. The latter has the same sharp hyperbolic tangent constant, as recorded in the source comparison below.

10.2 Exact modular distances

On $\mathbb{C}^P \otimes \mathbb{C}^F$ let

$$\rho = \operatorname{diag}_{p,f} \pi_{pf}, \quad L = \log \rho, \quad \mathcal{B} = M_P(\mathbb{C}) \otimes I_F.$$

Write $\rho_P = \operatorname{diag}_p \mu_p$ and $\rho_F = \operatorname{diag}_f \nu_f$ for the two marginal density matrices. The state is $\phi_\rho(A) = \operatorname{Tr}(\rho A)$, and its modular automorphism group is $\sigma_t(A) = \rho^{it} A \rho^{-it}$. Distances in this subsection use the ordinary operator norm on the matrix algebra, not a weighted Hilbert–Schmidt norm.

Theorem MM.2 (modular matrix-unit defect). For $p, p' \in P$ set

$$r_f^{pp'} = \log \pi_{pf} - \log \pi_{p'f}, \quad w_{pp'} = \max_f r_f^{pp'} - \min_f r_f^{pp'}.$$

Then

$$\text{dist}([L, E_{pp'} \otimes I], \mathcal{B}) = \frac{w_{pp'}}{2}, \quad (51)$$

$$d_{\text{mod}} := \max_{p, p'} \text{dist}([L, E_{pp'} \otimes I], \mathcal{B}) = \frac{\Delta_{\text{cr}}}{2}. \quad (52)$$

Consequently the sharp comparison is

$$\boxed{r(\pi) \leq \tanh(d_{\text{mod}}/2)}. \quad (53)$$

For the finite modular flow,

$$\max_{p, p'} \text{dist}(\sigma_t(E_{pp'} \otimes I), \mathcal{B}) \leq \min\{1, |t|\Delta_{\text{cr}}/2\}. \quad (54)$$

If $|t|\Delta_{\text{cr}} \leq \pi$, its left side equals $\sin(|t|\Delta_{\text{cr}}/2)$.

Proof. Direct multiplication gives

$$[L, E_{pp'} \otimes I] = E_{pp'} \otimes \text{diag}_f r_f^{pp'}.$$

For any $A \otimes I \in \mathcal{B}$, compress the difference to its (p, p') block. Its norm is at least $\max_f |r_f^{pp'} - A_{pp'}|$. A complex scalar cannot approximate a finite real interval better than its real midpoint, whose maximal error is $w_{pp'}/2$. Conversely, choosing $A = \frac{1}{2}(\max r^{pp'} + \min r^{pp'})E_{pp'}$ attains that error for the whole matrix. This proves (51). The equality $\max_{p, p'} w_{pp'} = \Delta_{\text{cr}}$ proves (52); MM.1 then proves (53).

The same block argument says that the flow distance is the smallest enclosing-disk radius of $\{e^{itr_f^{pp'}}\}_f$. It is at most one, by using the zero scalar. Approximating instead by the exponential of the midpoint of the real interval gives the upper bound $|t|w_{pp'}/2$. If $\alpha = |t|w_{pp'} \leq \pi$, rotate the arc to angles $[-\alpha/2, \alpha/2]$. The disk with center $\cos(\alpha/2)$ and radius $\sin(\alpha/2)$ contains this entire minor arc, while its two endpoints force a disk radius at least half their chord length, namely $\sin(\alpha/2)$. Taking the maximum over p, p' proves both assertions. $\square$

The matrix-unit maximum is specified deliberately. If instead one uses the full derivation defect

$$d_{\text{full}} = \sup_{b \in \mathcal{B}, \|b\| \leq 1} \text{dist}([L, b], \mathcal{B}),$$

then the safe dimension-free comparison is

$$d_{\text{mod}} \leq d_{\text{full}} \leq 2d_{\text{mod}} = \Delta_{\text{cr}}.$$

For the upper bound fix a reference row p_0 , and subtract from each row $L_{pf} - L_{p_0f}$ its midpoint. This writes $L = A \otimes I + I \otimes C + R$ with diagonal A, C and $\|R\| \leq \Delta_{\text{cr}}/2$. Modulo $\mathcal{B}$, $[L, b]$ is $[R, b]$, whose norm is at most $\Delta_{\text{cr}}\|b\|$. No equality between d_{full} and the matrix-unit maximum is needed.

Corollary MM.3 (the exact zero-defect endpoint). The following are equivalent: $\Delta_{\text{cr}} = 0$; $d_{\text{mod}} = 0$; $\pi = \mu \otimes \nu$; and $\sigma_t(\mathcal{B}) = \mathcal{B}$ for every $t \in \mathbb{R}$.

Proof. If the cross-ratios are one, every pair of rows of π is proportional, so their probability normalizations are the same vector ν . This is the product law. Conversely a product law has vanishing cross-ratios and $L = \log \rho_P \otimes I + I \otimes \log \rho_F$, which gives modular invariance. Invariance of the whole differentiable group implies $[L, \mathcal{B}] \subseteq \mathcal{B}$, and MM.2 then gives zero defect. $\square$

Invariance at one isolated nonzero modular time is insufficient because phases can coincide modulo 2π ; the group or a time interval is needed. Also, a classical table has been embedded in a full matrix algebra using the given past/future tensor factors. No noncommuting physical neuron observables are inferred from that representation.

10.3 Conditional expectations and a relative modular comparison

Theorem MM.4 (state-preserving expectation and its defect). Every unital positive $\mathcal{B}$ -bimodular projection $\mathcal{E} : M_P \otimes M_F \rightarrow \mathcal{B}$ has the form

$$\mathcal{E}(A \otimes C) = A\eta(C) \otimes I$$

for a state η on M_F . A projection of this kind preserving ϕ_ρ exists if and only if π is a product law. When it exists it is unique, with $\eta(C) = \text{Tr}(\rho_F C)$. The always available completely positive product expectation

$$\mathcal{E}_{\rho_F}(A \otimes C) = A \text{Tr}(\rho_F C) \otimes I$$

satisfies the exact identity and quantitative bound

$$\|\phi_\rho - \phi_\rho \circ \mathcal{E}_{\rho_F}\| = \|\rho - \rho_P \otimes \rho_F\|_1 = 2 \text{TV}(\pi, \mu \otimes \nu) \leq 2 \tanh(\Delta_{\text{cr}}/4). \quad (55)$$

Proof. Since $I \otimes C$ commutes with $\mathcal{B}$, bimodularity forces its image to commute with $\mathcal{B}$ while lying in $\mathcal{B}$. It is therefore a scalar multiple $\eta(C)I$ of the identity. Positivity and unitality make η a state, and bimodularity gives the displayed form. State preservation on $A \otimes C$ says $\phi_\rho(A \otimes C) = \phi_{\rho_P}(A)\eta(C)$. Setting $A = I$ fixes $\eta = \phi_{\rho_F}$, and equality for all simple tensors then forces $\rho = \rho_P \otimes \rho_F$. The converse is immediate.

For the product expectation the state $\phi_\rho \circ \mathcal{E}_{\rho_F}$ has density $\rho_P \otimes \rho_F$. Duality of the matrix operator norm and trace norm gives the first equality, and diagonality gives the total variation equality. Finally $\nu = \sum_{p'} \mu_{p'} K_{p'}$, so convexity of total variation and the row bound in MM.1 imply

$$\text{TV}(\pi, \mu \otimes \nu) = \sum_p \mu_p \text{TV}(K_p, \nu) \leq \sum_{p,p'} \mu_p \mu_{p'} \text{TV}(K_p, K_{p'}) \leq q(\pi).$$

$\square$

This obstruction concerns expectation onto the *full matrix factor* $\mathcal{B}$. Ordinary conditional expectation onto the diagonal past sigma-algebra exists for every joint law, including correlated ones. The two assertions must not be conflated.

Theorem MM.5 (exact weighted and relative modular diagnostics). Set

$$R_{pf} = \frac{\pi_{pf}}{\sqrt{\mu_p \nu_f}}, \quad u_p = \sqrt{\mu_p}, \quad v_f = \sqrt{\nu_f}.$$

Then

$$\begin{aligned} r(\pi) &= \|R - uv^\top\|_{\text{op}} = \sigma_2(R), \\ r(\pi)^2 &\leq \|R - uv^\top\|_F^2 = \chi^2(\pi \| \mu \otimes \nu). \end{aligned} \tag{56}$$

Set $\sigma_2(R) = 0$ if one alphabet has only one point. These identities and the inequality remain valid with structural zeros in π , provided zero-marginal rows and columns have first been removed.

For a strictly positive table let $\sigma = \rho_P \otimes \rho_F$, and use the relative modular convention on Hilbert–Schmidt matrices

$$\Delta_{\rho|\sigma}(X) = \rho X \sigma^{-1}, \quad \Omega_\sigma = \sigma^{1/2}.$$

The diagonal vector of $(\Delta_{\rho|\sigma} - I)\Omega_\sigma$, reshaped as a P-by-F matrix using the specified tensor labels, is exactly $R - uv^\top$. Its Hilbert–Schmidt norm squared is the chi-square divergence, and its reshaped rectangular operator norm is the maximal correlation.

Proof. Multiplication of a function by the square root of its marginal is a unitary map from weighted L^2 to the corresponding Euclidean space. Under these two unitaries, K becomes R . The identities $Rv = u$ and $R^\top u = v$ isolate a constant singular block of value one. The matrix $R - uv^\top$ is zero on that block and is precisely the centered conditional operator on its orthogonal complement. Jensen’s inequality gives $\|K\| \leq 1$, so this proves the two norm identities, including possible further singular values equal to one. The Frobenius norm bounds the operator norm, and direct expansion gives

$$\|R - uv^\top\|_F^2 = \sum_{p,f} \frac{(\pi_{pf} - \mu_p \nu_f)^2}{\mu_p \nu_f}.$$

None of this proof requires nonzero joint cells.

For completeness, the relative modular operator follows directly from the finite relative Tomita map, with its convention made explicit:

$$S_{\rho|\sigma}(A\sigma^{1/2}) = A^* \rho^{1/2}, \quad S_{\rho|\sigma}(Z) = \sigma^{-1/2} Z^* \rho^{1/2}.$$

Its squared norm is

$$\|S_{\rho|\sigma} Z\|_{\text{HS}}^2 = \text{Tr}(Z^* \rho Z \sigma^{-1}) = \langle Z, \Delta_{\rho|\sigma} Z \rangle_{\text{HS}},$$

so $S_{\rho|\sigma}^* S_{\rho|\sigma} = \Delta_{\rho|\sigma}$ with this convention. Finally,

$$(\Delta_{\rho|\sigma} - I)\Omega_\sigma = (\rho - \sigma)\sigma^{-1/2},$$

whose (p, f) diagonal entry is $(\pi_{pf} - \mu_p \nu_f) / \sqrt{\mu_p \nu_f}$. This is the claimed reshape. The displayed algebraic formulas extend to nonfaithful ρ while σ remains invertible; no logarithm of a zero cell is used. $\square$

The norm of the relative modular operator itself is *not* being identified with maximal correlation. The comparison uses its displacement of the specified product-state vector and the specified past/future reshape. This weighted diagnostic can be much less pessimistic than a maximum cross-ratio determined by rare cells.

10.4 Fisher contraction after conditioning a carried variable

Theorem MM.6 (conditional Fisher loss). Let S be a carried variable with fixed law, and suppose that conditional on $S = s$ the actual pair (P, F) has finite table π_s . Remove zero-marginal states in each fiber. For a square-integrable score $h = h(S, P)$ with $\mathbb{E}[h | S] = 0$, put

$$(Th)(S, F) = \mathbb{E}[h(S, P) | S, F].$$

Writing $r_s = r(\pi_s)$, one has

$$\|Th\|_2^2 \leq \mathbb{E}_S \left[r_s^2 \|h(s, \cdot)\|_{L^2(\mu_s)}^2 \right]. \quad (57)$$

In particular, if the tables are positive almost everywhere and $\text{ess sup}_s \Delta_{\text{cr},s} \leq D < \infty$, then

$$\|Th\|_2 \leq q \|h\|_2, \quad q = \tanh(D/4) < 1.$$

For a general score $h = a(S) + h_\perp$ with $\mathbb{E}[h_\perp | S] = 0$, the $a(S)$ component is preserved exactly, and

$$\|Th\|_2^2 \leq \|a\|_2^2 + q^2 \|h_\perp\|_2^2.$$

Proof. On each fiber the score map is K_s^* , whose centered norm is r_s by definition and MM.5. Square its norm inequality and integrate in s . MM.1 gives the uniform bound. The two components are orthogonal before and after transport because $\mathbb{E}[Th_\perp | S] = 0$. $\square$

If $\mu_\epsilon(p | s) = \mu(p | s)(1 + \epsilon h(s, p)) + o(\epsilon)$ and the conditional kernel is held fixed, differentiating the output probabilities shows that Th is exactly the output score. Thus squared L^2 norms here are Fisher information for that tangent perturbation. If the kernel itself changes, its derivative contributes an additional source; the theorem does not discard it. For an unbounded score, the identity is read for a differentiable probability path realizing that score, or by tangent-space completion, not by assuming $1 + \epsilon h$ stays positive for a uniform nonzero interval.

10.5 From the static defect to a controlled memory kernel

Theorem MM.7 (a matched Markov family and its memory tail). Let $X_0, X_1, \dots$ be an actual, possibly nonstationary finite-state Markov chain with transition kernels K_ℓ , marginals μ_ℓ , and joint tables

$$\pi_\ell(x, y) = \mu_\ell(x) K_\ell(x, y), \quad \mu_{\ell+1} = \mu_\ell K_\ell.$$

Use K_ℓ also for the conditional operator from observables at $\ell + 1$ to observables at ℓ , and let $T_\ell = K_\ell^* : H_\ell \rightarrow H_{\ell+1}$ on $H_\ell = L_0^2(\mu_\ell)$. Suppose $\|T_\ell\| \leq q_\ell \leq q < 1$. This follows, in particular, from positive joint tables with $\Delta_{\text{cr},\ell} \leq D < \infty$ and $q = \tanh(D/4)$; it can instead be checked directly by MM.5.

Let P_ℓ be arbitrary orthogonal retained projections on H_ℓ , write $Q_\ell = I - P_\ell$, and define the four blocks

$$A_\ell = P_{\ell+1} T_\ell P_\ell, \quad B_\ell = P_{\ell+1} T_\ell Q_\ell, \quad C_\ell = Q_{\ell+1} T_\ell P_\ell, \quad D_\ell = Q_{\ell+1} T_\ell Q_\ell.$$

For $z_{\ell+1} = T_\ell z_\ell$, $x_\ell = P_\ell z_\ell$, and $y_\ell = Q_\ell z_\ell$, elimination of the omitted variables gives

$$\begin{aligned} x_{\ell+1} &= A_\ell x_\ell + B_\ell D_{\ell-1} \cdots D_0 y_0 \\ &\quad + \sum_{k=0}^{\ell-1} B_\ell D_{\ell-1} \cdots D_{k+1} C_k x_k. \end{aligned} \quad (58)$$

Empty products are identities on the relevant space. Every memory kernel satisfies

$$\|B_\ell D_{\ell-1} \cdots D_{k+1} C_k\| \leq \prod_{j=k}^{\ell} q_j \leq q^{\ell-k+1}. \quad (59)$$

For an integer $m \geq 0$, if a retained history has $\|x_k\| \leq M$, deleting terms with lag $\ell - k > m$ from this exact right-hand side incurs a norm error at most

$$\frac{M q^{m+2}}{1 - q}. \quad (60)$$

The initial omitted term has norm at most $q^{\ell+1} \|y_0\|$.

Proof. Each of the four blocks has norm at most q_ℓ , because orthogonal projections have norm at most one. Iteration of $y_{\ell+1} = C_\ell x_\ell + D_\ell y_\ell$ and substitution into $x_{\ell+1} = A_\ell x_\ell + B_\ell y_\ell$ proves (58). Multiplying the block norm bounds proves (59). The omitted lags $r = \ell - k > m$ are bounded by the geometric series $M \sum_{r=m+1}^{\infty} q^{r+1}$, giving (60). Multiplying the bounds for $B_\ell, D_{\ell-1}, \dots, D_0$ gives the initial-state bound. $\square$

The assertion is a bound on exact memory operators and on their action on a specified history. A uniformly accurate nonlinear closure rollout needs its own stability and forcing estimates. For comparison, in the linear forced recursion $z_{\ell+1} = T_\ell z_\ell + b_\ell$ with $\|b_\ell\| \leq B$, keeping only the most recent m source insertions gives directly, for $m \leq \ell + 1$, the full-state bound

$$q^{\ell+1} \|z_0\| + \frac{Bq^m}{1-q}.$$

This follows by expanding the same variation-of-constants series. The conditional version of MM.7 holds after applying MM.6 fiberwise to a Markov family conditional on a fixed carried S , with a uniform bound over the fibers. Functions of S are retained separately.

10.6 Structural zeros and two falsifying examples

After removing zero-marginal rows and columns, a missing joint cell prevents a finite cross-ratio diameter: if $\pi_{pf} = 0$, choose a positive cell $\pi_{pf'}$ in its row and a positive cell $\pi_{p'f}$ in its column. The cross-product $\pi_{pf}\pi_{p'f'}$ is zero while $\pi_{pf'}\pi_{p'f} > 0$. The extended diameter is therefore infinite. MM.5 remains available. For instance,

$$\pi = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 \end{pmatrix}$$

has singular values 1 and $1/2$. Thus $r(\pi) = 1/2$ even though the cross-ratio criterion is unavailable. Structural zeros are not by themselves a proof of no contraction.

Counterexample MM.8 (a completion can manufacture a gap). For any joint table with positive marginal supports, its independent smoothing

$$\pi^\epsilon = (1 - \epsilon)\pi + \epsilon\mu \otimes \nu, \quad 0 < \epsilon \leq 1,$$

has the same marginals and satisfies the exact identity

$$r(\pi^\epsilon) = (1 - \epsilon)r(\pi).$$

Indeed, $R^\epsilon - uv^\top = (1 - \epsilon)(R - uv^\top)$. For the deterministic balanced binary table $\pi = \text{diag}(1/2, 1/2)$, the original maximal correlation is one, while the smoothed law is positive, has maximal correlation $1 - \epsilon$, and attains the sharp bound in MM.1. A favorable defect measured on a smoothed or completed table certifies that modified law; it does not establish a gap for the original transition.

Counterexample MM.9 (adjacent product tables do not imply memory decay). Let X_0, X_1 be independent fair bits and set $X_2 = X_0$. The pairs (X_0, X_1) and (X_1, X_2) both have product tables, so both adjacent modular defects and adjacent maximal correlations are zero. Nevertheless X_2 remembers X_0 perfectly: their maximal correlation is one. The pair kernels do not compose into the actual two-step conditional channel, because the process is not Markov on this proposed state. Thus even vanishing adjacent defects do not replace the Markov hypothesis.

An exact deterministic update of a full state likewise has a centered score channel of norm one whenever it has a nonconstant output observable; the memory route proves this coisometry

obstruction. Overlapping window states can have structural zeros from coordinates carried identically across an update. A genuinely positive multi-step transition can be used by rebuilding the Markov model at those block boundaries. A gap of an uncompressed block transition alone does not bound products with arbitrary intermediate retained/omitted projections inserted; MM.7 must be applied to the operators that actually occur in the chosen blocked recursion.

10.7 Source comparison and status of the network arrow

Primary-source checks. The finite modular formula was checked in Alain Connes, *Noncommutative Geometry* (1994), Chapter V, Section 4, printed page 475 (PDF page index 474), in the author-hosted copy <https://alainconnes.org/wp-content/uploads/book94bigpdf.pdf>. It gives $\phi = \text{Tr}(\rho \cdot)$, $\sigma_t(A) = e^{it \log \rho} A e^{-it \log \rho}$, and the corresponding matrix-unit eigenvalue ratios. The source theorem concerns a faithful matrix state. Here it is specialized to the explicitly specified diagonal joint-state representation; MM.2 supplies the distance calculation. MM.4 is proved directly in finite dimensions and does not invoke a general conditional-expectation theorem with unchecked weight or normality hypotheses.

The projective contraction comparison was checked in Bas Lemmens and Roger Nussbaum, *Birkhoff's version of Hilbert's metric and its applications in analysis*, arXiv:1304.7921v1, Theorem 2.9 and the positive matrix formula immediately following it, printed pages 9–10: <https://arxiv.org/pdf/1304.7921>. Their theorem identifies Birkhoff's projective contraction ratio as $\tanh(\Delta(L)/4)$, and their formula expresses the projective diameter of a positive matrix by its maximal cross-ratio. Row and column rescaling preserve that cross-ratio. Our target is instead the centered weighted L^2 norm. MM.1 proves that target by two-sided oscillation contraction and reversibilization, so it does not assume that distinct operator metrics are identical. This was a bounded check of the exact formulas and hypotheses, not a search for a global novelty claim. The relative Tomita convention and all finite relative-modular formulas used in MM.5 are explicitly derived above.

What this adds to the programme. Given a specified conditional table, there are now three checkable quantities: a sharp upper bound from the modular-generator defect, the exact weighted singular value, and a chi-square upper bound from a relative modular displacement. If the table is the transition law of a matched conditional Markov family, these quantities control propagation of a centered source and the norm of each omitted-observable memory kernel. The source may in particular be the normalized third-cumulant score, and the target may be a retained or omitted fourth-cumulant influence. Conditioning on a carried scale preserves that scale mode and allows the theorem to act on its orthogonal complement, when the actual conditional channel has the required gap.

The first unproved application arrow is the identification of the chosen coarse conditional table and its composed score channel with the process used by the network closure. No actual transition arrays or Markov sufficiency proof were supplied here. Even after that arrow is proved, a memory norm bound does not fix the signed amplitude of the missing third-to-fourth feed. The response pairing and Gram/residual calculations in the companion routes supply that separate observable information. The present result supplies a quantitative criterion for how far such unresolved information can persist, with exact constants and explicit counterexamples to the two tempting shortcuts.

11 A dynamically consistent reduction with a finite-horizon certificate

The earlier local width and Hankel-rank criteria identify what can be compressed at a cut. They do not, by themselves, give compatible updates between independently chosen small spaces. The construction below supplies those updates and an a priori error bound. All spaces are finite dimensional in this section; complex adjoints can replace real transposes without change.

Consider the nonautonomous input–output system

$$y_0 = E_0 a, \quad y_{\ell+1} = D_\ell y_\ell + G_\ell u_\ell, \quad z_\ell = B_\ell y_\ell, \quad 0 \leq \ell \leq L. \quad (61)$$

The update equation is used only for $\ell < L$. The input norm is the Euclidean direct-sum norm of $(a, u_0, \dots, u_{L-1})$, and the output norm is that of $(z_0, \dots, z_L)$. The source and readout scalings are part of the problem: rescaling a gain coordinate changes the certificate unless its metric is transformed with it. A single final readout is obtained by putting $B_\ell = 0$ for $\ell < L$.

Let

$$\Phi(\ell, k) = D_{\ell-1} \cdots D_k, \quad \Phi(k, k) = I,$$

and define the exact reachability Gramians and future observability maps by

$$W_0 = E_0 E_0^*, \quad W_{\ell+1} = D_\ell W_\ell D_\ell^* + G_\ell G_\ell^*, \quad (\text{MG.1})$$

$$O_\ell y = (B_j \Phi(j, \ell) y)_{j=\ell}^L, \quad Q_\ell = O_\ell^* O_\ell. \quad (\text{MG.2})$$

Thus $Q_L = B_L^* B_L$ and $Q_\ell = B_\ell^* B_\ell + D_\ell^* Q_{\ell+1} D_\ell$. If R_ℓ maps the past input blocks to y_ℓ , then $W_\ell = R_\ell R_\ell^*$ by direct expansion of the state equation. The nonzero singular values of the cross-cut input–output map $H_\ell = O_\ell R_\ell$ equal those of $O_\ell W_\ell^{1/2}$, because both products with their adjoints equal $O_\ell W_\ell O_\ell^*$.

Fix an integer $r_\ell \leq \text{rank } W_\ell$ at every cut. Let V_ℓ have orthonormal columns spanning the first r_ℓ right singular vectors of $O_\ell W_\ell^{1/2}$, chosen inside range W_ℓ and completed there if some retained singular values are zero. Define

$$P_\ell = W_\ell^{1/2} V_\ell V_\ell^* W_\ell^{\dagger/2}, \quad L_\ell = W_\ell^{1/2} V_\ell, \quad M_\ell = V_\ell^* W_\ell^{\dagger/2}. \quad (62)$$

Here $W^{\dagger/2}$ is the positive square root of the Moore–Penrose inverse. The symbol L_ℓ in this subsection denotes a synthesis matrix, not network depth. We have $M_\ell L_\ell = I$ and $P_\ell^2 = P_\ell$. These are generally oblique projections in the ambient Euclidean metric; they are orthogonal in the reachability metric $W_\ell^\dagger$ on range W_ℓ .

Theorem 11.1 (Compatible Gramian reduction and its error certificate). *The reduced model*

$$q_0 = M_0 E_0 a, \quad (63)$$

$$q_{\ell+1} = M_{\ell+1} D_\ell L_\ell q_\ell + M_{\ell+1} G_\ell u_\ell, \quad (64)$$

$$\hat{z}_\ell = B_\ell L_\ell q_\ell \quad (65)$$

is a causal realization with r_ℓ states at cut ℓ . If $\mathcal{T}$ and $\widehat{\mathcal{T}}$ are the exact and reduced operators from the full declared input sequence to the output sequence, then

$$\|\mathcal{T} - \widehat{\mathcal{T}}\|_{2 \rightarrow 2} \leq \sum_{\ell=0}^L \sigma_{r_\ell+1}(O_\ell W_\ell^{1/2}). \quad (66)$$

Singular values are padded by zero. A corresponding Hilbert–Schmidt bound is

$$\|\mathcal{T} - \widehat{\mathcal{T}}\|_{\text{HS}} \leq \sum_{\ell=0}^L \left(\sum_{j>r_\ell} \sigma_j(O_\ell W_\ell^{1/2})^2 \right)^{1/2}. \quad (67)$$

In particular, discarding only zero singular directions at every cut gives an exact dynamically consistent input–output realization.

Proof. Embed the reduced states as $\hat{y}_\ell = L_\ell q_\ell$. Then

$$\hat{y}_0 = P_0 E_0 a, \quad \hat{y}_{\ell+1} = P_{\ell+1} (D_\ell \hat{y}_\ell + G_\ell u_\ell).$$

The crucial identity is

$$P_\ell W_\ell P_\ell^* = W_\ell^{1/2} V_\ell V_\ell^* W_\ell^{1/2} \preceq W_\ell. \quad (68)$$

Let $\hat{R}_\ell$ map past inputs to $\hat{y}_\ell$, and let $\tilde{R}_\ell$ map past inputs to the reduced state immediately before the projection at cut ℓ . Since each new input occupies an orthogonal block of the input direct sum,

$$\tilde{R}_0 \tilde{R}_0^* = W_0, \quad \tilde{R}_{\ell+1} \tilde{R}_{\ell+1}^* = D_\ell \hat{R}_\ell \hat{R}_\ell^* D_\ell^* + G_\ell G_\ell^*.$$

Induction using (68) proves

$$\tilde{R}_\ell \tilde{R}_\ell^* \preceq W_\ell, \quad \hat{R}_\ell \hat{R}_\ell^* \preceq W_\ell. \quad (\text{MG.3})$$

The state error $e_\ell = y_\ell - \hat{y}_\ell$ obeys

$$e_0 = (I - P_0) E_0 a, \quad e_{\ell+1} = D_\ell e_\ell + \rho_{\ell+1}, \quad \rho_\ell = (I - P_\ell) \tilde{R}_\ell u_{<\ell},$$

where $u_{<\ell}$ includes the initial source block. Each residual therefore contributes $O_\ell \rho_\ell$ to the subsequent output sequence, with zeros padded before cut ℓ . Consequently the full error operator is the sum of these residual-injection operators.

For the summand at cut ℓ , positive-semidefinite domination gives

$$\begin{aligned} & [O_\ell (I - P_\ell) \tilde{R}_\ell] [O_\ell (I - P_\ell) \tilde{R}_\ell]^* \\ & \preceq O_\ell (I - P_\ell) W_\ell (I - P_\ell)^* O_\ell^*. \end{aligned}$$

But $(I - P_\ell) W_\ell^{1/2} = W_\ell^{1/2} (I - V_\ell V_\ell^*)$. Its product with O_ℓ is exactly the singular-value tail chosen at this cut. Taking operator norms or traces, then summing the embedded residual operators by the triangle inequality, proves both bounds. All ranges remain inside the corresponding reachable supports by (MG.1), so the pseudoinverses introduce no undefined coordinates. $\square$

Why the metric is essential. For arbitrary Euclidean projections chosen independently at each cut, $P_\ell W_\ell P_\ell^* \preceq W_\ell$ need not hold. The reduced past can then create a state that is large in the next cut’s exact reachability metric, invalidating a naive sum of the original Hankel tails. The oblique metric projection above prevents this failure. Moreover,

$$W_{\ell+1}^{\dagger/2} [D_\ell W_\ell^{1/2} G_\ell] [D_\ell W_\ell^{1/2} G_\ell]^* W_{\ell+1}^{\dagger/2} = \text{proj}_{\text{range } W_{\ell+1}}.$$

Thus the augmented update is contractive in reachability-normalized coordinates. This is a quantitative compatibility condition for the moving retained frames, separate from their topological dimension or bundle class.

The sum in (66) can grow with depth; the theorem claims a finite-horizon certificate, not a uniform asymptotic rate. It is also an existence and reduction theorem after W_ℓ and Q_ℓ are available. Computing these matrices, constructing the chosen observables, and controlling conditioning of their square roots are separate algorithmic obligations. No low state count alone implies a low construction cost.

Literature scope. Time-varying balanced realizations and error bounds are developed by H. Sandberg and A. Rantzer, *Balanced Truncation of Linear Time-Varying Systems*, IEEE Transactions on Automatic Control 49(2), 217–229 (2004), <https://lucris.lub.lu.se/ws/files/4812893/625601.pdf>. Their analysis uses time-varying reachability and observability Lyapunov inequalities. The finite-horizon estimate above is proved directly for the explicitly stated projections; it is not an invocation of an unspecified time-invariant “twice the tail” formula and makes no novelty claim.

12 Extracting a symmetry channel before measuring memory

Let a full tangent realization obey

$$s_{\ell+1} = T_\ell s_\ell + G_\ell u_\ell.$$

An exact scale or symmetry channel is specified by a nonzero moving vector n_ℓ with $T_\ell n_\ell = n_{\ell+1}$. Its conserved coefficient requires a dual covector, not just this right-vector identity.

Theorem 12.1 (Exact moving symmetry splitting). *Suppose covectors p_ℓ satisfy*

$$p_\ell n_\ell = 1, \quad p_{\ell+1} T_\ell = p_\ell.$$

Set $S_\ell = n_\ell p_\ell$ and $Q_\ell = I - S_\ell$. Then

$$T_\ell S_\ell = S_{\ell+1} T_\ell, \quad T_\ell Q_\ell = Q_{\ell+1} T_\ell.$$

Writing $s_\ell = n_\ell a_\ell + y_\ell$, where $a_\ell = p_\ell s_\ell$ and $y_\ell \in \ker p_\ell$, gives the exact system

$$a_{\ell+1} = a_\ell + p_{\ell+1} G_\ell u_\ell, \tag{69}$$

$$y_{\ell+1} = D_\ell y_\ell + \bar{G}_\ell u_\ell, \quad D_\ell = T_\ell|_{\ker p_\ell}, \quad \bar{G}_\ell = Q_{\ell+1} G_\ell, \tag{70}$$

$$B_\ell s_\ell = (B_\ell n_\ell) a_\ell + B_\ell y_\ell. \tag{71}$$

Thus the one-dimensional symmetry coordinate can be carried exactly while only the transverse subsystem is reduced by Theorem 11.1 or the memory theorem below. No reduction residual feeds the symmetry coordinate.

Proof. The first intertwining identity is $T_\ell n_\ell p_\ell = n_{\ell+1} p_{\ell+1} T_\ell$; subtraction from T_ℓ gives the second. Applying $p_{\ell+1}$ and $Q_{\ell+1}$ to the state update gives the two coordinate equations. $\square$

Over a fixed finite horizon, a dual family always exists for a prescribed nonvanishing transported n_ℓ : choose $p_L n_L = 1$ and put $p_\ell = p_L T_{L-1} \cdots T_\ell$. Then $p_\ell n_\ell = p_L n_L = 1$. This existence assertion gives no uniform bound on $\|p_\ell\|$ and no inexpensive way of constructing it. A robust depth-uniform reduction needs well-conditioned frames, for example a uniform bound on $\|n_\ell\| \|p_\ell\|$, as well as a transverse estimate.

Application dictionary for a ReLU network. For a bias-free positively homogeneous network, the exact radial law family intertwines with the network maps. If its derivative is represented in an exact score or tangent realization, it supplies a transported n_ℓ . The previously proved Ward identity supplies its moment and gain components, including reference drift and normalization. It does not, by itself, produce a uniformly bounded dual p_ℓ or a transverse contraction rate. On a fixed gate cell, the ordinary input Jacobian also transports the radial activation vector by Euler homogeneity, but that single-trajectory Jacobian is not automatically the operator transporting probability-law cumulants. The realization and reference law must be matched before applying the theorem.

13 Uniform finite memory from an observable transverse gap

After the exact symmetry split, write the transverse subsystem as

$$y_0 = E_0 u_{-1}, \quad y_{\ell+1} = D_\ell y_\ell + G_\ell u_\ell, \quad z_\ell = \beta_\ell a_\ell + B_\ell y_\ell.$$

The initial transverse source is included as the input block u_{-1} . Define the impulse-response blocks

$$K_{\ell,-1} = B_\ell \Phi(\ell, 0) E_0, \tag{72}$$

$$K_{\ell k} = B_\ell \Phi(\ell, k+1) G_k, \quad 0 \leq k < \ell. \tag{73}$$

The key hypothesis is the *observable* bound

$$\boxed{\|K_{\ell k}\| \leq M\rho^{\ell-k-1}, \quad -1 \leq k < \ell, \quad 0 < \rho < 1.} \quad (74)$$

Constants independent of L are required when a claim is uniform in depth. This hypothesis controls precisely the declared source-to-readout paths. It does not require $\|D_\ell\| < 1$, nor contraction of every omitted state.

Theorem 13.1 (Causal finite-memory approximation with a uniform bound). *Assume (74), and carry a_ℓ exactly. For an integer $m \geq 0$, define*

$$\hat{z}_\ell = \beta_\ell a_\ell + \sum_{\substack{-1 \leq k < \ell \\ k \geq \ell - m}} K_{\ell k} u_k. \quad (75)$$

This has a causal realization storing the last m input blocks and the one exact symmetry coordinate. If each input block has dimension at most d , its state dimension is at most $1 + md$. The error bounds are

$$\sup_{0 \leq \ell \leq L} \|z_\ell - \hat{z}_\ell\| \leq \frac{M\rho^m}{1 - \rho} \sup_{-1 \leq k < L} \|u_k\|, \quad (76)$$

$$\|(z_\ell - \hat{z}_\ell)_{\ell=0}^L\|_{\ell^2} \leq \frac{M\rho^m}{1 - \rho} \|(u_k)_{k=-1}^{L-1}\|_{\ell^2}, \quad (77)$$

$$\|z_L - \hat{z}_L\| \leq \frac{M\rho^m}{\sqrt{1 - \rho^2}} \|(u_k)_{k=-1}^{L-1}\|_{\ell^2}. \quad (78)$$

For a prescribed single final readout, a sharper exact coefficient is the operator norm of the horizontal block matrix $[K_{L,-1} \cdots K_{L,L-m-1}]$, omitting an empty range.

Proof. Unrolling the exact recurrence gives $z_\ell = \beta_\ell a_\ell + \sum_{k=-1}^{\ell-1} K_{\ell k} u_k$. The approximate state is the queue $(u_{\ell-1}, \dots, u_{\ell-m})$, padded by zeros before u_{-1} ; append u_ℓ and discard the oldest block at each step. This proves causality and the stated dimension. All omitted terms have lag $j = \ell - k - 1 \geq m$, so

$$\|z_\ell - \hat{z}_\ell\| \leq \sum_{j \geq m} M\rho^j \|u_{\ell-j-1}\|.$$

The geometric sum proves (76). Extend the finite input sequence by zero; the triangle inequality in ℓ^2 and invariance of its norm under shifts prove (77). Cauchy–Schwarz in the omitted input index at the final layer gives (78). The exact final error is the displayed horizontal block operator applied to the old input blocks, which proves the sharper statement. $\square$

Thus a bounded input of size U needs

$$m \geq \max \left\{ 0, \left\lceil \frac{\log(MU/((1 - \rho)\varepsilon))}{|\log \rho|} \right\rceil \right\}$$

for output error at most ε in the supremum norm. This is a dimension bound independent of width once the source dimension d and observable constants are controlled. Forming all coefficients $K_{\ell k}$ may still require the unreduced operator; economical coefficient generation is a separate theorem, not hidden inside the memory count.

13.1 A sufficient gap on the reachable–observable quotient

Let $\mathcal{R}_\ell$ be the reachable transverse subspace and

$$\mathcal{N}_\ell = \mathcal{R}_\ell \cap \bigcap_{j=\ell}^L \ker(B_j \Phi(j, \ell)).$$

As proved in the exact realization analysis, $D_\ell \mathcal{N}_\ell \subseteq \mathcal{N}_{\ell+1}$ and the dynamics descends to $\mathcal{Z}_\ell = \mathcal{R}_\ell / \mathcal{N}_\ell$. Suppose norms on these quotients obey

$$\|\bar{\Phi}(\ell, k)\| \leq C\rho^{\ell-k}, \quad \|\bar{G}_k\| \leq g, \quad \|\bar{B}_\ell\| \leq b,$$

including the corresponding initial-source bound. Factoring each $K_{\ell k}$ through the quotient proves (74) with $M = bCg$. All constants and quotient norms must be controlled uniformly for a uniform conclusion.

This is strictly weaker than contraction of the full omitted operator. For example, $D = \text{diag}(\rho, 2)$, $G = (1, 0)^\top$, and $B = (1, 0)$ give $BD^j G = \rho^j$ despite $\|D\| = 2$. The growing second coordinate is absent from every declared source-to-readout path. Conversely, deleting it would not be justified if either the source family or required future readouts were enlarged to observe it.

13.2 Closing a memory approximation around a retained recursion

The input in the preceding theorem is a declared exogenous signal. In a Mori–Zwanzig application it can contain the evolving retained state itself. A small open-loop memory tail then needs a closed-loop amplification bound. This can be stated without assuming stationarity.

Let x be the stacked transverse retained sequence and suppose

$$x = \mathcal{V}f + \mathcal{V}\mathcal{K}x, \quad \hat{x} = \mathcal{V}f + \mathcal{V}\mathcal{K}_m\hat{x}.$$

Here $\mathcal{V}$ is the direct retained response and $\mathcal{K}$ is the omitted feedback memory; the symmetry coordinate has already been carried exactly. These are finite causal sequence operators. Suppose, in a fixed signal norm,

$$\|\mathcal{V}\| \leq v, \quad \|\mathcal{K}\|, \|\mathcal{K}_m\| \leq k, \quad \|\mathcal{K} - \mathcal{K}_m\| \leq \delta_m, \quad vk < 1.$$

Then

$$\|x - \hat{x}\| \leq \frac{v^2 \delta_m}{(1 - vk)^2} \|f\|. \quad (79)$$

Indeed, the geometric Neumann series bounds both inverses $(I - \mathcal{V}\mathcal{K})^{-1}$ and $(I - \mathcal{V}\mathcal{K}_m)^{-1}$ by $(1 - vk)^{-1}$. Subtracting the two equations gives

$$x - \hat{x} = (I - \mathcal{V}\mathcal{K})^{-1} \mathcal{V}(\mathcal{K} - \mathcal{K}_m)\hat{x},$$

which proves the bound. If the feedback blocks have the geometric envelope of (74), then $\delta_m \leq M\rho^m/(1 - \rho)$ in the supremum and ℓ^2 norms.

The small-gain hypothesis is sufficient, not necessary. It may be replaced by separately proved uniform bounds on the two exact causal inverses. For a scalar final functional c^*x , only

$$\|c^*(I - \mathcal{V}\mathcal{K})^{-1}\mathcal{V}\|$$

is needed on the left of the residual. This readout-weighted resolvent can be small even if the full-state inverse is large. Carrying one neutral mode does not itself establish any of these transverse bounds.

14 Conditional information contraction as a route to the observable gap

There is a precise bridge from a conditional gate-process description to the preceding operator estimates. It also identifies an obstruction to applying a mixing theorem directly to the deterministic network.

Let S be a carried global variable and let (X_ℓ) , conditional on $S = s$, be a nonautonomous Markov chain with reference laws $\pi_{\ell,s}$ and kernels $K_{\ell,s}$ satisfying $\pi_{\ell,s}K_{\ell,s} = \pi_{\ell+1,s}$. Assume regular conditional laws exist, as they do for finite gate alphabets and standard Borel state spaces. The conditional-centered score space is

$$\mathcal{H}_\ell^\circ = \{f(S, X_\ell) : \mathbb{E}f^2 < \infty, \mathbb{E}[f | S] = 0\}.$$

Its forward score operator is

$$\mathcal{T}_\ell f = \mathbb{E}[f(S, X_\ell) | S, X_{\ell+1}].$$

For each fiber, let $\rho_\ell(s)$ be the supremum of $|\mathbb{E}[fg | S = s]|$ over conditionally centered functions of X_ℓ and $X_{\ell+1}$ of conditional variance one. This is the maximal correlation of the conditional input–output pair.

Theorem 14.1 (Conditional maximal correlation implies memory decay). *If $\rho_\ell(s) \leq \rho < 1$ for almost every s and every layer, then*

$$\|\mathcal{T}_\ell f\|_2 \leq \rho \|f\|_2 \quad (f \in \mathcal{H}_\ell^\circ), \quad \|\mathcal{T}_{j-1} \cdots \mathcal{T}_k\| \leq \rho^{j-k}.$$

Consequently bounded conditional-centered source maps of norm at most g and readout maps of norm at most b satisfy (74) with $M = bg$ for the associated score system.

A sufficient condition is the reference-law minorization

$$K_{\ell,s}(x, A) \geq \varepsilon \pi_{\ell+1,s}(A) \tag{80}$$

for all measurable A , almost every admissible (s, x) , and a fixed $\varepsilon > 0$. Under this condition one may take $\rho = 1 - \varepsilon$.

Proof. For $f \in \mathcal{H}_\ell^\circ$ and $g \in \mathcal{H}_{\ell+1}^\circ$,

$$\begin{aligned} |\langle \mathcal{T}_\ell f, g \rangle| &= |\mathbb{E}[fg]| \\ &\leq \mathbb{E} \left[\rho_\ell(S) \sqrt{\mathbb{E}[f^2 | S] \mathbb{E}[g^2 | S]} \right] \leq \rho \|f\|_2 \|g\|_2. \end{aligned}$$

The score remains conditionally centered by the tower property. Hilbert duality proves the norm bound and multiplication proves the product bound. Composing with the specified source and readout maps gives the kernel bound.

For $0 < \varepsilon < 1$, minorization gives $K_{\ell,s} = \varepsilon \Pi_{\ell+1,s} + (1 - \varepsilon)R_{\ell,s}$, where $\Pi_{\ell+1,s}$ ignores its input and samples the reference output, and $R_{\ell,s}$ is a Markov kernel. The marginal identity implies $\pi_{\ell,s}R_{\ell,s} = \pi_{\ell+1,s}$. On centered output functions the backward kernel is therefore $(1 - \varepsilon)R_{\ell,s}$. Jensen’s inequality, followed by integration against $\pi_{\ell,s}$, proves that $R_{\ell,s}$ has L^2 norm at most one. Taking its adjoint gives the same bound for the forward score operator. The case $\varepsilon = 1$ is complete conditional independence and has zero centered operator. $\square$

Fisher and χ^2 interpretation. For a bounded score f with $\mathbb{E}[f | S] = 0$, the positive local density family $1 + tf$ preserves the marginal of S . Its output density is $1 + t\mathcal{T}_\ell f$. Its input and output χ^2 divergences are exactly $t^2 \|f\|_2^2$ and $t^2 \|\mathcal{T}_\ell f\|_2^2$. Thus the squared norm bound is contraction of the local distinguishability remaining after the global variable has been specified. The relation between maximal correlation and the χ^2 contraction coefficient is treated in M. Raginsky, *Strong data processing inequalities and Φ -Sobolev inequalities for discrete channels*, Theorem 3.2 and Remark 3.4, <https://arxiv.org/pdf/1411.3575>. The conditional finite-depth argument used here is proved above and does not assume that a given gate closure satisfies its Markov or minorization hypotheses.

Theorem 14.2 (No strict full conditional L^2 gap for a deterministic layer). *Suppose $Y = F(X, S)$ deterministically, with the exact pushed-forward reference law, and suppose there is a nonzero $g(Y, S) \in L^2$ with $\mathbb{E}[g | S] = 0$. Then the forward score operator*

$$Tf = \mathbb{E}[f(X, S) | Y, S]$$

has norm exactly one between the full conditional-centered L^2 spaces.

Proof. Conditional expectation has norm at most one. Set $f(X, S) = g(F(X, S), S)$. Exact pushforward gives $\|f\|_2 = \|g\|_2$ and $\mathbb{E}[f | S] = 0$. Since $f = g(Y, S)$, its conditional expectation given (Y, S) is g , so equality in the norm bound is attained. $\square$

Therefore scale conditioning alone cannot produce a strict contraction on all score directions of an exact deterministic ReLU layer. A gap must be proved on the declared source/readout class, or for a genuinely stochastic coarse conditional state. If the state is the entire gate history, it retains the previous history as a deterministic component, which can itself force maximal correlation one. A finite-memory gate state needs its Markov property and any truncation error justified separately. This full- L^2 operator-norm obstruction does not contradict weak mixing, decay for selected observables, or summable variations of a g -function; those are different estimates, so it does not by itself refute a measured or restricted-mode statement in Note XIX.

Conditioning on one random variable S removes the whole function space $L^2(S)$ from the conditional-centered problem. That space is generally infinite dimensional. Carrying one scalar gain coefficient removes only one chosen tangent direction unless the required dependence on S is otherwise represented exactly. The distinction between one global variable and one linear moment is consequential for any proposed conditional-gap proof.

15 Homogeneity and a spectral gap do not imply short memory

Fix $m \geq 1$ and consider coordinates $(r, z_1, \dots, z_m)$ with the update

$$r' = r, \quad z'_1 = r, \quad z'_j = z_{j-1} \quad (2 \leq j \leq m). \quad (81)$$

Its matrix P_m has nonnegative entries and every row sums to one. It fixes the all-ones vector, and on the positive cone it is exactly the bias-free ReLU layer $h \mapsto \text{ReLU}(P_m h) = P_m h$. Set $y_j = z_j - r$. Then

$$r' = r, \quad y' = S_m y, \quad S_m e_j = e_{j+1} \quad (j < m), \quad S_m e_m = 0.$$

Thus P_m has one simple eigenvalue 1 and all its other eigenvalues are zero. Its exact neutral channel is already separated. Nevertheless,

$$\|S_m^j\| = 1 \quad (0 \leq j < m), \quad e_m^* S_m^{m-1} e_1 = 1.$$

An arbitrarily small perturbation in the first transverse coordinate of a strictly positive reference propagates unchanged to the last coordinate after $m - 1$ layers. No constants C and $\rho < 1$, independent of m , can satisfy $\|S_m^j\| \leq C\rho^j$: use $j = m - 1$.

If a scalar source enters by $G = e_1$ and the output is $B = e_m^*$, then $BS_m^j G$ equals one at $j = m - 1$ and zero at every other $j \geq 0$. This is an exact m -step delay. Across a sufficiently long cut, its past-to-future matrix contains an $m \times m$ permutation matrix, so the declared causal linear source-to-output family requires at least m memory coordinates. Indeed, the m input impulses immediately preceding the cut produce the m coordinate basis vectors as states, and their subsequent outputs form the corresponding permuted basis. This last minimal-dimension assertion uses the specified independently variable input family; it is not inferred from a single initial probe.

This example has stronger row homogeneity than a two-gain row-sum check, a single exact neutral scale direction, and a maximal eigenvalue gap. It still has arbitrarily long observable memory. What fails is a uniform product or source-to-readout decay estimate; the nilpotent transverse block is increasingly nonnormal. A spectrum without control of the cocycle and its observable residues cannot certify the required memory bound.

Why a carried neutral coordinate is not enough for feedback. Even if a transverse omitted block has $\|D\| < 1$, the retained feedback can destabilize the combined transverse dynamics. For example,

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

has omitted block $D = 0$ but eigenvalues 2 and -2 . Appending an independent exact coordinate $a' = a$ changes neither instability. The example is an algebraic warning about the logical implication, not a model of the reported networks: stable omitted propagation alone does not bound the amplification of a truncation error after re-closing the retained loop.

Precisely what remains for the reported networks. The new constructive route needs one matched tangent realization, the exact source family and readout metric, and a conditioned symmetry frame. One may then prove either small finite-horizon singular tails in $O_\ell W_\ell^{1/2}$ and use Theorem 11.1, or a uniform observable kernel bound and use Theorem 13.1. If retained variables feed the missing channel, a transverse closed-loop response bound is also required. Homogeneity, a fitted 2×2 matrix, and agreement on two source probes establish none of these conditions by themselves. The propositions identify a quantitative theorem to seek; no unprovided network arrays have been treated as evidence for its hypotheses.

16 When conditional heat smoothing does and does not compose through depth

The Mehler multipliers in the factor calculation are exact within-layer conditional expectation operators. To turn them into a depth-memory bound, one needs a compatible stochastic transition, a controlled source/readout class, and a comparison with the exact dynamics. The next results state that bridge and two concrete obstructions to using it without those data.

16.1 The exact bridge for a Gaussian stochastic channel

Let $X \sim N(0, I_d)$ and

$$Y = AX + (I - AA^*)^{1/2}\xi, \quad \xi \sim N(0, I_e), \quad \xi \perp X, \quad \|A\|_{\text{op}} < 1,$$

where $A : \mathbb{R}^d \rightarrow \mathbb{R}^e$. Both reference marginals are standard Gaussian. The forward score map is $\mathcal{T}_A f = \mathbb{E}[f(X) | Y]$.

Proposition CL.1 (second quantization and the source-sector gap). On homogeneous Hermite chaos of degree $q \geq 1$, $\mathcal{T}_A$ is the symmetric tensor map $A^{\otimes q}$, with its Gaussian Hilbert normalization. Consequently

$$\|\mathcal{T}_A|_{\mathcal{H}_q}\| = \|A\|_{\text{op}}^q, \quad \|\mathcal{T}_A|_{L_0^2}\| = \|A\|_{\text{op}}. \quad (\text{CL.1})$$

If an actual nonautonomous Gaussian Markov process has such transitions A_ℓ and compatible Gaussian reference marginals, then

$$\mathcal{T}_{A_{j-1}} \cdots \mathcal{T}_{A_k} = \Gamma(A_{j-1} \cdots A_k), \quad \|\mathcal{T}_{A_{j-1}} \cdots \mathcal{T}_{A_k}|_{\mathcal{H}_q}\| \leq \prod_{\ell=k}^{j-1} \|A_\ell\|^q. \quad (\text{CL.2})$$

Here Γ denotes this explicitly defined action on each Hermite degree. In particular, a third-chaos source has exponent three in this genuine Gaussian channel, as long as the intervening dynamics preserves that degree.

Proof. The conditional Gaussian mean is $\mathbb{E}[X | Y] = A^*Y$, and its conditional covariance is $I - A^*A$. Therefore

$$\mathbb{E}[e^{t^*X - \|t\|^2/2} | Y] = e^{(At)^*Y - \|At\|^2/2}.$$

Comparing homogeneous coefficients gives the tensor action. A singular-value decomposition of A reduces it to independent scalar correlated Gaussian pairs. A tensor-product Hermite basis then has multipliers given by products of singular values, whose largest value in degree q is $\|A\|^q$. A degree- q Hermite polynomial in a leading singular direction attains it. The centered space is the orthogonal direct sum of degrees $q \geq 1$, so its norm is $\|A\|$. Tensor functoriality proves the composition identity. Conditional Markovness is what identifies this composition with the true multistep score map of the process. $\square$

The same proof applies fiberwise after specifying a global variable S , provided the standardized conditional Gaussian laws and the transition matrices are measurable in S . Uniform bounds must hold for almost every fiber. A nonlinear ReLU operation can mix Hermite degrees, so degree-three contraction cannot be imported across it without an invariant-degree or comparison theorem. The full centered estimate has exponent one.

16.2 Unary smoothing does not give a width-uniform vector gap

For the factor representation $Z = \mu + BX + D^{1/2}\epsilon$ with $D > 0$ diagonal, the standardized whole-vector channel has matrix

$$A = \Sigma^{-1/2}B, \quad \Sigma = D + BB^*.$$

Its maximal correlation is

$$\rho_{\text{vector}}^2 = \|A\|^2 = \frac{\lambda_{\max}(B^*D^{-1}B)}{1 + \lambda_{\max}(B^*D^{-1}B)}. \quad (82)$$

Indeed, direct multiplication proves $\Sigma^{-1}B = D^{-1}B(I + B^*D^{-1}B)^{-1}$, so $A^*A = C(I + C)^{-1}$ for $C = B^*D^{-1}B$. Proposition CL.1 identifies its largest square-root eigenvalue with maximal correlation. For zero factor dimension, the centered factor space is zero and the channel norm is defined to be zero; the eigenvalue formula is used when the factor dimension is positive.

The unary factors used in the conditional-moment expansion are only $r_i = \|B_i\|/\sqrt{\|B_i\|^2 + D_{ii}}$. They measure individual channels; (82) measures all observations jointly.

Proposition CL.2 (replicated-factor counterexample, including ReLU). Fix $s > 0$ and take

$$Z_i = X + s\epsilon_i, \quad H_i = (Z_i)_+, \quad X, \epsilon_1, \epsilon_2, \dots \text{ independent standard Gaussians.}$$

Every unary Mehler factor is the same fixed number $r_i = (1 + s^2)^{-1/2} < 1$. Nevertheless

$$\rho(X, (Z_1, \dots, Z_n))^2 = \frac{n}{n + s^2} \longrightarrow 1. \quad (\text{CL.3})$$

The maximal correlation with the ReLU observations also tends to one. More quantitatively, set

$$m_s(x) = \mathbb{E}_\epsilon(x + s\epsilon)_+, \quad \eta(X) = \arctan(m_s(X)), \quad V_\eta = \text{Var}(\eta) > 0.$$

Then

$$\rho(X, (H_1, \dots, H_n))^2 \geq \max \left\{ 0, 1 - \frac{s^2}{nV_\eta} \right\} \longrightarrow 1. \quad (83)$$

Proof. For the unrectified vector, B is the all-ones column and $D = s^2 I_n$. Substitution into (82) gives (CL.3).

For ReLU, $m'_s(x) = \Phi(x/s) > 0$, so m_s is strictly increasing and $V_\eta > 0$. Let $\bar{H}_n = n^{-1} \sum_i H_i$. Conditional independence gives

$$\mathbb{E}|\bar{H}_n - m_s(X)|^2 = \frac{1}{n} \mathbb{E} \text{Var}(H_1 | X) \leq \frac{s^2}{n}.$$

For the last inequality, at each fixed $X = x$ use an independent residual copy H'_1 and the ReLU contraction: $\text{Var}(H_1 | x) = \frac{1}{2} \mathbb{E}[(H_1 - H'_1)^2 | x] \leq \frac{1}{2} s^2 \mathbb{E}(\epsilon_1 - \epsilon'_1)^2 = s^2$. The function $\arctan$ is bounded and one-Lipschitz, so

$$\mathbb{E}|\arctan(\bar{H}_n) - \eta(X)|^2 \leq s^2/n.$$

Conditional expectation is the best square-integrable predictor from the whole observation vector. For $g = \eta - \mathbb{E}\eta$,

$$\|g - \mathbb{E}[g | H_1, \dots, H_n]\|_2^2 \leq s^2/n.$$

Orthogonality of the conditional expectation gives $\|\mathbb{E}[g | H_1, \dots, H_n]\|_2^2 \geq V_\eta - s^2/n$. Dividing by V_η gives a lower bound on the squared maximal-correlation norm. Its universal upper bound is one. $\square$

Thus geometric tails for every unary conditional moment can coexist with almost perfect recovery of the common factor from a wide activation vector. There is no contradiction: the first claim bounds individual queries; the second permits all joint observation functions. A memory theorem must specify which of those classes it uses.

16.3 One-step Gaussian kernels can hide a non-Markov global mode

Proposition CL.3 (a contracting fitted kernel with persistent true memory). Let $0 < a < 1$, and define for all integer layers

$$Y_\ell = aS + \sqrt{1 - a^2} \epsilon_\ell, \quad S, (\epsilon_\ell)_\ell \text{ independent standard Gaussians.}$$

Each marginal is standard Gaussian. Every adjacent conditional law is

$$\mathcal{L}(Y_{\ell+1} \mid Y_\ell = y) = N(a^2 y, 1 - a^4). \quad (\text{CL.4})$$

Its centered score operator has norm $a^2 < 1$. But for every $j \neq k$,

$$\text{Corr}(Y_j, Y_k) = a^2, \quad \|f \mapsto \mathbb{E}[f(Y_k) \mid Y_j]\|_{L_0^2 \rightarrow L_0^2} = a^2. \quad (\text{CL.5})$$

In particular, multiplying the adjacent kernels predicts the wrong multistep score operator.

Proof. The variables are jointly Gaussian and have covariance a^2 at every distinct pair. Gaussian regression gives (CL.4) and the same formula for every positive separation. Proposition CL.1 gives the score norms. A Markov process with the adjacent kernel (CL.4) would instead have lag- m covariance a^{2m} . Here

$$\text{Cov}(Y_0, Y_2 \mid Y_1) = a^2 - a^4 > 0,$$

which directly proves failure of the Markov property. $\square$

The actual conditional process given S is independent across layers: $Y_\ell \mid S = s$ has mean as and variance $1 - a^2$, and every conditional centered transition has norm zero. This differs from fitting unconditioned one-step kernels and multiplying them. Carrying the random global variable S exactly removes this example's persistent information; carrying only one coefficient describing its marginal distribution does not in general represent all its conditional dependence.

16.4 The exact comparison term needed for deterministic layers

Let $T_\ell : \mathcal{H}_\ell \rightarrow \mathcal{H}_{\ell+1}$ be the exact score maps, with $\|T_\ell\| \leq 1$. Let $C_\ell : \mathcal{H}_\ell \rightarrow \mathcal{K}_\ell$ be specified coarse-observable maps with $\|C_\ell\| \leq 1$. These maps are part of a matched realization; they are not identified across different reference laws without justification. Suppose a proposed stochastic coarse model has $S_\ell : \mathcal{K}_\ell \rightarrow \mathcal{K}_{\ell+1}$ with $\|S_\ell\| \leq \rho$ for $0 < \rho < 1$. Define its intertwining defect

$$E_\ell = C_{\ell+1}T_\ell - S_\ell C_\ell.$$

Proposition CL.4 (composition error and its nondecaying floor). For $j > k$,

$$\begin{aligned} & C_j T_{j-1} \cdots T_k - S_{j-1} \cdots S_k C_k \\ &= \sum_{\ell=k}^{j-1} S_{j-1} \cdots S_{\ell+1} E_\ell T_{\ell-1} \cdots T_k. \end{aligned} \quad (84)$$

Empty products are identities. In particular, if $\|E_\ell\| \leq \delta$,

$$\|C_j T_{j-1} \cdots T_k - S_{j-1} \cdots S_k C_k\| \leq \frac{\delta(1 - \rho^{j-k})}{1 - \rho}. \quad (\text{CL.6})$$

The bound approaches an error floor; it does not by itself prove decay of the exact coarse response with source age.

For a specified source map G_k , a sharper statement uses

$$\varepsilon_{\ell k} = \|E_\ell T_{\ell-1} \cdots T_k G_k\|.$$

For a readout B_j with $\|B_j\| \leq b$,

$$\|B_j C_j T_{j-1} \cdots T_k G_k\| \leq b \left(\rho^{j-k} \|C_k G_k\| + \sum_{\ell=k}^{j-1} \rho^{j-\ell-1} \varepsilon_{\ell k} \right). \quad (\text{CL.7})$$

Thus an actual geometric observable-memory bound follows if these source-restricted defects themselves decay with the age of the source, or vanish on the relevant reachable queries.

Proof. Insert $C_{\ell+1} T_\ell = S_\ell C_\ell + E_\ell$ repeatedly, beginning at the last layer. The resulting telescoping sum is (84). Taking operator norms and using the exact contractions proves (CL.6). Apply the identity to G_k before taking norms to obtain (CL.7). $\square$

For example, if $\varepsilon_{\ell k} \leq g\delta\theta^{\ell-k}$ with $0 < \theta < 1$, the added convolution for $m = j - k$ is bounded by

$$bg\delta \begin{cases} (\rho^m - \theta^m)/(\rho - \theta), & \rho \neq \theta, \\ m\rho^{m-1}, & \rho = \theta. \end{cases}$$

It then decays geometrically at any slightly larger rate than $\max(\rho, \theta)$. A width-independent bound on the one-step defect alone does not supply this source-age decay.

The shared-factor example makes the floor concrete even in two linear score coordinates. For the exact full state (S, ϵ_ℓ) its first-chaos score map is $T = \text{diag}(1, 0)$. The coarse observation has $C = (a, \sqrt{1-a^2})$ and its adjacent scalar model is $S_c = a^2$. Hence

$$\|CT - S_c C\| = a\sqrt{1-a^2}, \quad CT^m e_1 = a, \quad S_c^m C e_1 = a^{2m+1} \quad (m \geq 1).$$

The one-step defect can be arbitrarily small as $a \downarrow 0$, but its response to the persistent hidden source does not decay with source age. All these operators are contractions in their declared Gaussian score metrics.

The within-layer conditional factor theorem controls certain C_ℓ queries and their approximation. An exact depth theorem additionally needs the transition realization, the compatibility of its conditional laws, and control of the defects in (84). These are the precise comparison data separating a useful Gaussian heat calculus from an unjustified mixing assertion about the fixed deterministic network.

17 The exact source observable behind the quadratic residual

The relative determinant route identifies a residual norm whose derivative is additional fourth-order information. The following envelope identity removes an otherwise unnecessary derivative of the regression coefficients.

Theorem 17.1 (Source derivative of the optimal quadratic residual). *Let $H = (H_1, \dots, H_n)$ be fixed real observables, and let μ_ϵ be a differentiable probability family with centered score h at zero. Assume the mixed moments below can be differentiated and the covariance remains positive definite near zero. Fix a weight row w . Recompute in each law*

$$X_\epsilon = H - \mathbb{E}_\epsilon H, \quad Y_\epsilon = w^\top X_\epsilon, \quad v_\epsilon = \mathbb{E}_\epsilon Y_\epsilon^2, \quad C_\epsilon = \mathbb{E}_\epsilon[X_\epsilon X_\epsilon^\top],$$

$$t_\epsilon = \mathbb{E}_\epsilon[X_\epsilon Y_\epsilon^2], \quad R_\epsilon = Y_\epsilon^2 - v_\epsilon - X_\epsilon^\top C_\epsilon^{-1} t_\epsilon.$$

Then

$$\left. \frac{d}{d\epsilon} \|R_\epsilon\|_{L^2(\mu_\epsilon)}^2 \right|_0 = \mathbb{E}[hR_0^2]. \quad (\text{CR.1})$$

Consequently, putting $\alpha = C^{-1}t$ and evaluating all unmarked symbols at zero,

$$\delta_h \kappa_4(Y) = 2\dot{t}^\top \alpha - \alpha^\top \dot{C} \alpha + \mathbb{E}[hR^2] - 4v\dot{v}. \quad (\text{CR.2})$$

Here, if $b = \mathbb{E}[hX]$ and $\beta = w^\top b$,

$$\dot{C} = \mathbb{E}[hXX^\top], \quad \dot{v} = w^\top \dot{C} w, \quad \dot{t} = \mathbb{E}[hXY^2] - bv - 2\beta C w. \quad (\text{CR.3})$$

Proof. The normal equations imply $\mathbb{E}R = 0$ and $\mathbb{E}[XR] = 0$. Write $\alpha_\epsilon = C_\epsilon^{-1}t_\epsilon$. Since the underlying observables H and w are fixed,

$$\dot{X} = -b, \quad \dot{Y} = -\beta, \quad \dot{R} = -2\beta Y - \dot{v} + b^\top \alpha - X^\top \dot{\alpha}.$$

This is an affine linear combination of the components of X . Therefore $\mathbb{E}[R\dot{R}] = 0$, by the same normal equations. Differentiating the squared norm gives $\mathbb{E}[hR^2] + 2\mathbb{E}[R\dot{R}] = \mathbb{E}[hR^2]$. Differentiate the exact Schur identity $\kappa_4(Y) = t^\top C^{-1}t + \|R\|_2^2 - 2v^2$ to obtain (CR.2). Differentiating the definitions of C, v, t gives (CR.3); the centered first moments remove the other terms. $\square$

The essential additional source query is therefore hR^2 . There is no need to differentiate a fitted regression map to evaluate this term. It can be accumulated for the rows actually used by a future readout. For the omitted contraction, subtract the complete retained-sector derivative from (CR.2). Any outgoing gain normalization must still be differentiated.

This identity does not force the added term to have either sign: $R^2 \geq 0$, but the centered score h is signed. It also explains why knowing only the residual's unweighted norm is insufficient. The relevant object is its alignment with the specified incoming source. For example,

$$|\mathbb{E}[hR^2]| \leq \|h\|_2 \|R\|_4^2,$$

and for an approximation $\hat{R} \in L^4$,

$$|\mathbb{E}[h(R^2 - \hat{R}^2)]| \leq \|h\|_2 \|R - \hat{R}\|_4 (\|R\|_4 + \|\hat{R}\|_4). \quad (\text{CR.4})$$

These follow by Cauchy–Schwarz and Hölder. An L^2 approximation to R alone does not supply this L^4 certificate. The clipped conditional smoothing construction supplies one controlled way to obtain the required higher-norm accuracy for its specified conditional primitives.

18 How the three geometric constructions join

There is an exact common tangent between the Witten calculation and the relative determinant. For a bounded centered score h , the normalized positive harmonic state has $\dot{\Omega} = h/2$. Its rank-two skew-adjoint generator

$$K = |\dot{\Omega}\rangle\langle\Omega| - |\Omega\rangle\langle\dot{\Omega}|$$

satisfies $K\Omega = \dot{\Omega}$ and $\dot{P} = [K, P]$ for $P = |\Omega\rangle\langle\Omega|$. The Witten theorem gives

$$\boxed{K\Omega = \frac{h}{2} = -\mathcal{D}_{\perp}^{-1}\dot{\Omega}.} \quad (\text{CR.5})$$

This same K moves the probability vacuum in the relative counting determinant. Its fourth logarithmic derivative is the source response of the fourth cumulant. Thus one continuous displacement has three exact descriptions: a Gaussian Dirac reduced-resolvent response, a change of probability state, and a cumulant-generating determinant variation.

For $h = S_T$ understood through the proved cutoff limit, $\dot{\Omega} = -dS_T/2$. Its coefficients have Hermite degree two and exterior degree one, hence total energy three. Therefore

$$\|\mathcal{D}_{\perp}^{-1}\dot{\Omega}\|^2 = \frac{\|S_T\|_2^2}{4} = \frac{\|T\|_F^2}{24}.$$

This is precisely the Fisher metric computed from the moving state. The equality proves the comparison rather than just assigning similar names to separate expressions.

The conditional-cumulant route gives the probabilistic contents of the same response. For a Gaussian factor reference, the pair source creates d_{11}^{ij} , a conditional covariance response. Its connected correlations with the other conditional fields produce the omitted multiplicity classes. The quadratic Gram route packages those same fourth-order correlations into positive matrices and identifies their linearly predictable third-order part. Neither representation changes the declared incoming tensor or supplies unmeasured coefficients.

18.1 What the duality framework contributes

For a finite collection of gates, the Boolean algebra generated by their events is represented by indicator multiplication on the reference probability space. Passing from a gate history to a smaller admissible state is represented on L^2 by conditional expectation. This statement is exact once the state and reference law are specified. The lattice description organizes which observables belong to a state; the probability law supplies their Hilbert metric.

The polynomial span of two gain influences is generally not an algebra: its products need not remain in that span. Its orthogonal projection therefore differs from conditional expectation onto a gate sigma-algebra. The failure of multiplication to survive compression is measured by

$$PM_f^*M_fP - (PM_fP)^*(PM_fP) = PM_f^*(I - P)M_fP \succeq 0.$$

Here the equality holds for bounded f and any orthogonal projection P ; for polynomial observables it holds as a quadratic-form identity on the specified finite moment domain. It follows by inserting $I = P + (I - P)$. The quadratic residual and relative determinant are concrete descendants of this product defect.

This distinguishes the useful bridges from several unsupported shortcuts. Classical gate indicators commute. A noncommutative algebra appears when operators implementing transitions, conditional expectations, or compressions are included. A skew Boolean representation of partial sections requires its own concrete operations; an order-dependent approximation alone does not establish its axioms. Likewise, the set partition lattice used in cumulant inversion is not a distributive lattice:

the lattice of partitions of a three-element set is the five-element diamond. One cannot directly apply a distributive-lattice duality theorem to that partition lattice.

Topological duality, as developed by Gehrke and van Gool [10], organizes the state spaces and observable algebras. The present work then adds a measure, a source, and an operator metric. The quantitative modular theorem in this note is a further bridge: the failure of a finite prefix algebra to be preserved by the modular generator controls conditional information contraction, provided the tables define the matched Markov process. That is additional analytic information beyond the topology.

19 A constructive route from the missing feed to a certified deep model

The proved results separate three operations that must have the same source and readout conventions.

First specify the tangent to be transported. At a Gaussian reference, a pure third tensor gives the Hermite score S_T . A unit scale-mixture direction can also contain reference drift and a fourth-tensor component; those contributions remain separate and must be calibrated to the same gain coordinate as the ledger. The all-distinct formula includes both pair and fully distinct incoming triples when the full tensor is intended.

Next evaluate the additional observables. One may use the exact weak gradient formula, the conditional fields d_{ab}^{ij} , or the source-weighted quadratic residual hR_w^2 . These are different exact representations of explicitly declared responses. The covariance-path estimates certify changes between Gaussian references; the conditional Hermite and clipping estimates certify numerical replacement of the conditional primitives. Neither certificate by itself bounds the error between a Gaussian reference and the actual network law.

Finally form a matched dynamic realization. The compatible Gramian projection theorem constructs updates between moving reduced spaces and bounds the entire causal source-to-readout error. Its rank is selected by the singular values of $O_\ell W_\ell^{1/2}$, using the declared source and output metrics. A bounded number of scalar output channels does not make these past-to-future maps low rank.

There are two justified closure regimes. Over the finite depth of the experiment, one can use the computable sum of discarded singular tails. For a depth-uniform theorem, one needs observable kernel decay. Uniform conditional modular diameter supplies such a decay estimate for an actual conditional Markov family. A comparison with deterministic network scores must then control the explicitly displayed intertwining defects, including their dependence on source age.

If separate arguments give transfer-operator errors $\varepsilon_{\text{reference}}$ for the chosen law/model comparison and $\varepsilon_{\text{eval}}$ for coefficient evaluation in the same input and output norms, then applying the compatible reduction to that reference realization gives

$$\|\mathcal{T}_{\text{true}} - \widehat{\mathcal{T}}\| \leq \varepsilon_{\text{reference}} + \varepsilon_{\text{eval}} + \sum_{\ell=0}^L \sigma_{r_{\ell}+1}(O_\ell W_\ell^{1/2}). \quad (\text{CR.6})$$

This is the triangle inequality plus the proved reduction theorem. Its premise is about full transfer-operator errors; a small error in one moment table must not be substituted for either epsilon without a propagation argument. The composition and feedback theorems specify such arguments and the amplification factors they require.

The remaining network-specific implication. The exact unknown can now be stated as the source-observable geometry of the conditional covariance responses and quadratic residuals over

depth. For the reported networks, the source calibration, conditional reference arrays, weight rows, and output projection would determine these quantities. They would also determine whether the extra observable family has small reachable–observable rank or a controlled modular/conditional contraction. None of those numeric properties follows from the ordinary index, from a single scale register, or from a two-gain fit alone.

The general theoretical development has therefore produced a complete response mechanism and conditional reduction theorems. The actual deep g_4/g_3 coefficient remains unevaluated, because the network-specific comparison data were not supplied. This note makes no change to the fitted chain table and treats the previously reported percentages as reports in Note XIX, not as newly reproduced experiments.

20 Proof review, computation scope, and sources

The four initial routes were developed separately and reviewed by another participant against their raw proofs. The review reconstructed the conditional partition multiplicities, the fourth determinant coefficient, the quadratic susceptibility, the compatible Gramian reduction, and the Witten domain and sign conventions. It corrected two material qualifications: the reduced-resolvent expectation formula is stated for real observables, and the outgoing normalized gain includes the derivative of its variance denominator. Later cross-route additions were reviewed separately. This is internal mathematical review, not external peer review.

The conditional-route exact computation enumerated the 15 partitions of four labels to check the marked-block coefficients and the constants 11 and 38 in the reference-derivative bound. It also checked the Mehler jet coefficients in 36 rational polynomial cases, of degrees zero through eight and derivative orders zero through three. These finite computations support transcription and combinatorial checks. The general proofs in the text, rather than those bounded computations, establish the theorems. No new network Monte Carlo experiment was performed.

The bibliography identifies primary mathematical sources used to verify the adjacent theorem conventions. The main operator, probability, and approximation claims are derived in this note. No comprehensive novelty claim is made. The project evidence is the user-supplied Note XIX and accompanying notes; a source repository and raw network state arrays were not available.

References

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